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Brozy et al.(10) **Pub. No.: US 2025/0277024 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **MOLECULES WITH MULTIPLE BINDING DOMAINS**(71) Applicants: **AMGEN INC.**, Thousand Oaks, CA (US); **AMGEN RESEARCH (MUNICH) GMBH**, Munich (DE)(72) Inventors: **Johannes Brozy**, Munich (DE); **Pavan Ghattyvenkatakrishna**, Thousand Oaks, CA (US); **Brendan Amer**, Thousand Oaks, CA (US)(21) Appl. No.: **18/038,886**(22) PCT Filed: **Dec. 2, 2021**(86) PCT No.: **PCT/US21/61587**

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(57)

ABSTRACT

New formats of multispecific molecules are described, as well as their methods of making. Additionally, uses in therapeutic indications are also described.

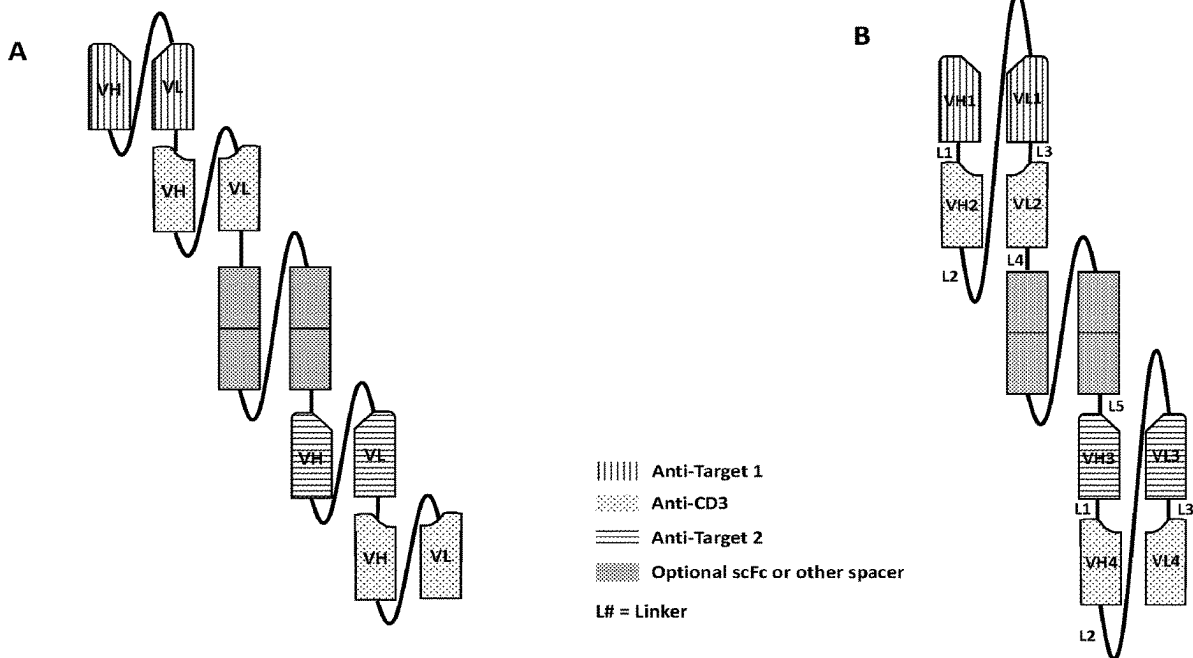
Specification includes a Sequence Listing.

FIGURE 1

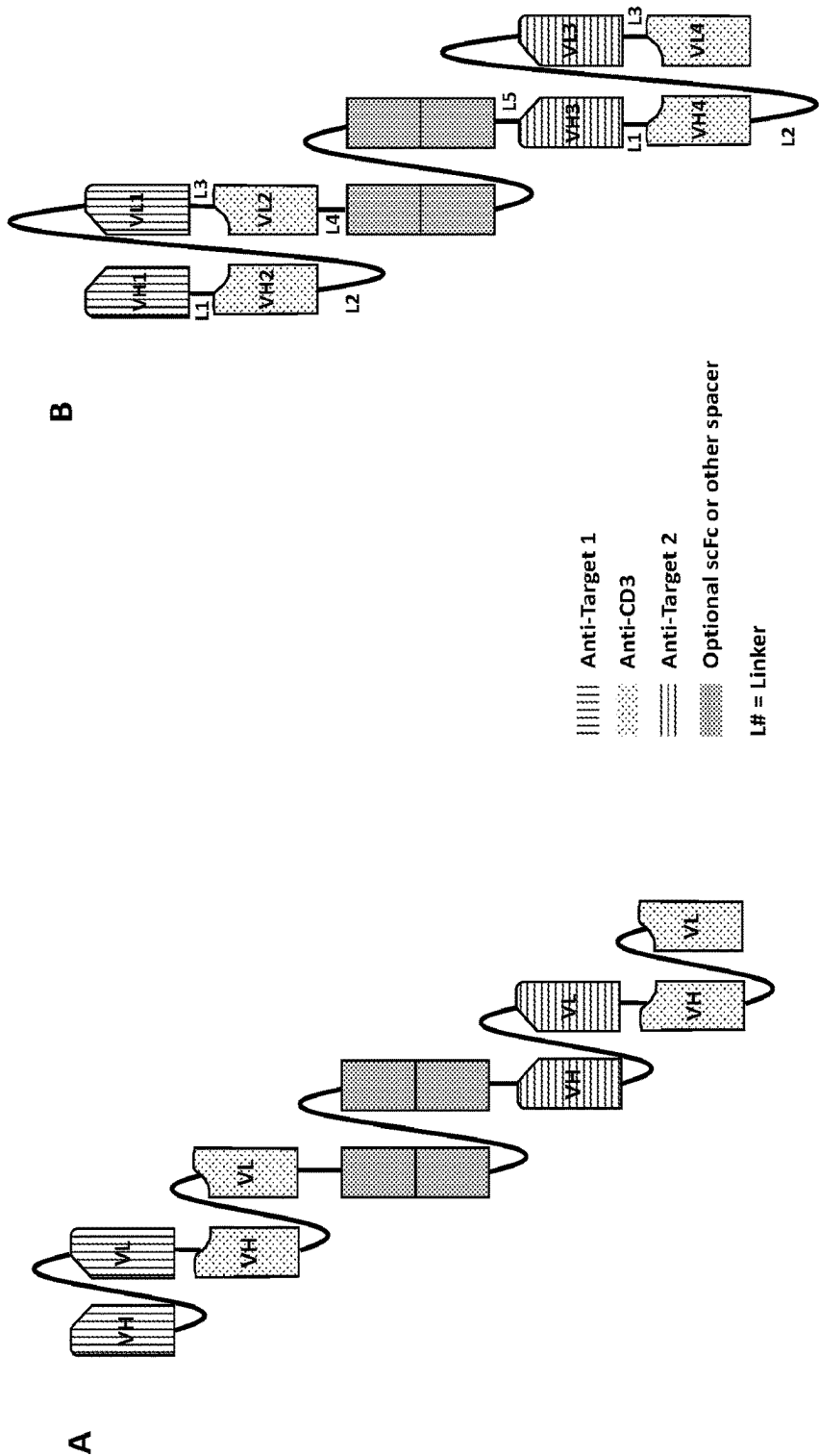


FIGURE 2

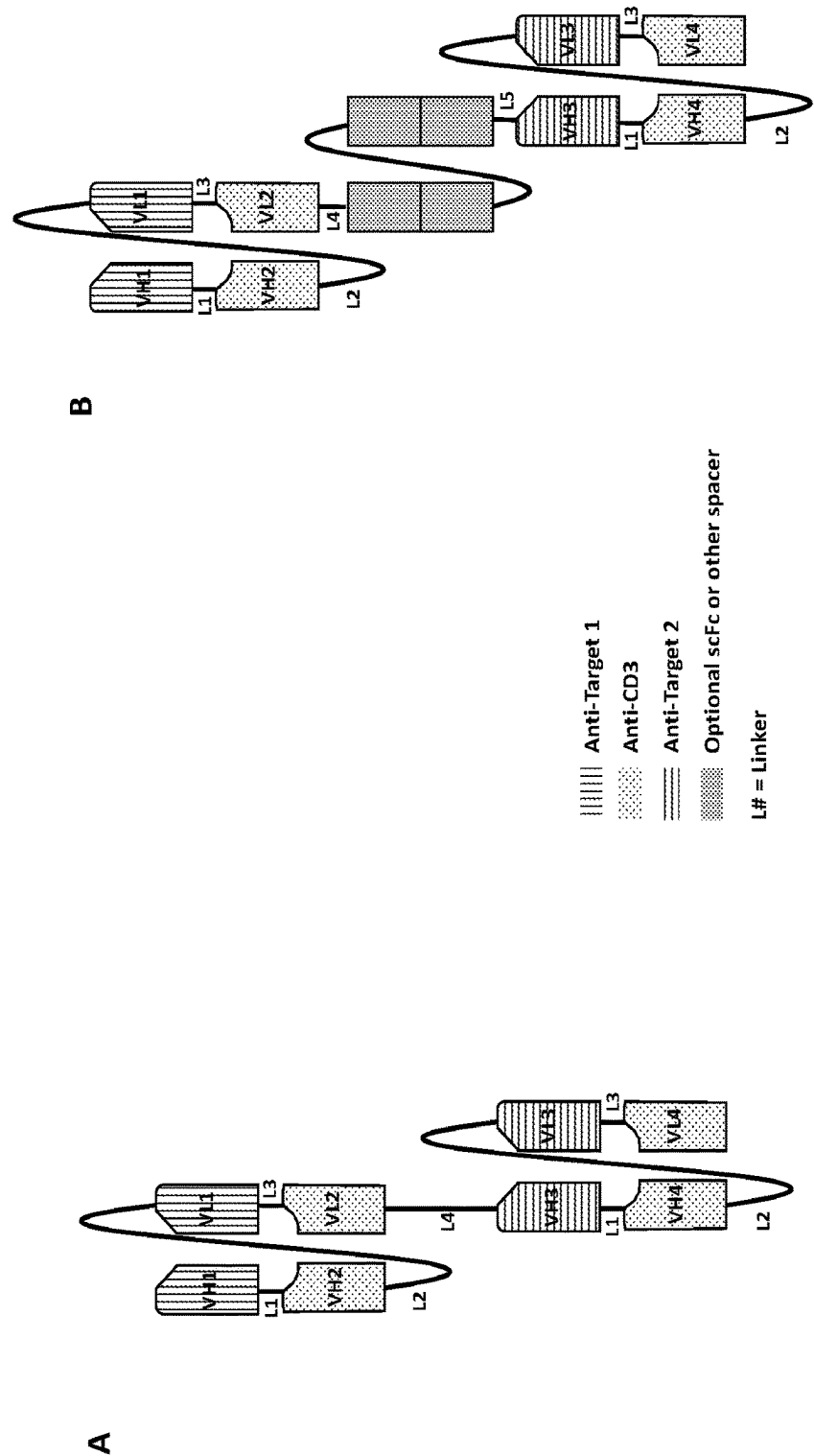


FIGURE 3

Preparative SEC of (HLHL)² and (HLL)² Formats Shows Comparable Profiles

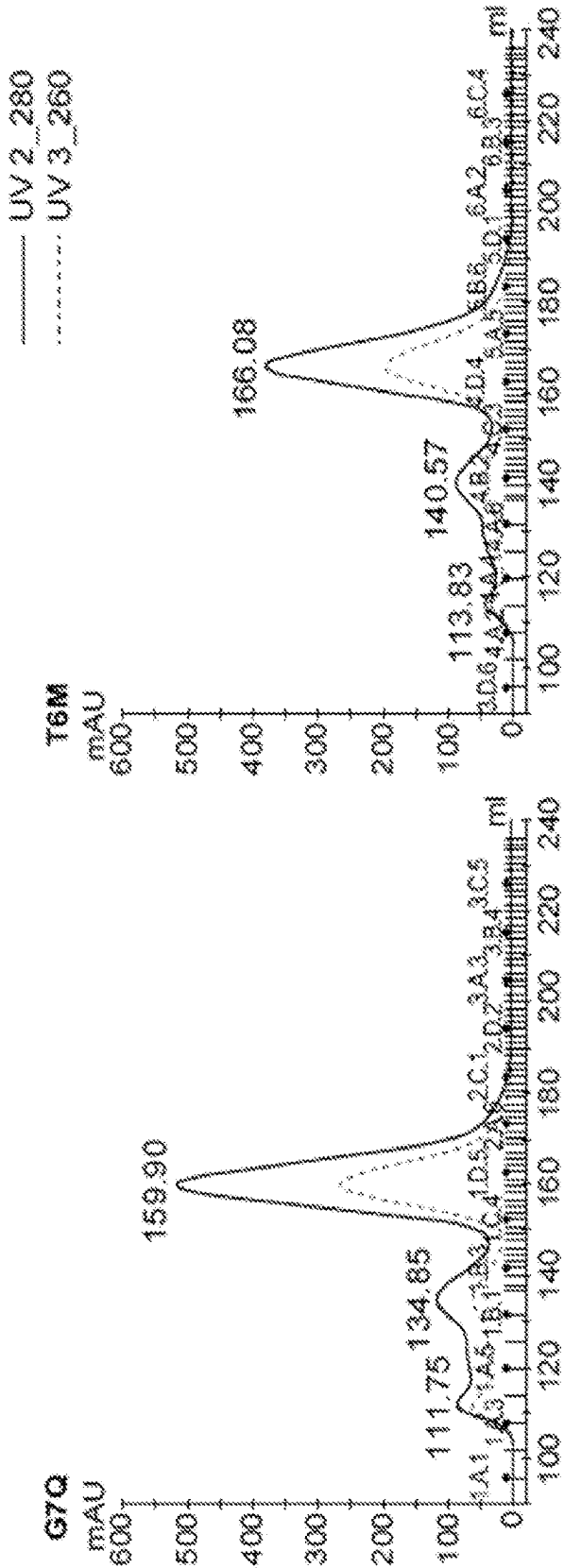


FIGURE 4

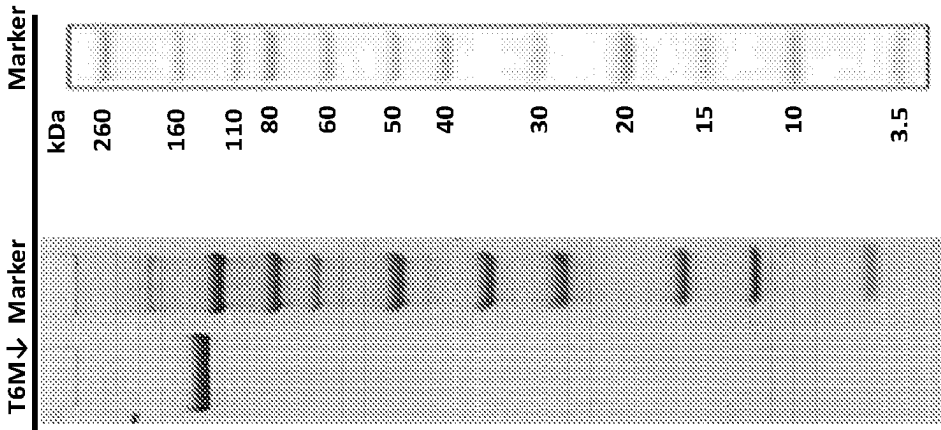


FIGURE 5

HHLL BiTE² Format Shows Superior TDCC Activity on Mono-Positive Target Cells at 48 Hours

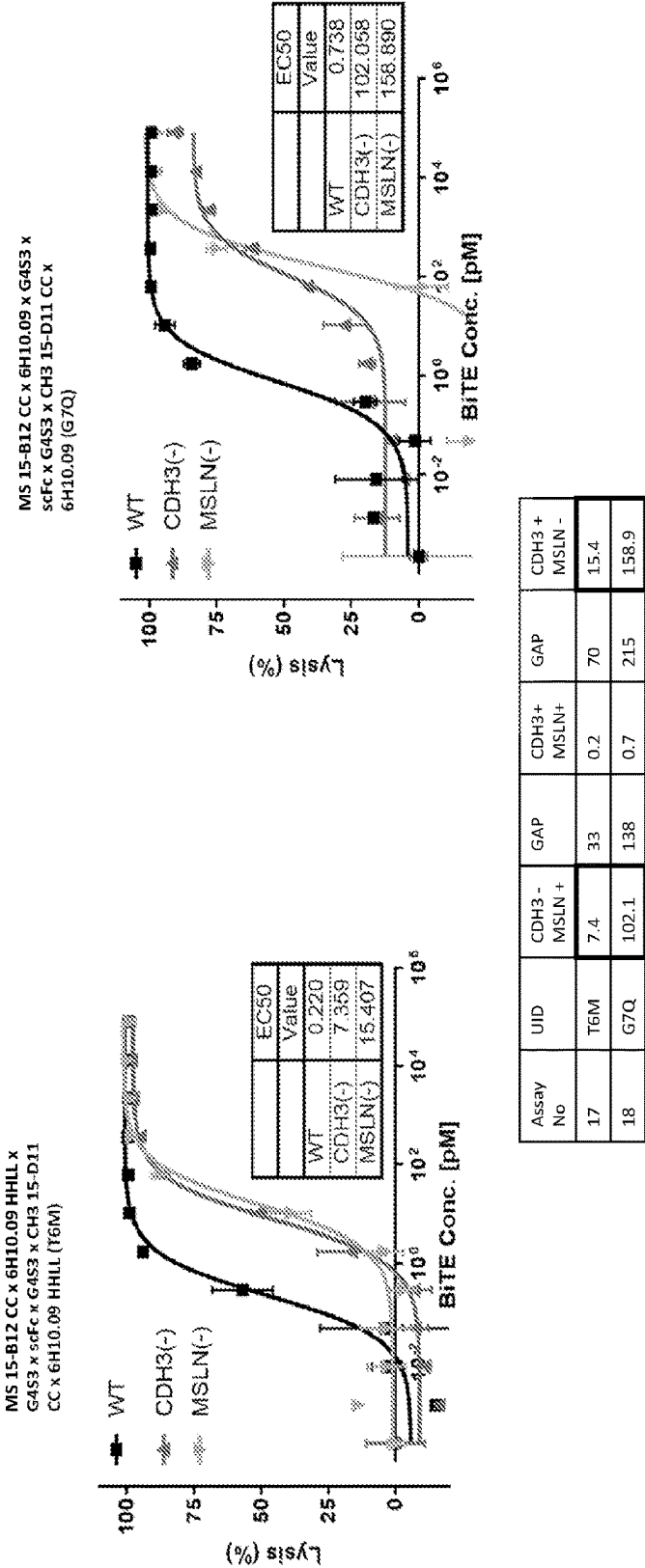
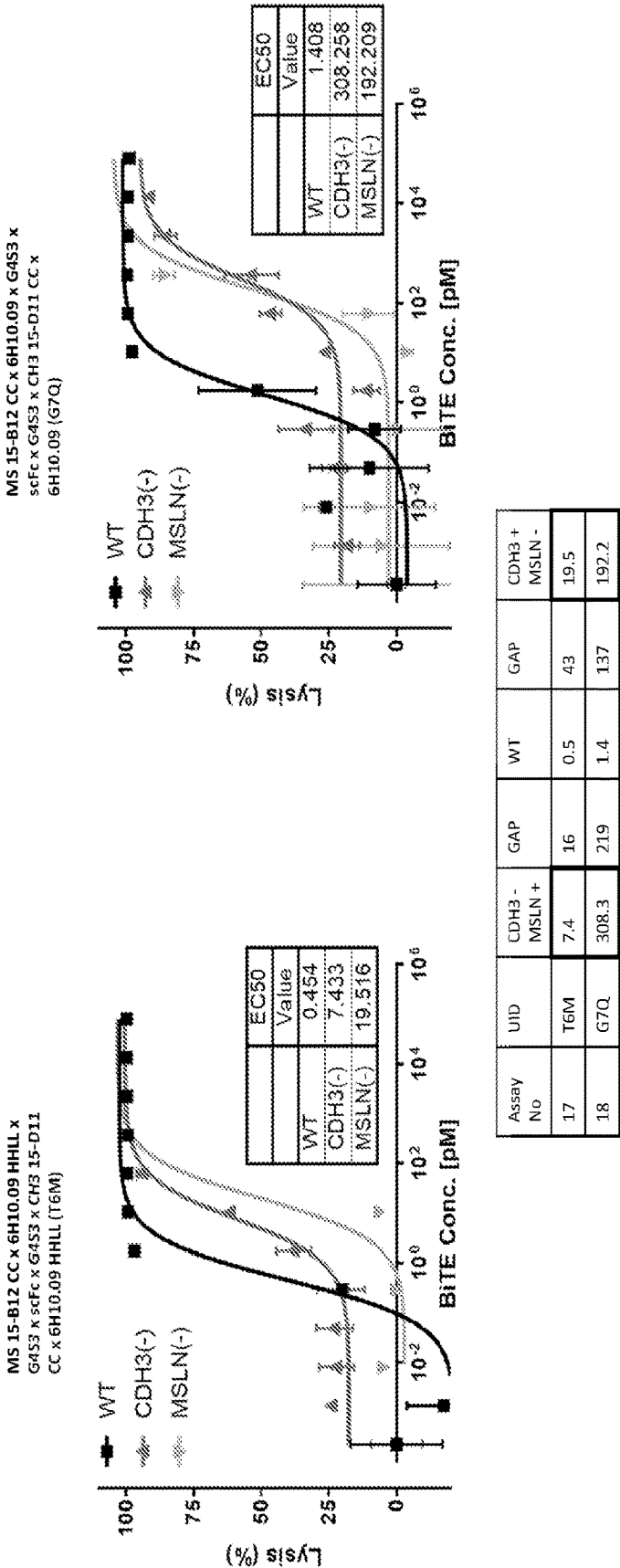


FIGURE 6

HHLL BiTE² Format Shows Superior TDCC Activity on Mono-Positive Target Cells at 72 Hours



MOLECULES WITH MULTIPLE BINDING DOMAINS

[0001] This application claims priority to U.S. Provisional Application No. 63/121,166, filed Dec. 3, 2020. The above-identified application is hereby incorporated herein by reference for all purposes.

[0002] This application includes a sequence listing submitted electronically by the file name 57674_Seqlisting.txt; Size: 116,880 bytes; Created: Apr. 27, 2023, and is incorporated herein by reference.

FIELD OF THE INVENTION

[0003] The invention is in the field of protein engineering.

BACKGROUND

[0004] Bispecific binding molecules have shown therapeutic promise in recent years. For example, a bispecific molecule that targets both CD3 and CD19 in a Bispecific T cell Engager (BiTE®) format has shown impressive efficacy at low doses. Bargou et al. (2008), Science 321:974-978. This BiTE® format comprises two scFv's, one of which targets CD3 and one of which targets a tumor antigen, CD19, joined by a flexible linker. This unique design allows the bispecific molecule to bring activated T-cells into proximity with target cells, resulting in cytolytic killing of the target cells. See, for example, WO 99/54440A1 (U.S. Pat. No. 7,112,324 B1) and WO 2005/040220 (U.S. Patent Appl. Publ. No. 2013/0224205A1). Later developments were bispecific constructs binding to a context independent epitope at the N-terminus of the CD3ε chain (see WO 2008/119567; U.S. Patent Appl. Publ. No. 2016/0152707A1).

[0005] In certain therapeutic indications, it may be desirable to target more than two targets. For example, in an immunooncology therapeutic indication tumor escape is a known mechanism where, through mutation and selective pressure of the treatment, tumors lose expression of the targeted antigen. When this occurs, the immunooncology therapeutic loses efficacy against the tumor cells. Adding additional antigen targets associated with the tumor is one manner that can address this type of tumor escape.

[0006] In addition to addressing tumor escape, a molecule comprising multiple target binding sites can also be useful for those targets that are expressed in relatively low levels by a cell. In this type of scenario, likely due to avidity effects, multiple binding sites on a single molecule to the same target can help overcome this low expression and improve target binding. Some examples of molecules that utilize multiple binding sites are provided in U.S. Provisional Patent Appl. No. 63/110,957.

[0007] In the biopharmaceutical industry, molecules are typically produced in a large-scale fashion in order to meet the commercial needs of supplying a large number of patients and can be assessed for a number of attributes to mitigate the risk that the molecule is not amenable to large-scale production and purification. Efficient expression of these complex, recombinant polypeptides can be an ongoing challenge. Further, even once expressed the polypeptides are often not as stable as desired for a pharmaceutical composition. Various attempts have been made to successfully alter molecule formats to address these challenges. See, for example, International Patent Application No. PCT/US20/36464 entitled "Bispecific Binding Constructs," and PCT/US20/36474 entitled "Bispecific Binding

Constructs with Selectively Cleavable Linkers." However, challenges still exist for molecules with multiple binding sites. Accordingly, there is a need in the art for therapeutic molecules with multiple binding sites and with favorable pharmacokinetic properties, therapeutic efficacy, and a format that provides efficient production and increased stability.

SUMMARY

[0008] Described herein are novel formats of binding molecules that comprise multiple binding domains. In one embodiment, the invention provides a molecule comprising a polypeptide chain having the structure:

[0009] VH1-L1-VH2-L2-VL1-L3-VL2-L4-VH3-L1-VH4-L2-VL3-L3-VL4, or

[0010] VH1-L1-VH2-L2-VL1-L3-VL2-L4-Half Life Extending moiety-L5-VH3-L1-VH4-L2-VL3-L3-VL4,

[0011] wherein VH1, VH2, VH3, and VH4 are immunoglobulin heavy chain variable regions, VL1, VL2, VL3, and VL4 are immunoglobulin light chain variable regions, and L1, L2, L3, L4, and L5 are linkers, wherein L1 is at least 10 amino acids, L2 is at least 15 amino acids and L3 is at least 10 amino acids, and wherein the molecule can bind to an immune effector cell and a target cell.

[0012] In another embodiment, the invention provides a molecule comprising a polypeptide chain having the structure:

[0013] VH1-L1-VH2-L2-VL1-L3-VL2-L4-VH3-L1-VH4-L2-VL3-L3-VL4, or

[0014] VH1-L1-VH2-L2-VL1-L3-VL2-L4-Half-Life Extending moiety-L5-VH3-L1-VH4-L2-VL3-L3-VL4,

[0015] wherein VH1, VH2, VH3, and VH4 are immunoglobulin heavy chain variable regions, VL1, VL2, VL3, and VL4 are immunoglobulin light chain variable regions, and L1, L2 and L3 are linkers, wherein L1 is at least 10 amino acids, L2 is at least 10 amino acids and L3 is at least 10 amino acids, and wherein the total amino acids of L1, L2 and L3 is at least 35 amino acids, and wherein the molecule can bind to an immune effector cell and a target cell.

[0016] The invention further provides nucleic acids encoding the molecule described herein, vectors comprising these nucleic acids, and host cells comprising these vectors.

[0017] In yet other embodiments, the invention provides methods of manufacturing the molecules describe herein comprising (1) culturing a host cell under conditions so as to express the molecule and (2) recovering the molecule from the cell mass or cell culture supernatant, wherein the host cell comprises one or more nucleic acid(s) encoding any of the molecules provided herein.

[0018] In yet other embodiments, the invention provides a method of treating a cancer patient comprising administering to the patient a therapeutically effective amount of the molecules provided herein.

[0019] In other embodiments, the invention provides a method for treating a patient having an infectious disease comprising administering to the patient a therapeutically effective dose of the molecules provided herein.

[0020] In a further embodiment, the invention provides a method for treating a patient having an autoimmune, inflammatory, or fibrotic condition comprising administering to the patient a therapeutically effective dose of the molecules provided herein.

[0021] In another embodiment, the invention further provides a pharmaceutical composition comprising the molecules provided herein.

[0022] In another embodiment, the invention provides for the use of the molecules provided herein in the manufacture of a medicament for the prevention, treatment or amelioration of a disease.

BRIEF DESCRIPTION OF DRAWINGS

[0023] FIG. 1. This figure depicts a comparison between structures of two different exemplary binding molecules—an exemplary (HLHL)² molecule and an exemplary (HHLL)² molecule.

[0024] FIG. 2. This figure depicts an exemplary (HHLL)² molecule with VH/VL domains that bind to two different therapeutic targets and CD3, where both the VH2/VL2 and VH4/VL4 domains bind to CD3. The different exemplary linkers are represented by L # (e.g., L1, L2, L3, etc. . . .) and structure A depicts a molecule with a linker as the spacer moiety between the two (HHLL) components, and structure B depicts a molecule with an optional scFc as the spacer moiety.

[0025] FIG. 3. This figure is a chromatography readout indicating proper expression of the T6M (HHLL)² molecule as compared to the G7Q (HLHL)² molecule.

[0026] FIG. 4. This figure is an image of an SDS PAGE analysis to assay for purity and whether the molecules have the correct molecular weight, and indicates the T6M molecule is expressed at the correct molecular weight.

[0027] FIG. 5. This figure provides graphical representations of in vitro TDCC assay results, demonstrating functionality of the T6M (HHLL)² molecule and superior target cell killing as compared to the G7Q (HLHL)² molecule at 48 hours.

[0028] FIG. 6. This figure provides graphical representations of in vitro TDCC assay results, demonstrating functionality of the T6M (HHLL)² molecule and superior target cell killing as compared to the G7Q (HLHL)² molecule at 72 hours.

DETAILED DESCRIPTION

[0029] Described herein are novel formats for molecules with four distinct binding domains. These molecules comprise a single polypeptide chain that comprises four immunoglobulin variable heavy chain (VH) domains, four immunoglobulin variable light chain (VL) domains, and optionally, an Fc region (e.g., an scFc), arranged in the following order: VH-linker-VH-linker-VL-linker-VL-linker-VH-linker-VH-linker-VL-linker-VL, or VH-linker-VH-linker-VL-linker-VL-linker-scFc-VH-linker-VH-linker-VL-linker-VL, referred to hereinafter as “(HHLL)?” or a “squared format” with an exemplary format of this depicted in FIG. 2 herein.

[0030] This (HHLL)² format may provide both enhanced stability and increased in vitro expression as compared to, for example, an (HLHL)² format, yet it maintains the intended function of binding the desired targets on the immune effector cell and the target cell. Accordingly, the present (HHLL)² format provides molecules that can be produced more efficiently and have greater stability, characteristics that are sought after in a pharmaceutical composition.

[0031] The present invention provides a molecule with four distinct binding domains and the molecule comprises at least one polypeptide and is characterized by comprising at least five distinctive structural entities that together form the (HHLL)² molecule, i.e. (i.) a first domain binding comprising a VH and VL, (ii.) a second binding domain comprising a VH and VL, (iii.) a spacer which connects but also sufficiently spaces apart the first (HHLL) domain from a second (HHLL) domain, (iv.) a third binding domain, and (v.) a fourth binding domain. Preferably, the domains are comprised of VH linked to VH linked to VL linked to VL domains in amino to carboxyl orientation, respectively, with flexible peptide linkers as depicted in FIG. 2 herein. In a specific embodiment of the invention, the first binding domain binds to an extracellular target other than CD3 (e.g., a tumor associated antigen, “TAA”), the second binding domain binds to an extracellular epitope of the human and non-human (e.g., *Macaca*) CD3δ chain, the third binding domain binds to an extracellular target other than CD3 that is the same or different than the target which the first binding domain binds, and a fourth binding domain that binds to an extracellular epitope of the human and non-human (e.g., *Macaca*) CD3δ chain.

[0032] It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the invention as claimed. In this application, the use of the singular includes the plural unless specifically stated otherwise. In this application, the use of “or” means “and/or” unless stated otherwise. Furthermore, the use of the term “including”, as well as other forms, such as “includes” and “included”, is not limiting. Also, terms such as “element” or “component” encompass both elements and components comprising one unit and elements and components that comprise more than one subunit unless specifically stated otherwise. Also, the use of the term “portion” can include part of a moiety or the entire moiety.

[0033] Unless otherwise defined herein, scientific and technical terms used in connection with the present invention shall have the meanings that are commonly understood by those of ordinary skill in the art. Further, unless otherwise required by context, singular terms shall include pluralities and plural terms shall include the singular. Generally, nomenclatures used in connection with, and techniques of, cell and tissue culture, molecular biology, immunology, microbiology, genetics and protein and nucleic acid chemistry and hybridization described herein are those well-known and commonly used in the art. The methods and techniques of the present invention are generally performed according to conventional methods well known in the art and as described in various general and more specific references.

[0034] Polynucleotide and polypeptide sequences are indicated using standard one- or three-letter abbreviations. Unless otherwise indicated, polypeptide sequences have their amino termini at the left and their carboxy termini at the right, and single-stranded nucleic acid sequences, and the top strand of double-stranded nucleic acid sequences, have their 5' termini at the left and their 3' termini at the right. A particular section of a polypeptide can be designated by amino acid residue number such as amino acids 1 to 50, or by the actual residue at that site such as asparagine to

proline. A particular polypeptide or polynucleotide sequence also can be described by explaining how it differs from a reference sequence.

Definitions

[0035] The term “isolated” in reference to a molecule (where the molecule is, for example, a polypeptide, a polynucleotide, multispecific molecule, bispecific molecule, or an antibody) is a molecule that by virtue of its origin or source of derivation (1) is not associated with naturally associated components that accompany it in its native state, (2) is substantially free of other molecules from the same species (3) is expressed by a cell from a different species, or (4) does not occur in nature. Thus, a molecule that is chemically synthesized, or expressed in a cellular system different from the cell from which it naturally originates, will be “isolated” from its naturally associated components. A molecule also may be rendered substantially free of naturally associated components by isolation, using purification techniques well known in the art. Molecule purity or homogeneity may be assayed by a number of means well known in the art. For example, the purity of a polypeptide sample may be assayed using polyacrylamide gel electrophoresis and staining of the gel to visualize the polypeptide using techniques well known in the art. For certain purposes, higher resolution may be provided by using HPLC or other means well known in the art for purification.

[0036] The terms “polynucleotide,” “oligonucleotide” and “nucleic acid” are used interchangeably throughout and include DNA molecules (e.g., cDNA or genomic DNA), RNA molecules (e.g., mRNA), analogs of the DNA or RNA generated using nucleotide analogs (e.g., peptide nucleic acids and non-naturally occurring nucleotide analogs), and hybrids thereof. The nucleic acid molecule can be single-stranded or double-stranded. In one embodiment, the nucleic acid molecules of the invention comprise a contiguous open reading frame encoding a binding molecule, or a fragment, derivative, mutein, or variant thereof, of the invention.

[0037] A “vector” is a nucleic acid that can be used to introduce another nucleic acid linked to it into a cell. One type of vector is a “plasmid,” which refers to a linear or circular double stranded DNA molecule into which additional nucleic acid segments can be ligated. Another type of vector is a viral vector (e.g., replication defective retroviruses, adenoviruses and adeno-associated viruses), wherein additional DNA segments can be introduced into the viral genome. Certain vectors are capable of autonomous replication in a host cell into which they are introduced (e.g., bacterial vectors comprising a bacterial origin of replication and episomal mammalian vectors). Other vectors (e.g., non-episomal mammalian vectors) are integrated into the genome of a host cell upon introduction into the host cell, and thereby are replicated along with the host genome. An “expression vector” is a type of vector that can direct the expression of a chosen polynucleotide.

[0038] A nucleotide sequence is “operably linked” to a regulatory sequence if the regulatory sequence affects the expression (e.g., the level, timing, or location of expression) of the nucleotide sequence. A “regulatory sequence” is a nucleic acid that affects the expression (e.g., the level, timing, or location of expression) of a nucleic acid to which it is operably linked. The regulatory sequence can, for example, exert its effects directly on the regulated nucleic acid, or through the action of one or more other molecules

(e.g., polypeptides that bind to the regulatory sequence and/or the nucleic acid). Examples of regulatory sequences include promoters, enhancers and other expression control elements (e.g., polyadenylation signals).

[0039] A “host cell” is a cell that can be used to express a nucleic acid, e.g., a nucleic acid of the invention. A host cell can be a prokaryote, for example, *E. coli*, or it can be a eukaryote, for example, a single-celled eukaryote (e.g., a yeast or other fungus), a plant cell (e.g., a tobacco or tomato plant cell), an animal cell (e.g., a human cell, a monkey cell, a hamster cell, a rat cell, a mouse cell, or an insect cell) or a hybridoma. Typically, a host cell is a cultured cell that can be transformed or transfected with a polypeptide-encoding nucleic acid, which can then be expressed in the host cell. The phrase “recombinant host cell” can be used to denote a host cell that has been transformed or transfected with a nucleic acid to be expressed. A host cell also can be a cell that comprises the nucleic acid but does not express it at a desired level unless a regulatory sequence is introduced into the host cell such that it becomes operably linked with the nucleic acid. It is understood that the term host cell refers not only to the particular subject cell but to the progeny or potential progeny of such a cell. Because certain modifications may occur in succeeding generations due to, e.g., mutation or environmental influence, such progeny may not, in fact, be identical to the parent cell, but are still included within the scope of the term as used herein.

[0040] A “single-chain variable fragment” (“scFv”) is a fusion protein in which a VL and a VH region are joined via a linker (e.g., a synthetic sequence of amino acid residues) to form a continuous protein chain wherein the linker is long enough to allow the protein chain to fold back on itself and form a monovalent antigen binding site or binding domain (see, e.g., Bird et al., *Science* 242:423-26 (1988) and Huston et al., 1988, *Proc. Natl. Acad. Sci. USA* 85:5879-83 (1988)). When in the context of other additional moieties (e.g., an Fc region), the scFv can be arranged VH-linker-VL, or VL-linker-VH, for example.

[0041] The term “CDR” refers to the complementarity determining region (also termed “minimal recognition units” or “hypervariable region”) within antibody variable sequences, and the molecules of the present invention comprises heavy and/or light chain CDRs. The CDRs permit the binding molecule to specifically bind to a particular antigen of interest. There are three heavy chain variable region CDRs (CDRH1, CDRH2 and CDRH3) and three light chain variable region CDRs (CDRL1, CDRL2 and CDRL3). The CDRs in each of the two chains typically are aligned by the framework regions to form a structure that binds specifically to a specific epitope or domain on the target protein. From N-terminus to C-terminus, naturally-occurring light and heavy chain variable regions both typically conform to the following order of these elements: FR1, CDR1, FR2, CDR2, FR3, CDR3 and FR4. A numbering system has been devised for assigning numbers to amino acids that occupy positions in each of these domains. This numbering system is defined in Kabat Sequences of Proteins of Immunological Interest (1987 and 1991, NIH, Bethesda, MD), or Chothia & Lesk, 1987, *J. Mol. Biol.* 196:901-917; Chothia et al., 1989, *Nature* 342:878-883. Complementarity determining regions (CDRs) and framework regions (FR) of a given antibody may be identified using this system. Other numbering systems for the amino acids in immunoglobulin chains include IMGT® (the international ImMunoGeneTics information

system; Lefranc et al, *Dev. Comp. Immunol.* 29:185-203; 2005) and AHO (Honegger and Pluckthun, *J. Mol. Biol.* 309 (3): 657-670; 2001). One or more CDRs may be incorporated into a molecule either covalently or noncovalently to make it a binding molecule.

[0042] The “binding domain” of a binding molecule according to the invention may, e.g., comprise the above referred groups of CDRs. Preferably, those CDRs are comprised in the framework of an antibody light chain variable region (VL) and an antibody heavy chain variable region (VH) comprised by the molecules of the invention. Or in the terminology used herein, the “L” and “H” variable regions (e.g., “HHLL”).

[0043] The term “human antibody” includes antibodies having antibody regions such as variable and constant regions or domains which correspond substantially to human germline immunoglobulin sequences known in the art, including, for example, those described by Kabat et al. (1991) (*loc. cit.*). The human antibodies referred to herein may include amino acid residues not encoded by human germline immunoglobulin sequences (e.g., mutations introduced by random or site-specific mutagenesis *in vitro* or by somatic mutation *in vivo*), for example in the CDRs, and in particular, in CDR3. The human antibodies can have at least one, two, three, four, five, or more positions replaced with an amino acid residue that is not encoded by the human germline immunoglobulin sequence. The definition of human antibodies as used herein also contemplates fully human antibodies, which include only non-artificially and/or genetically altered human sequences of antibodies as those can be derived by using technologies or systems known in the art, such as for example, phage display technology or transgenic mouse technology, including but not limited to the Xenomouse®. In the context of the present invention, the variable regions from a human antibody can be used in the molecule formats contemplated.

[0044] A humanized antibody has a sequence that differs from the sequence of an antibody derived from a non-human species by one or more amino acid substitutions, deletions, and/or additions, such that the humanized antibody is less likely to induce an immune response, and/or induces a less severe immune response, as compared to the non-human species antibody, when it is administered to a human subject. In one embodiment, certain amino acids in the framework and constant domains of the heavy and/or light chains of the non-human species antibody are mutated to produce the humanized antibody. In another embodiment, the constant domain(s) from a human antibody are fused to the variable domain(s) of a non-human species. In another embodiment, one or more amino acid residues in one or more CDR sequences of a non-human antibody are changed to reduce the likely immunogenicity of the non-human antibody when it is administered to a human subject, wherein the changed amino acid residues either are not critical for immunospecific binding of the antibody or binding molecule to its antigen, or the changes to the amino acid sequence that are made are conservative changes, such that the binding of the humanized antibody to the antigen is not significantly worse than the binding of the non-human antibody to the antigen. Examples of how to make humanized antibodies may be found in U.S. Pat. Nos. 6,054,297, 5,886,152 and 5,877,293. In the context of the present invention, the variable regions from a humanized antibody can be used in the molecule formats contemplated.

[0045] The term “chimeric antibody” refers to an antibody that contains one or more regions from one antibody and one or more regions from one or more other antibodies. In one embodiment, one or more of the CDRs are derived from a human antibody. In another embodiment, all of the CDRs are derived from a human antibody. In another embodiment, the CDRs from more than one human antibodies are mixed and matched in a chimeric antibody. For instance, a chimeric antibody may comprise a CDR1 from the light chain of a first human antibody, a CDR2 and a CDR3 from the light chain of a second human antibody, and the CDRs from the heavy chain from a third antibody. Further, the framework regions may be derived from one of the same antibodies, from one or more different antibodies, such as a human antibody, or from a humanized antibody. In one example of a chimeric antibody, a portion of the heavy and/or light chain is identical with, homologous to, or derived from an antibody from a particular species or belonging to a particular antibody class or subclass, while the remainder of the chain(s) is/are identical with, homologous to, or derived from an antibody or antibodies from another species or belonging to another antibody class or subclass. Also included are fragments of such antibodies that exhibit the desired biological activity. In the context of the present invention, the variable regions from a chimeric antibody can be used in the molecule formats contemplated.

[0046] In the context of the present (HHLL)² molecules of the invention, a “spacer” domain is positioned between each of the two (HHLL) subunits that together comprise the (HHLL)² molecule. In some embodiments, the spacer is a half-life extending moiety. In other embodiments, the spacer is a polypeptide linker.

[0047] For further examples of spacer domains, see U.S. Provisional Patent Appl. No. 63/110,957. In certain embodiments of the invention, the spacer domain has a molecular weight of more than about 3.2 kDa, preferably 10 kDa, more even preferably at least 15 kDa, 20 kDa or even 50 kDa, and/or wherein said spacer domain comprises an amino acid sequence which comprises at least 50 amino acids, preferably 75 amino acids, more preferably at least 150 amino acids, and even more preferably at least 500 amino acids.

[0048] In other embodiments, the spacer domain that sufficiently spaces apart the first and the second (HHLL) domains is selected from a group consisting of a programmed cell death protein 1 (PD1) domain, human serum albumin (HSA) or a derivative thereof, a multimer of a rigid linker, e.g. (EAAAK)₁₀, and a Fc domain comprising two polypeptide monomers comprising each a hinge, a CH2 and a CH3 domain a hinge and a further CH2 and a CH3 domain, wherein said two polypeptide monomers are fused to each other via a peptide linker or wherein the two polypeptide monomers are linked together by non-covalent CH3-CDH3 interactions and/or covalent disulfide bonds to form a heterodimer.

[0049] In a specific embodiment, the spacer entity is at least one domain, preferably one domain or two covalently linked domains, which or each of which comprises in an amino to carboxyl order:

[0050] hinge-CH2-CH3-linker-hinge-CH2-CH3.

[0051] In particular embodiments, it is also envisaged that the CH2 domain in the spacer comprises an intra domain cysteine disulfide bridge.

[0052] According to the present invention, it is preferable in certain embodiments that the two bispecific entities must

be spaced apart a certain distance, preferably more than 35 Å, more preferably at least 40, 50, 60, 70, 80, 90 or at least 100 Å. The distance can be readily determined by crystallography, cryo electron microscopy, or nuclear magnetic resonance analytic technology. This distance is facilitated by a spacer entity between the two (HHLL) domains which spaces the two domains apart and maintains them in a desired conformation and prevents an undesired interaction of the two separated (HHLL) domains. In general, the more rigid the linker is, the less the minimal distance required between the two (HHLL) domains.

[0053] The composition and arrangement of these amino acids preferably confer a certain rigidity and are not characterized by high flexibility. In this regard, spacers of amino acids rich in proline and less rich in serine and glycine are preferred. Especially envisaged are spacers which are folded polypeptides e.g. of secondary order (e.g. helical structures) or of ternary order forming e.g. three dimensional domains which in turn ensure a certain rigidity by their constitution and preferably confer further advantageous effects such as in vivo half-life extension of the multitargeting bispecific molecule as a therapeutic. In the context of the present invention, spacers comprising an Fc domain or parts thereof are envisaged.

[0054] In certain embodiments where the (HHLL)² spacer is a half-life extending moiety, the half-life extending moiety is an Fc polypeptide chain. In other embodiments, the half-life extending moiety is a single-chain Fc (see, e.g., SEQ ID NOs: 45-53). In yet other embodiments, the half-life extending moiety is a hetero-Fc (see, e.g., SEQ ID NOs: 55 and 56). In yet other embodiments, the half-life extending moiety is human albumin or human serum albumin (see, e.g., see SEQ ID NO: 57). In other embodiments, the half-life extending moiety is an albumin binding domain. Further specific examples and sequences of half-life extending moieties and spacers are provided in U.S. Provisional Patent Appl. No. 63/110,957 (e.g., see Table 17).

Linkers

[0055] Between the immunoglobulin variable regions is a peptide linker, which can be the same linker or different linkers of different lengths. In additional embodiments, the (HHLL)² molecule further comprises a spacer moiety between (HHLL) domains where the spacer, in particular embodiments, is a linker. The linkers can play a critical role in the structure of the binding molecule and the invention described herein provides not only the appropriate linker sequences, but also the appropriate linker lengths for each position in the binding molecules of the invention. If the linker is too short, it will not allow enough flexibility for the appropriate variable regions on a single polypeptide chain to interact to form an antigen binding site (or “binding domain”). If the linker is the appropriate length, it will allow a variable region to interact with another variable region on the same polypeptide chain to form an antigen binding site. In certain embodiments, the HHLL format comprises disulfide bonds—both intra-domain (within H1, L1) and inter-domain (between H1 and L1). In order to achieve proper expression and conformation of the molecules of the invention, in certain embodiments specific linkers are used between the various immunoglobulin regions (see, e.g., FIG. 1 herein). Exemplary linkers are provided in Table 1 herein. In certain embodiments, increasing linker length might result in increased protein clipping, an undesirable property.

Accordingly, it is desirable to achieve the appropriate balance between linker length to allow proper polypeptide structure and activity, yet not result in increased clipping.

[0056] A “linker,” as meant herein, is a peptide that links two polypeptides. In certain embodiments, a linker can link more than one immunoglobulin variable regions in the context of a molecule. A linker can be from 2-30 amino acids in length. In some embodiments, a linker can be 2-25, 2-20, or 3-18 amino acids long. In some embodiments, a linker can be a peptide no more than 14, 13, 12, 11, 10, 9, 8, 7, 6, or 5 amino acids long. In other embodiments, a linker can be 5-25, 5-15, 4-11, 10-20, or 20-30 amino acids long. In other embodiments, a linker can be about, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 amino acids long. Exemplary linkers include, for example, the amino acid sequences GGGGS (SEQ ID NO: 1), GGGGSGGGGS (SEQ ID NO: 2), GGGGSGGGGS (SEQ ID NO: 3), GGGGSGGGGS (SEQ ID NO: 4), GGGGSGGGGS (SEQ ID NO: 5), GGGGQ (SEQ ID NO: 6), GGGGQGGGGQ (SEQ ID NO: 7), GGGGQGGGGQGGGGQ (SEQ ID NO: 8), GGGGQGGGGQGGGGQGGGGQ (SEQ ID NO: 9), GGGGQGGGGQGGGGQGGGGQGGGGQ (SEQ ID NO: 10), GGGGSAAA (SEQ ID NO: 11), TVAAP (SEQ ID NO: 12), ASTKGP (SEQ ID NO: 13), AAA (SEQ ID NO: 14), SGGGS (SEQ ID NO: 17), and SGGGGQ (SEQ ID NO: 18), among others, including repeats of the aforementioned amino acid sequences or subunits of amino acid sequences (e.g., GGGGS (SEQ ID NO: 1) or GGGGQ (SEQ ID NO: 6) repeats).

[0057] In certain embodiments in the context of the (HHLL)² molecules of the invention, the linker sequence of Linker 1 is at least 10 amino acids. In other embodiments, Linker 1 is at least 15 amino acids. In other embodiments, Linker 1 is at least 20 amino acids. In other embodiments, Linker 1 is at least 25 amino acids. In other embodiments, Linker 1 is at least 30 amino acids. In other embodiments, Linker 1 is 10-30 amino acids. In other embodiments, Linker 1 is 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 amino acids. In yet other embodiments, Linker 1 is greater than 30 amino acids.

[0058] In certain embodiments in the context of the (HHLL)² molecules of the invention, the linker sequence of Linker 2 is at least 15 amino acids. In other embodiments, Linker 2 is at least 20 amino acids. In other embodiments, Linker 2 is at least 25 amino acids. In other embodiments, Linker 2 is at least 30 amino acids. In other embodiments, Linker 2 is 15-30 amino acids. In other embodiments, Linker 2 is 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 amino acids. In yet other embodiments, Linker 2 is greater than 30 amino acids.

[0059] In certain embodiments in the context of the (HHLL)² molecules of the invention, the linker sequence of Linker 3 is at least 15 amino acids. In other embodiments, Linker 3 is at least 20 amino acids. In other embodiments, Linker 3 is at least 25 amino acids. In other embodiments, Linker 3 is at least 30 amino acids. In other embodiments, Linker 3 is 15-30 amino acids. In other embodiments, Linker 3 is 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 amino acids. In yet other embodiments, Linker 3 is greater than 30 amino acids.

[0060] In certain embodiments in the context of the (HHLL)² molecules of the invention where molecules do not

have a spacer moiety such as an scFc and instead only comprise a linker at L4, the linker sequence of L4 is at least five amino acids. In preferred embodiments, the linker sequence of L4 in this context is SGGGS. In other embodiments, the linker sequence of L4 in this context is at least 10 amino acids. In other embodiments in this context, Linker 4 is at least 15 amino acids. In other embodiments in this context, Linker 4 is at least 20 amino acids. In other embodiments in this context, Linker 4 is at least 25 amino acids. In other embodiments in this context, Linker 4 is at least 30 amino acids. In other embodiments in this context, Linker 4 is 5-30 amino acids. In other embodiments in this context, Linker 4 is 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 amino acids. In yet other embodiments in this context, Linker 4 is greater than 30 amino acids.

[0061] In certain embodiments in the context of the (HHLL)² molecules of the invention where molecules comprise a spacer moiety such as an scFc, the linker sequence of Linker 4 is at least 5 amino acids. In other embodiments in this context, Linker 4 is at least 10 amino acids. In other embodiments in this context, Linker 4 is at least 15 amino acids. In a specific embodiment, Linker 4 in this context is (GGGGS)₃. In other embodiments in this context, Linker 4 is at least 20 amino acids. In other embodiments in this context, Linker 4 is at least 25 amino acids. In other embodiments in this context, Linker 4 is at least 30 amino acids. In other embodiments in this context, Linker 4 is 5-30 amino acids. In other embodiments in this context, Linker 4 is 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 amino acids. In yet other embodiments in this context, Linker 4 is greater than 30 amino acids.

[0062] In certain embodiments in the context of the (HHLL)² molecules of the invention where molecules comprise a spacer moiety such as an scFc, the linker sequence of L5 is at least 5 amino acids. In other embodiments in this context, Linker 5 is at least 10 amino acids. In other

embodiments in this context, Linker 5 is at least 15 amino acids. In a specific embodiment, Linker 5 in this context is (GGGGS)₃. In other embodiments in this context, Linker 5 is at least 20 amino acids. In other embodiments in this context, Linker 5 is at least 25 amino acids. In other embodiments in this context, Linker 5 is at least 30 amino acids. In other embodiments in this context, Linker 5 is 5-30 amino acids. In other embodiments in this context, Linker 5 is 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 amino acids. In yet other embodiments in this context, Linker 5 is greater than 30 amino acids.

[0063] For further guidance on linker lengths in the context of each HHLL subunit within the (HHLL)² molecules of the present invention, see FIGS. 4A-4D of International Patent Application No. PCT/US20/36464 entitled "Bispecific Binding Constructs," which depict molecular models in various orientations of the HHLL molecule and show how the linkers of particular lengths are necessary in order for the HHLL molecule to take on proper conformation and allow both H-L binding domains to function. A, B, and C represent the distance between C-alpha atoms of the terminal residue of one domain and starting residue of another domain. Using this information, a skilled practitioner could model a contemplated HHLL molecule and adjust linker lengths as needed for the particular H-L binding domains so that the HHLL molecule, and accordingly an (HHLL)² molecule, expresses and functions as desired.

[0064] In certain embodiments in the context of the (HHLL)² molecules of the invention that comprises a spacer such as an scFc between the two (HHLL) subunits, a sampling of representative linker sequences and positions are set forth in the following Table 1, with linker positions corresponding to those set forth in FIGS. 1B and 2B. In a specific embodiment in this context, Linkers 1, 2, and 3 are (GGGGS)₄, and Linkers 4 and 5 are (GGGGS)₃. In other embodiments, although the linker between the two scFc subunits is not given a numbered linker designation, this linker is preferably (GGGGS)₆.

TABLE 1

Representative Linkers							
Linker 1	SEQ ID NO:	Linker 2	SEQ ID NO:	Linker 3	SEQ ID NO:	Linkers 4 and 5	SEQ ID NO:
(GGGGS) ₂	2	(GGGGS) ₃	3	(GGGGS) ₃	3	GGGG	19
(GGGGS) ₂	2	(GGGGS) ₃	3	(GGGGS) ₃	3	GGGGS	1
(GGGGS) ₄	4	(GGGGS) ₄	4	(GGGGS) ₄	4	(GGGG) ₃	21
(GGGGS) ₄	4	(GGGGS) ₄	4	(GGGGS) ₄	4	(GGGGS) ₃	3
(GGGGS) ₅	5	(GGGGS) ₅	5	(GGGGS) ₅	5	GGGG	19
(GGGGS) ₅	5	(GGGGS) ₅	5	(GGGGS) ₅	5	GGGGS	1
(GGGGS) ₃	3	(GGGGS) ₅	5	(GGGGS) ₅	5	GGGG	56
(GGGGS) ₃	3	(GGGGS) ₅	5	(GGGGS) ₅	5	GGGGS	1
(GGGGS) ₃	3	(GGGGS) ₃	3	(GGGGS) ₂	2	GGGG	19
(GGGGS) ₃	3	(GGGGS) ₃	3	(GGGGS) ₂	2	GGGGS	1
(GGGGS) ₂₋₁₀	28	(GGGGS) ₃₋₁₀	29	(GGGGS) ₃₋₁₀	29	(GGGG) ₁₋₁₀	24
(GGGGS) ₂₋₁₀	28	(GGGGS) ₃₋₁₀	29	(GGGGS) ₃₋₁₀	29	(GGGGS) ₁₋₁₀	27
(GGGGQ) ₂	7	(GGGGQ) ₃	8	(GGGGQ) ₃	8	GGGG	19
(GGGGQ) ₂	7	(GGGGQ) ₃	8	(GGGGQ) ₃	8	GGGGQ	6
(GGGGQ) ₄	9	(GGGGQ) ₄	9	(GGGGQ) ₄	9	GGGG	19
(GGGGQ) ₄	9	(GGGGQ) ₄	9	(GGGGQ) ₄	9	GGGGQ	6
(GGGGQ) ₅	10	(GGGGQ) ₅	10	(GGGGQ) ₅	10	GGGG	19
(GGGGQ) ₅	10	(GGGGQ) ₅	10	(GGGGQ) ₅	10	GGGGQ	6
(GGGGQ) ₃	8	(GGGGQ) ₅	10	(GGGGQ) ₅	10	GGGG	19

TABLE 1-continued

Representative Linkers							
Linker 1	SEQ ID NO:	Linker 2	SEQ ID NO:	Linker 3	SEQ ID NO:	Linkers 4 and 5	SEQ ID NO:
(GGGGQ) ₃	8	(GGGGQ) ₅	10	(GGGGQ) ₅	10	GGGGQ	6
(GGGGQ) ₂₋₁₀	31	(GGGGQ) ₃₋₁₀	32	(GGGGQ) ₃₋₁₀	32	(GGGG) ₁₋₁₀	24
(GGGGQ) ₂₋₁₀	31	(GGGGQ) ₃₋₁₀	32	(GGGGQ) ₃₋₁₀	32	(GGGGQ) ₁₋₁₀	30

*numerical subscript indicates the number of repeats, e.g., (GGGS)₂ = GGGSGGG (SEQ ID NO: 2)

Amino Acid Sequences of Binding Regions

[0065] In the exemplary embodiments described herein, the molecules maintain desired binding to the various desired targets which results from their assuming the proper conformation to allow this binding. The immunoglobulin variable region comprises a VH and a VL domain, which associate to form the variable domain which binds the desired target.

[0066] The variable domains can be obtained from any immunoglobulin with the desired characteristics, and the methods to accomplish this are further described herein. In one embodiment, VH1 and VL1 associate and bind CD3 ϵ , VH3 and VL3 associate and bind CD3 ϵ , and VH2 and VL2 associate and bind a different target, e.g., a TAA, and the VH4 and VL4 associate and bind a different target, e.g., a same or different TAA.

[0067] In another embodiment, the VH2 and VL2 associate and bind CD3 ϵ , the VH4 and VL4 associate and bind CD3 ϵ , the VH1 and VL1 associate and bind a different target, and the VH3 and VL3 associate and bind a different target.

[0068] In a specific embodiment, the VH1 and VL1 associate and bind to mesothelin, the VH2 and VL2 associate and bind to CD3 ϵ , the VH3 and VL3 associate and bind to CDH3, and the VH4 and VL4 associate and bind to CD3 ϵ .

[0069] In a further specific embodiment, the VH1 and VL1 associate and bind to CDH3, the VH2 and VL2 associate and bind to CD3 ϵ , the VH3 and VL3 associate and bind to mesothelin, and the VH4 and VL4 associate and bind to CD3 ϵ .

[0070] In yet another specific embodiment, the VH1 and VL1 associate and bind to CD3 ϵ , the VH2 and VL2 associate and bind to mesothelin, the VH3 and VL3 associate and bind to CD3 ϵ , and the VH4 and VL4 associate and bind to CDH3.

[0071] In yet another specific embodiment, the VH1 and VL1 associate and bind to CD3 ϵ , the VH2 and VL2 associate and bind to CDH3, the VH3 and VL3 associate and bind to CD3 ϵ , and the VH4 and VL4 associate and bind to mesothelin.

[0072] In a specific embodiment, the VH1 (SEQ ID NO: 39) and VL1 (SEQ ID NO: 40) associate and bind to mesothelin, the VH2 (SEQ ID NO: 41) and VL2 (SEQ ID NO: 42) associate and bind to CD36, the VH3 (SEQ ID NO: 43) and VL3 (SEQ ID NO: 44) associate and bind to CDH3, and the VH4 (SEQ ID NO: 41) and VL4 (SEQ ID NO: 42) associate and bind to CD3 ϵ .

[0073] In a further specific embodiment, the VH1 (SEQ ID NO: 43) and VL1 (SEQ ID NO: 44) associate and bind to CDH3, the VH2 (SEQ ID NO: 41) and VL2 (SEQ ID NO: 42) associate and bind to CD3 ϵ , the VH3 (SEQ ID NO: 39) and VL3 (SEQ ID NO: 40) associate and bind to mesothelin, and the VH4 (SEQ ID NO: 41) and VL4 (SEQ ID NO: 42) associate and bind to CD3 ϵ .

[0074] In yet another specific embodiment, the VH1 (SEQ ID NO: 41) and VL1 (SEQ ID NO: 42) associate and bind to CD3 ϵ , the VH2 (SEQ ID NO: 39) and VL2 (SEQ ID NO: 40) associate and bind to mesothelin, the VH3 (SEQ ID NO: 41) and VL3 (SEQ ID NO: 42) associate and bind to CD3 ϵ , and the VH4 (SEQ ID NO: 43) and VL4 (SEQ ID NO: 44) associate and bind to CDH3.

[0075] In yet another specific embodiment, the VH1 (SEQ ID NO: 41) and VL1 (SEQ ID NO: 42) associate and bind to CD36, the VH2 (SEQ ID NO: 43) and VL2 (SEQ ID NO: 44) associate and bind to CDH3, the VH3 (SEQ ID NO: 41) and VL3 (SEQ ID NO: 42) associate and bind to CD3 ϵ , and the VH4 (SEQ ID NO: 39) and VL4 (SEQ ID NO: 40) associate and bind to mesothelin.

[0076] In a further specific embodiment, a representative (HHLL)² molecule amino acid sequence is set forth in SEQ ID NO: 37.

[0077] Further specific examples of sequences that can be incorporated in molecule binding domains that bind CD3 ϵ are provided herein in SEQ NOs: 58-97, including the specified VH, VL and CDRs.

[0078] In another embodiment, the light-chain variable domain comprises a sequence of amino acids that is at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical to the sequence of a light chain variable domain set forth herein.

[0079] In another embodiment, the light chain variable domain comprises a sequence of amino acids that is encoded by a nucleotide sequence that is at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical to the polynucleotide sequence set forth herein. In another embodiment, the light chain variable domain comprises a sequence of amino acids that is encoded by a polynucleotide that hybridizes under moderately stringent conditions to the complement of a polynucleotide that encodes a light chain variable domain selected from the sequences set forth herein. In another embodiment, the light chain variable domain comprises a sequence of amino acids that is encoded by a polynucleotide that hybridizes under stringent conditions to the complement of a polynucleotide that encodes a light chain variable domain selected from the group consisting of the sequences set forth herein.

[0080] In another embodiment, the heavy chain variable domain comprises a sequence of amino acids that is at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical to the sequence of a heavy chain variable

domain selected from the sequences set forth herein. In another embodiment, the heavy chain variable domain comprises a sequence of amino acids that is encoded by a nucleotide sequence that is at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical to a nucleotide sequence that encodes a heavy chain variable domain selected from the sequences set forth herein. In another embodiment, the heavy chain variable domain comprises a sequence of amino acids that is encoded by a polynucleotide that hybridizes under moderately stringent conditions to the complement of a polynucleotide that encodes a heavy chain variable domain selected from the sequences set forth herein. In another embodiment, the heavy chain variable domain comprises a sequence of amino acids that is encoded by a polynucleotide that hybridizes under stringent conditions to the complement of a polynucleotide that encodes a heavy chain variable domain selected from the sequences set forth herein.

Substitutions

[0081] It will be appreciated that a molecule of the present invention may have at least one amino acid substitution, providing that the molecule retains the same or better desired binding specificity (e.g., binding to CD3). Therefore, modifications to the binding molecule structures are encompassed within the scope of the invention. In one embodiment, the binding molecule comprises sequences that each independently differ by 5, 4, 3, 2, 1, or 0 single amino acid additions, substitutions, and/or deletions from a CDR sequence of those set forth herein. As used herein, a CDR sequence that differs by no more than a total of, for example, four amino acid additions, substitutions and/or deletions from a CDR sequence set forth herein refers to a sequence with 4, 3, 2, 1 or 0 single amino acid additions, substitutions, and/or deletions compared with the sequences set forth herein. These may include amino acid substitutions, which may be conservative or non-conservative that do not destroy the desired binding capability of a binding molecule. Conservative amino acid substitutions may encompass non-naturally occurring amino acid residues, which are typically incorporated by chemical peptide synthesis rather than by synthesis in biological systems. These include peptidomimetics and other reversed or inverted forms of amino acid moieties. A conservative amino acid substitution may also involve a substitution of a native amino acid residue with a normative residue such that there is little or no effect on the polarity or charge of the amino acid residue at that position.

[0082] Non-conservative substitutions may involve the exchange of a member of one class of amino acids or amino acid mimetics for a member from another class with different physical properties (e.g. size, polarity, hydrophobicity, charge). In certain embodiments, such substituted residues may be introduced into regions of a human antibody that are homologous with non-human antibodies, or into the non-homologous regions of the molecule, which can be used to generate the binding molecules of the invention.

[0083] Moreover, one skilled in the art may generate test variants containing a single amino acid substitution at each desired amino acid residue. The variants can then be screened using activity assays known to those skilled in the art. Such variants could be used to gather information about suitable variants. For example, if one discovered that a change to a particular amino acid residue resulted in destroyed, undesirably reduced, or unsuitable activity, vari-

ants with such a change may be avoided. In other words, based on information gathered from such routine experiments, one skilled in the art can readily determine the amino acids where further substitutions should be avoided either alone or in combination with other mutations.

[0084] A skilled artisan will be able to determine suitable variants of the binding molecule as set forth herein using well-known techniques. In certain embodiments, one skilled in the art may identify suitable areas of the molecule that may be changed without destroying activity by targeting regions not believed to be important for activity. In certain embodiments, one can identify residues and portions of the molecules that are conserved among similar polypeptides as has been describe above. In certain embodiments, even areas that may be important for biological activity or for structure may be subject to conservative amino acid substitutions without destroying the biological activity or without adversely affecting the polypeptide structure.

[0085] Additionally, one skilled in the art can review structure-function studies identifying residues in similar polypeptides that are important for activity or structure. In view of such a comparison, one can predict the importance of amino acid residues in a protein that correspond to amino acid residues which are important for activity or structure in similar proteins. One skilled in the art may opt for chemically similar amino acid substitutions for such predicted important amino acid residues.

[0086] In some embodiments, one skilled in the art may identify residues that may be changed that result in enhanced properties as desired. For example, an amino acid substitution (conservative or non-conservative) may result in enhanced binding affinity to a desired target.

[0087] One skilled in the art can also analyze the three-dimensional structure and amino acid sequence in relation to that structure in similar polypeptides. In view of such information, one skilled in the art may predict the alignment of amino acid residues of an antibody with respect to its three-dimensional structure. In certain embodiments, one skilled in the art may choose not to make radical changes to amino acid residues predicted to be on the surface of the protein, since such residues may be involved in important interactions with other molecules. A number of scientific publications have been devoted to the prediction of secondary structure. See Moult J., *Curr. Op. in Biotech.*, 7(4): 422-427 (1996), Chou et al., *Biochemistry*, 13(2): 222-245 (1974); Chou et al., *Biochemistry*, 113(2): 211-222 (1974); Chou et al., *Adv. Enzymol. Relat. Areas Mol. Biol.*, 47:45-148 (1978); Chou et al., *Ann. Rev. Biochem.*, 47:251-276 (1978); Chou et al., *Biophys. J.*, 26:367-384 (1979). Moreover, computer programs are currently available to assist with predicting secondary structure. One method of predicting secondary structure is based upon homology modeling. For example, two polypeptides or proteins which have a sequence identity of greater than 30%, or similarity greater than 40% often have similar structural topologies. The growth of the protein structural database (PDB) has provided enhanced predictability of secondary structure, including the potential number of folds within a polypeptide's or protein's structure. See Holm et al., *Nucl. Acid. Res.*, 27 (1): 244-247 (1999). Additional methods of predicting secondary structure include "threading" (Jones, D., *Curr. Opin. Struct. Biol.*, 7 (3): 377-87 (1997); Sippl et al., *Structure*, 4 (1): 15-19 (1996)), "profile analysis" (Bowie et al., *Science*, 253:164-170 (1991); Gribskov et al., *Meth. Enzym.*, 183:

146-159 (1990); Gribskov et al., *Proc. Nat. Acad. Sci.*, 84(13): 4355-4358 (1987)), and “evolutionary linkage” (See Holm, *supra* (1999), and Brenner, *supra* (1997)).

[0088] In certain embodiments, variants of the binding molecule include glycosylation variants wherein the number and/or type of glycosylation site has been altered compared to the amino acid sequences of a parent polypeptide. In certain embodiments, variants comprise a greater or a lesser number of N-linked glycosylation sites than the native protein. Alternatively, substitutions which eliminate this sequence will remove an existing N-linked carbohydrate chain. Also provided is a rearrangement of N-linked carbohydrate chains wherein one or more N-linked glycosylation sites (typically those that are naturally occurring) are eliminated and one or more new N-linked sites are created. Additional variants include cysteine variants wherein one or more cysteine residues are deleted from or substituted for another amino acid (e.g., serine) as compared to the parent amino acid sequence. Cysteine variants may be useful when antibodies or other polypeptide molecules must be refolded into a biologically active conformation such as after the isolation of insoluble inclusion bodies. Cysteine variants generally have fewer cysteine residues than the native protein, and typically have an even number to minimize interactions resulting from unpaired cysteines.

[0089] Desired amino acid substitutions (whether conservative or non-conservative) can be determined by those skilled in the art at the time such substitutions are desired. In certain embodiments, amino acid substitutions can be used to identify important residues of binding molecules to the target of interest, or to increase or decrease the affinity of the binding molecules to the target of interest described herein.

[0090] According to certain embodiments, desired amino acid substitutions are those which: (1) reduce susceptibility to proteolysis, (2) reduce susceptibility to oxidation, (3) alter binding affinity for forming protein complexes, (4) alter binding affinities, and/or (4) confer or modify other physiochemical or functional properties on such polypeptides. According to certain embodiments, single or multiple amino acid substitutions (in certain embodiments, conservative amino acid substitutions) may be made in the naturally-occurring sequence (in certain embodiments, in the portion of the polypeptide outside the domain(s) forming intermolecular contacts). In certain embodiments, a conservative amino acid substitution typically may not substantially change the structural characteristics of the parent sequence (e.g., a replacement amino acid should not tend to break a helix that occurs in the parent sequence, or disrupt other types of secondary structure that characterizes the parent sequence). Examples of art-recognized polypeptide secondary and tertiary structures are described in *Proteins, Structures and Molecular Principles* (Creighton, Ed., W. H. Freeman and Company, New York (1984)); *Introduction to Protein Structure* (C. Branden and J. Tooze, eds., Garland Publishing, New York, N.Y. (1991)); and Thornton et al. *Nature* 354:105 (1991), which are each incorporated herein by reference.

Half-Life Extension and Fc Regions

[0091] In certain embodiments, it is desirable to extend the in vivo half-life of the molecules of the invention. This can be accomplished by including a half-life extending moiety as part of the molecule. Nonlimiting examples of half-life extending moieties include an Fc polypeptide, albumin, an albumin fragment, a moiety that binds to albumin or to the neonatal Fc receptor (FcRn), a derivative of fibronectin that has been engineered to bind albumin or a fragment thereof, a peptide, a single domain protein fragment, or other polypeptide that can increase serum half-life. In alternate embodiments, a half-life-extending moiety can be a non-polypeptide molecule such as, for example, polyethylene glycol (PEG).

[0092] The term “Fc polypeptide” as used herein includes native and mutein forms of polypeptides derived from the Fc region of an antibody. Truncated forms of such polypeptides containing the hinge region that promotes dimerization also are included. In addition to other properties described herein, polypeptides comprising Fc moieties offer the advantage of purification by affinity chromatography over, e.g., Protein A or Protein G columns.

[0093] In certain embodiments, the half-life extending moiety is an Fc region of an antibody. The Fc region can be located at the N-terminal end of the (HHLL)² molecule, or it can be located at the C-terminal end of the (HHLL)² molecule. There can be, but need not be, a linker between the (HHLL)² molecule and the Fc region. As explained above, an Fc polypeptide chain may comprise all or part of a hinge region followed by a CH2 and a CH3 region. The Fc polypeptide chain can be of mammalian (for example, human, mouse, rat, rabbit, dromedary, or new or old world monkey), avian, or shark origin. In addition, as explained above, an Fc polypeptide chain can include a limited number of alterations. For example, an Fc polypeptide chain can comprise one or more heterodimerizing alterations, one or more alteration that inhibits or enhances binding to FcγR, or one or more alterations that increase binding to FcRn.

[0094] In a specific embodiment, the Fc utilized for half-life extension is a single chain Fc (“scFc”).

[0095] In some embodiments the amino acid sequences of the Fc polypeptides can be mammalian, for example a human, amino acid sequences. The isotype of the Fc polypeptide can be IgG, such as IgG1, IgG2, IgG3, or IgG4, IgA, IgD, IgE, or IgM. Table 2 below shows an alignment of the amino acid sequences of human IgG1, IgG2, IgG3, and IgG4 Fc polypeptide chains.

[0096] Sequences of human IgG1, IgG2, IgG3, and IgG4 Fc polypeptides that could be used are provided in SEQ ID NOs: 45-48. Variants of these sequences containing one or more heterodimerizing alterations, one or more Fc alteration that extends half-life, one or more alteration that enhances ADCC, and/or one or more alteration that inhibits Fc gamma receptor (FcγR) binding are also contemplated, as are other close variants containing not more than 10 deletions, insertions, or substitutions of a single amino acid per 100 amino acids of sequence.

TABLE 2

Amino acid sequences of human IgG Fc polypeptide chains						
IgG1	-----					
IgG2	-----					
IgG3	ELKTPGLGDTHTHTCPRCPEPKSCDTPPPCPRCPEPKSCDTPPPCPRCP					
IgG4	-----					
	225	235	245	255	265	275
	*	*	*	*	*	*
IgG1	EPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKF					
IgG2	ERKCCVE--CPPCAPPVA-GPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVQF					
IgG3	EPKSCDTPPPCPRCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVQF					
IgG4	ESKYG---PPCPSCPAPEFLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSDQEDPEVQF					
	285	295	305	315	325	335
	*	*	*	*	*	*
IgG1	NWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT					
IgG2	NWYVDGMEVHNAKTKPREEQFNSTFRVSVLTVVHQDWLNGKEYKCKVSNKGLPAPIEKT					
IgG3	KWYVDGVEVHNAKTKPREEQYNSTFRVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT					
IgG4	NWYVDGVEVHNAKTKPREEQFNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKGLPSSIEKT					
	345	355	365	375	385	395
	*	*	*	*	*	*
IgG1	ISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTP					
IgG2	ISKTKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTP					
IgG3	ISKTKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESGQPENNYNTTP					
IgG4	ISKAKGQPREPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTP					
	405	415	425	435	445	
	*	*	*	*	*	
IgG1	PVLDSGSGFFLYSKLTVDKSRWQQGNVFSQSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 45)					
IgG2	PMLDSGSGFFLYSKLTVDKSRWQQGNVFSQSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 46)					
IgG3	PMLDSGSGFFLYSKLTVDKSRWQQGNIFSCSVMEALHNRFQKSLSLSPGK (SEQ ID NO: 47)					
IgG4	PVLDSGSGFFLYSRLTVDKSRWQEGNVFSQSVMEALHNHYTQKSLSLSLGK (SEQ ID NO: 48)					

[0097] The numbering shown in Table 2 is according to the EU system of numbering, which is based on the sequential numbering of the constant region of an IgG1 antibody. Edelman et al. (1969), Proc. Natl. Acad. Sci. 63:78-85. Thus, it does not accommodate the additional length of the IgG3 hinge well. It is nonetheless used here to designate positions in an Fc region because it is still commonly used in the art to refer to positions in Fc regions. The hinge regions of the IgG1, IgG2, and IgG4 Fc polypeptides extend from about position 216 to about 230. It is clear from the alignment that the IgG2 and IgG4 hinge regions are each three amino acids shorter than the IgG1 hinge. The IgG3 hinge is much longer, extending for an additional 47 amino acids upstream. The CH2 region extends from about position 231 to 340, and the CH3 region extends from about position 341 to 447.

[0098] Naturally occurring amino acid sequences of Fc polypeptides can be varied slightly. Such variations can include no more than 10 insertions, deletions, or substitutions of a single amino acid per 100 amino acids of sequence of a naturally occurring Fc polypeptide chain. If there are substitutions, they can be conservative amino acid substitutions, as defined above. The Fc polypeptides on the first and second polypeptide chains can differ in amino acid sequence. In some embodiments, they can include “heterodimerizing alterations,” for example, charge pair substitutions, as defined above, that facilitate heterodimer formation. Further, the Fc polypeptide portions of the PABP can also contain alterations that inhibit or enhance FcγR binding. Such mutations are described above and in Xu et al. (2000), Cell Immunol. 200 (1): 16-26, the relevant portions of which

are incorporated herein by reference. The Fc polypeptide portions can also include an “Fc alteration that extends half-life,” as described above, including those described in, e.g., U.S. Pat. Nos. 7,037,784, 7,670,600, and 7,371,827, US Patent Application Publication 2010/0234575, and International Application PCT/US2012/070146, the relevant portions of all of which are incorporated herein by reference. Further, an Fc polypeptide can comprise “alterations that enhance ADCC,” as defined above.

[0099] Another suitable Fc polypeptide, described in PCT application WO 93/10151 (hereby incorporated by reference), is a single chain polypeptide extending from the N-terminal hinge region to the native C-terminus of the Fc region of a human IgG1 antibody. Another useful Fc polypeptide is the Fc mutein described in U.S. Pat. No. 5,457,035 and in Baum et al., 1994, EMBO J. 13:3992-4001. The amino acid sequence of this mutein is identical to that of the native Fc sequence presented in WO 93/10151, except that amino acid 19 has been changed from Leu to Ala, amino acid 20 has been changed from Leu to Glu, and amino acid 22 has been changed from Gly to Ala. The mutein exhibits reduced affinity for Fc receptors.

[0100] The effector function of an antibody can be increased, or decreased, by introducing one or more mutations into the Fc. Embodiments of the invention include IL-2 mutein Fc fusion proteins having an Fc engineered to increase effector function (U.S. Pat. No. 7,317,091 and Strohl, Curr. Opin. Biotech., 20:685-691, 2009; both incorporated herein by reference in its entirety). For certain therapeutic indications, it may be desirable to increase effector function. For other therapeutic indications, it may be desirable to decrease effector function.

[0101] Exemplary IgG1 Fc molecules having increased effector function include those having the following substitutions:

- [0102] S239D/I332E
- [0103] S239D/A330S/I332E
- [0104] S239D/A330L/I332E
- [0105] S298A/D333A/K334A
- [0106] P2471/A339D
- [0107] P2471/A339Q
- [0108] D280H/K290S
- [0109] D280H/K290S/S298D
- [0110] D280H/K290S/S298V
- [0111] F243L/R292P/Y300L
- [0112] F243L/R292P/Y300L/P396L
- [0113] F243L/R292P/Y300L/V3051/P396L
- [0114] G236A/S239D/I332E
- [0115] K326A/E333A
- [0116] K326W/E333S
- [0117] K290E/S298G/T299A
- [0118] K290N/S298G/T299A
- [0119] K290E/S298G/T299A/K326E
- [0120] K290N/S298G/T299A/K326E

[0121] Another method of increasing effector function of IgG Fc-containing proteins is by reducing the fucosylation of the Fc. Removal of the core fucose from the biantennary complex-type oligosaccharides attached to the Fc greatly increased ADCC effector function without altering antigen binding or CDC effector function. Several ways are known for reducing or abolishing fucosylation of Fc-containing molecules, e.g., antibodies. These include recombinant expression in certain mammalian cell lines including a FUT8 knockout cell line, variant CHO line Lec13, rat hybridoma cell line YB2/0, a cell line comprising a small interfering RNA specifically against the FUT8 gene, and a cell line coexpressing α -1,4-N-acetylglucosaminyltransferase III and Golgi α -mannosidase II. Alternatively, the Fc-containing molecule may be expressed in a non-mammalian cell such as a plant cell, yeast, or prokaryotic cell, e.g., *E. coli*.

[0122] In certain embodiments of the invention, the molecules comprise an Fc engineered to decrease effector function. Exemplary Fc molecules having decreased effector function include those having the following substitutions:

- [0123] N297A or N297Q (IgG1)
- [0124] L234A/L235A (IgG1)
- [0125] V234A/G237A (IgG2)
- [0126] L235A/G237A/E318A (IgG4)
- [0127] H268Q/V309L/A330S/A331S (IgG2)
- [0128] C220S/C226S/C229S/P238S (IgG1)
- [0129] C226S/C229S/E233P/L234V/L235A (IgG1)
- [0130] L234F/L235E/P331S (IgG1)
- [0131] S267E/L328F (IgG1)

[0132] It is known that human IgG1 has a glycosylation site at N297 (EU numbering system) and glycosylation contributes to the effector function of IgG1 antibodies. An exemplary IgG1 sequence is provided in SEQ ID NO: 45. N297 can be mutated to make aglycosylated antibodies. For example, mutations can substitute N297 with amino acids that resemble asparagine in physiochemical nature such as glutamine (N297Q), or with alanine (N297A), which mimics asparagines without polar groups.

[0133] In certain embodiments, mutation of amino acid N297 of human IgG1 to glycine, i.e., N297G, provides far superior purification efficiency and biophysical properties

over other amino acid substitutions at that residue. See, for example, U.S. Pat. Nos. 9,546,203 and 10,093,711. In a specific embodiment, the molecules of the invention comprise a human IgG1 Fc having an N297G substitution.

[0134] A molecule of the invention comprising a human IgG1 Fc having the N297G mutation may also comprise further insertions, deletions, and substitutions. In certain embodiments the human IgG1 Fc comprises the N297G substitution and is at least 90% identical, at least 91% identical, at least 92% identical, at least 93% identical, at least 94% identical, at least 95% identical, at least 96% identical, at least 97% identical, at least 98% identical, or at least 99% identical to the amino acid sequence set forth in SEQ ID NO: 45. In a particularly preferred embodiment, the C-terminal lysine residue is substituted or deleted.

[0135] In certain instances, aglycosylated IgG1 Fc-containing molecules can be less stable than glycosylated IgG1 Fc-containing molecules. Accordingly, the Fc region may be further engineered to increase the stability of the aglycosylated molecule. In some embodiments, one or more amino acids are substituted to cysteine so to form di-sulfide bonds in the dimeric state. In specific embodiments, residues V259, A287, R292, V302, L306, V323, or I332 of the amino acid sequence set forth in SEQ ID NO: 45 may be substituted with cysteine. In other embodiments, specific pairs of residues are substituted such that they preferentially form a di-sulfide bond with each other, thus limiting or preventing di-sulfide bond scrambling. In specific embodiments, pairs include, but are not limited to, A287C and L306C, V259C and L306C, R292C and V302C, and V323C and I332C.

[0136] As discussed herein above in the Linker section, the molecules of the invention comprise linkers between the various domains and moieties that make up the (HHLL)² molecule and as depicted in, e.g., FIG. 2 herein. In certain embodiments, the linkers are glycosylated when expressed in the appropriate cells and such glycosylation may help stabilize the protein in solution and/or when administered in vivo. Accordingly, in certain embodiments, a molecule of the invention comprises at least one glycosylated linker between domains of the (HHLL)² polypeptide.

Nucleic Acids Encoding the Molecules

[0137] In another embodiment, the present invention provides isolated nucleic acid molecules that encode the molecules of the present invention. In addition, provided are vectors comprising the nucleic acids, cell comprising the nucleic acids, and methods of making the binding molecules of the invention. The nucleic acids comprise, for example, polynucleotides that encode all or part of molecule, for example, or a fragment, derivative, mutein, or variant thereof, polynucleotides sufficient for use as hybridization probes, PCR primers or sequencing primers for identifying, analyzing, mutating or amplifying a polynucleotide encoding a polypeptide, anti-sense nucleic acids for inhibiting expression of a polynucleotide, and complementary sequences of the foregoing. The nucleic acids can be any length as appropriate for the desired use or function, and can comprise one or more additional sequences, for example, regulatory sequences, and/or be part of a larger nucleic acid, for example, a vector. The nucleic acids can be single-stranded or double-stranded and can comprise RNA and/or DNA nucleotides, and artificial variants thereof (e.g., peptide nucleic acids).

[0138] Nucleic acids encoding polypeptides (e.g., heavy or light chain, variable domain only, or full length) may be isolated from B-cells of mice that have been immunized with antigen. The nucleic acid may be isolated by conventional procedures such as polymerase chain reaction (PCR).

[0139] Nucleic acid sequences encoding the variable regions of the heavy and light chain variable regions are included herein. The skilled artisan will appreciate that, due to the degeneracy of the genetic code, each of the polypeptide sequences disclosed herein is encoded by a large number of other nucleic acid sequences. The present invention provides each degenerate nucleotide sequence encoding each binding molecule of the invention.

[0140] The invention further provides nucleic acids that hybridize to other nucleic acids under particular hybridization conditions. Methods for hybridizing nucleic acids are well-known in the art. See, e.g., Current Protocols in Molecular Biology, John Wiley & Sons, N.Y. (1989), 6.3.1-6.3.6. As defined herein, for example, a moderately stringent hybridization condition uses a prewashing solution containing 5× sodium chloride/sodium citrate (SSC), 0.5% SDS, 1.0 mM EDTA (pH 8.0), hybridization buffer of about 50% formamide, 6×SSC, and a hybridization temperature of 55° C. (or other similar hybridization solutions, such as one containing about 50% formamide, with a hybridization temperature of 42° C.), and washing conditions of 60° C., in 0.5×SSC, 0.1% SDS. A stringent hybridization condition hybridizes in 6×SSC at 45° C., followed by one or more washes in 0.1×SSC, 0.2% SDS at 68° C. Furthermore, one of skill in the art can manipulate the hybridization and/or washing conditions to increase or decrease the stringency of hybridization such that nucleic acids comprising nucleotide sequences that are at least 65, 70, 75, 80, 85, 90, 95, 98 or 99% identical to each other typically remain hybridized to each other. The basic parameters affecting the choice of hybridization conditions and guidance for devising suitable conditions are set forth by, for example, Sambrook, Fritsch, and Maniatis (1989, Molecular Cloning: A Laboratory Manual, Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y., chapters 9 and 11; and Current Protocols in Molecular Biology, 1995, Ausubel et al., eds., John Wiley & Sons, Inc., sections 2.10 and 6.3-6.4), and can be readily determined by those having ordinary skill in the art based on, for example, the length and/or base composition of the DNA. Changes can be introduced by mutation into a nucleic acid, thereby leading to changes in the amino acid sequence of a polypeptide (e.g., a binding molecule) that it encodes. Mutations can be introduced using any technique known in the art. In one embodiment, one or more particular amino acid residues are changed using, for example, a site-directed mutagenesis protocol. In another embodiment, one or more randomly selected residues is changed using, for example, a random mutagenesis protocol. However, it is made, a mutant polypeptide can be expressed and screened for a desired property.

[0141] Mutations can be introduced into a nucleic acid without significantly altering the biological activity of a polypeptide that it encodes. For example, one can make nucleotide substitutions leading to amino acid substitutions at non-essential amino acid residues. In one embodiment, a nucleotide sequence provided herein for of the binding molecules of the present invention, or a desired fragment, variant, or derivative thereof, is mutated such that it encodes an amino acid sequence comprising one or more deletions or

substitutions of amino acid residues that are shown herein for the light chains of the binding molecules of the present invention or the heavy chains of the binding molecules of the present invention to be residues where two or more sequences differ. In another embodiment, the mutagenesis inserts an amino acid adjacent to one or more amino acid residues shown herein for the light chains of the binding molecules of the present invention or the heavy chains of the binding molecules of the present invention to be residues where two or more sequences differ. Alternatively, one or more mutations can be introduced into a nucleic acid that selectively change the biological activity of a polypeptide that it encodes.

[0142] In another embodiment, the present invention provides vectors comprising a nucleic acid encoding a polypeptide of the invention or a portion thereof. Examples of vectors include, but are not limited to, plasmids, viral vectors, non-episomal mammalian vectors and expression vectors, for example, recombinant expression vectors.

[0143] The recombinant expression vectors of the invention can comprise a nucleic acid of the invention in a form suitable for expression of the nucleic acid in a host cell. The recombinant expression vectors include one or more regulatory sequences, selected on the basis of the host cells to be used for expression, which is operably linked to the nucleic acid sequence to be expressed. Regulatory sequences include those that direct constitutive expression of a nucleotide sequence in many types of host cells (e.g., SV40 early gene enhancer, Rous sarcoma virus promoter and cytomegalovirus promoter), those that direct expression of the nucleotide sequence only in certain host cells (e.g., tissue-specific regulatory sequences, see Voss et al., 1986, Trends Biochem. Sci. 11:287; Maniatis et al., 1987, Science 236:1237, incorporated by reference herein in their entireties), and those that direct inducible expression of a nucleotide sequence in response to particular treatment or condition (e.g., the metallothionin promoter in mammalian cells and the tet-responsive and/or streptomycin responsive promoter in both prokaryotic and eukaryotic systems (see id.)). It will be appreciated by those skilled in the art that the design of the expression vector can depend on such factors as the choice of the host cell to be transformed, the level of expression of protein desired, etc. The expression vectors of the invention can be introduced into host cells to thereby produce proteins or peptides, including fusion proteins or peptides, encoded by nucleic acids as described herein.

[0144] In another embodiment, the present invention provides host cells into which a recombinant expression vector of the invention has been introduced. A host cell can be any prokaryotic cell or eukaryotic cell. Prokaryotic host cells include gram negative or gram positive organisms, for example *E. coli* or bacilli. Higher eukaryotic cells include insect cells, yeast cells, and established cell lines of mammalian origin. Examples of suitable mammalian host cell lines include Chinese hamster ovary (CHO) cells or their derivatives such as Veggie CHO and related cell lines which grow in serum-free media (see Rasmussen et al., 1998, Cytotechnology 28:31) or CHO strain DXB-11, which is deficient in DHFR (see Urlaub et al., 1980, Proc. Natl. Acad. Sci. USA 77:4216-20). Additional CHO cell lines include CHO-K1 (ATCC #CCL-61), EM9 (ATCC #CRL-1861), and UV20 (ATCC #CRL-1862). Additional host cells include the COS-7 line of monkey kidney cells (ATCC CRL 1651) (see Gluzman et al., 1981, Cell 23:175), L cells, C127 cells, 3T3

cells (ATCC CCL 163), AM-1/D cells (described in U.S. Pat. No. 6,210,924), HeLa cells, BHK (ATCC CRL 10) cell lines, the CV1/EBNA cell line derived from the African green monkey kidney cell line CV1 (ATCC CCL 70) (see McMahon et al., 1991, EMBO J. 10:2821), human embryonic kidney cells such as 293, 293 EBNA or MSR 293, human epidermal A431 cells, human Colo205 cells, other transformed primate cell lines, normal diploid cells, cell strains derived from in vitro culture of primary tissue, primary explants, HL-60, U937, HaK or Jurkat cells. Appropriate cloning and expression vectors for use with bacterial, fungal, yeast, and mammalian cellular hosts are described by Pouwels et al. (Cloning Vectors: A Laboratory Manual, Elsevier, New York, 1985).

[0145] Vector DNA can be introduced into prokaryotic or eukaryotic cells via conventional transformation or transfection techniques. For stable transfection of mammalian cells, it is known that, depending upon the expression vector and transfection technique used, only a small fraction of cells may integrate the foreign DNA into their genome. In order to identify and select these integrants, a gene that encodes a selectable marker (e.g., for resistance to antibiotics) is generally introduced into the host cells along with the gene of interest. Additional selectable markers include those which confer resistance to drugs, such as G418, hygromycin and methotrexate. Cells stably transfected with the introduced nucleic acid can be identified by drug selection (e.g., cells that have incorporated the selectable marker gene will survive, while the other cells die), among other methods.

[0146] The transformed cells can be cultured under conditions that promote expression of the polypeptide, and the polypeptide recovered by conventional protein purification procedures. Polypeptides contemplated for use herein include substantially homogeneous recombinant mammalian polypeptides substantially free of contaminating endogenous materials.

[0147] Cells containing the nucleic acid encoding the molecules of the present invention also include hybridomas. The production and culturing of hybridomas are discussed herein.

[0148] In some embodiments, a vector comprising a nucleic acid molecule as described herein is provided. In some embodiments, the invention comprises a host cell comprising a nucleic acid molecule as described herein.

[0149] In some embodiments, a nucleic acid molecule encoding the molecules as described herein is provided.

[0150] In some embodiments, a pharmaceutical composition comprising at least one molecule described herein is provided.

Methods of Producing

[0151] The molecules of the invention can be produced by any method known in the art for the synthesis of proteins (e.g., antibodies), in particular, by chemical synthesis or preferably, by recombinant expression techniques.

[0152] Recombinant expression of the molecules requires construction of an expression vector containing a polynucleotide that encodes the molecule. Once a polynucleotide encoding the molecule has been obtained, the vector for the production of the molecule may be produced by recombinant DNA technology. An expression vector is constructed containing the molecule coding sequences and appropriate transcriptional and translational control signals. These meth-

ods include, for example, in vitro recombinant DNA techniques, synthetic techniques, and in vivo genetic recombination.

[0153] The expression vector is transferred to a host cell by conventional techniques and the transfected cells are then cultured by conventional techniques to produce a molecule of the invention.

[0154] A variety of host-expression vector systems may be utilized to express the molecules of the invention. Such host-expression systems represent vehicles by which the coding sequences of interest may be produced and subsequently purified, but also represent cells which may, when transformed or transfected with the appropriate nucleotide coding sequences, express a molecule of the invention in situ. Bacterial cells such as *E. coli*, and eukaryotic cells are commonly used for the expression of a recombinant binding molecule, especially for the expression of whole recombinant binding molecule. For example, mammalian cells such as Chinese hamster ovary cells (CHO), in conjunction with a vector such as the major intermediate early gene promoter element from human cytomegalovirus is an effective expression system for antibodies (Foecking et al., Gene 45:101 (1986); Cockett et al., Bio/Technology 8:2 (1990)).

[0155] In addition, a host cell strain may be chosen which modulates the expression of the inserted sequences, or modifies and processes the gene product in the specific fashion desired. Such modifications (e.g., glycosylation) and processing (e.g., cleavage) of protein products may be important for the function of the protein. Different host cells have characteristic and specific mechanisms for the post-translational processing and modification of proteins and gene products. Appropriate cell lines or host systems can be chosen to ensure the correct modification and processing of the foreign protein expressed. To this end, eukaryotic host cells which possess the cellular machinery for proper processing of the primary transcript, glycosylation, and phosphorylation of the gene product may be used. Such mammalian host cells include, but are not limited to, CHO, COS, 293, 3T3, or myeloma cells.

[0156] For long-term, high-yield production of recombinant proteins, stable expression is preferred. For example, cell lines which stably express the binding molecule may be engineered. Rather than using expression vectors which contain viral origins of replication, host cells can be transformed with DNA controlled by appropriate expression control elements (e.g., promoter, enhancer, sequences, transcription terminators, polyadenylation sites, etc.), and a selectable marker. Following the introduction of the foreign DNA, engineered cells may be allowed to grow for 1-2 days in an enriched media, and then are switched to a selective media. The selectable marker in the recombinant plasmid confers resistance to the selection and allows cells to stably integrate the plasmid into their chromosomes and grow to form foci which in turn can be cloned and expanded into cell lines. This method may advantageously be used to engineer cell lines which express the binding molecule. Such engineered cell lines may be particularly useful in screening and evaluation of compounds that interact directly or indirectly with the binding molecule.

[0157] A number of selection systems may be used, including but not limited to the herpes simplex virus thymidine kinase (Wigler et al., Cell 11:223 (1977)), hypoxanthine-guanine phosphoribosyltransferase (Szybalska & Szybalski, Proc. Natl. Acad. Sci. USA 48:202 (1992)), and

adenine phosphoribosyltransferase (Lowy et al., Cell 22:817 (1980)) genes can be employed in tk, hgp^rt or ap^rt-cells, respectively. Also, antimetabolite resistance can be used as the basis of selection for the following genes: dhfr, which confers resistance to methotrexate (Wigler et al., Proc. Natl. Acad. Sci. USA 77:357 (1980); O'Hare et al., Proc. Natl. Acad. Sci. USA 78:1527 (1981)); gpt, which confers resistance to mycophenolic acid (Mulligan & Berg, Proc. Natl. Acad. Sci. USA 78:2072 (1981)); neo, which confers resistance to the aminoglycoside G-418 (Wu and Wu, Biotherapy 3:87-95 (1991)); and hyg^r, which confers resistance to hygromycin (Santerre et al., Gene 30:147 (1984)). Methods commonly known in the art of recombinant DNA technology may be routinely applied to select the desired recombinant clone, and such methods are described, for example, in Ausubel et al. (eds.), Current Protocols in Molecular Biology, John Wiley & Sons, NY (1993); Kriegler, Gene Transfer and Expression, A Laboratory Manual, Stockton Press, NY (1990); and in Chapters 12 and 13, Dracopoli et al. (eds.), Current Protocols in Human Genetics, John Wiley & Sons, NY (1994); Colberre-Garapin et al., J. Mol. Biol. 150:1 (1981), which are incorporated by reference herein in their entireties.

[0158] The expression levels of a binding molecule can be increased by vector amplification (for a review, see Bebbington and Hentschel, "The use of vectors based on gene amplification for the expression of cloned genes in mammalian cells" (DNA Cloning, Vol. 3. Academic Press, New York, 1987)). When a marker in the vector system expressing binding is amplifiable, increase in the level of inhibitor present in culture of host cell will increase the number of copies of the marker gene. Since the amplified region is associated with the gene, production of the protein will also increase (Crouse et al., Mol. Cell. Biol. 3:257 (1983)).

[0159] The host cell may be co-transfected with multiple expression vectors of the invention. The vectors may contain identical selectable markers which enable equal expression of the expressed polypeptides. Alternatively, a single vector may be used which encodes, and is capable of expressing, for example, the polypeptides of the invention. The coding sequences may comprise cDNA or genomic DNA.

[0160] Once a binding molecule of the invention has been produced by an animal, chemically synthesized, or recombinantly expressed, it may be purified by any method known in the art for purification of an immunoglobulin molecule, for example, by chromatography (e.g., ion exchange, affinity, particularly by affinity for the specific antigen after Protein A, and size-exclusion chromatography), centrifugation, differential solubility, or by any other standard technique for the purification of proteins. In addition, the binding molecules of the present invention or fragments thereof can be fused to heterologous polypeptide sequences described herein or otherwise known in the art, to facilitate purification. The purification techniques may be varied, depending on whether an Fc region (e.g., an scFC) is attached to the molecules of the invention.

[0161] In some embodiments, the present invention encompasses binding molecules recombinantly fused or chemically conjugated (including both covalently and non-covalently conjugations) to a polypeptide. Fused or conjugated binding molecules of the present invention may be used for case in purification. See e.g., Harbor et al., supra, and PCT publication WO 93/21232; EP 439,095; Naramura et al., Immunol. Lett. 39:91-99 (1994); U.S. Pat. No. 5,474,

981; Gillies et al., Proc. Natl. Acad. Sci. 89:1428-1432 (1992); Fell et al., J. Immunol. 146:2446-2452 (1991).

[0162] Moreover, the binding molecules or fragments thereof of the present invention can be fused to marker sequences, such as a peptide to facilitate purification. In preferred embodiments, the marker amino acid sequence is a hexa-histidine peptide (SEQ ID NO: 58), such as the tag provided in a pQE vector (QIAGEN, Inc., 9259 Eton Avenue, Chatsworth, Calif., 91311), among others, many of which are commercially available. As described in Gentz et al., Proc. Natl. Acad. Sci. USA 86:821-824 (1989), for instance, hexa-histidine (SEQ ID NO: 58) provides for convenient purification of the fusion protein. Other peptide tags useful for purification include, but are not limited to, the "HA" tag, which corresponds to an epitope derived from the influenza hemagglutinin protein (Wilson et al., Cell 37:767 (1984)) and the "flag" tag.

Generation of Molecules

[0163] The molecules of the invention, in a general sense, are constructed by selecting VH and VL regions from desired antibodies and linking them using polypeptide linkers as described herein to form the (HHLL)² molecule, optionally with an Fc region attached. More specifically, the nucleic acids encoding the VH, VL and linkers, and optionally the Fc, are combined to create the (HHLL)² nucleic acid constructs that encode the molecules of the invention.

Generation of Antibodies

[0164] In certain embodiments, prior to generation of the molecules of the invention, monospecific antibodies are first generated with binding specificities to desired targets.

[0165] Antibodies useful for generating the molecules of the invention may be prepared by techniques that are well known to those skilled in the art. For example, by immunizing an animal (e.g., a mouse or rat or rabbit) and then by immortalizing spleen cells harvested from the animal after completion of the immunization schedule. The spleen cells can be immortalized using any technique known in the art, e.g., by fusing them with myeloma cells to produce hybridomas. See, for example, Antibodies; Harlow and Lane, Cold Spring Harbor Laboratory Press, 1st Edition, e.g. from 1988, or 2nd Edition, e.g. from 2014).

[0166] In one embodiment, a humanized monoclonal antibody comprises the variable domain of a murine antibody (or all or part of the antigen binding site thereof) and a constant domain derived from a human antibody. Alternatively, a humanized antibody fragment may comprise the antigen binding site of a murine monoclonal antibody and a variable domain fragment (lacking the antigen-binding site) derived from a human antibody. Procedures for the production of engineered monoclonal antibodies include those described in Riechmann et al., 1988, Nature 332:323; Liu et al., 1987, Proc. Nat. Acad. Sci. USA 84:3439; Larrick et al., 1989, Bio/Technology 7:934, and Winter et al., 1993, TIPS 14:139. In one embodiment, the chimeric antibody is a CDR grafted antibody. Techniques for humanizing antibodies are discussed in, e.g., U.S. Pat. Nos. 5,869,619; 5,225,539; 5,821,337; 5,859,205; 6,881,557, Padlan et al., 1995, FASEB J. 9:133-39; Tamura et al., 2000, J. Immunol. 164:1432-41; Zhang, W., et al., Molecular Immunology. 42 (12): 1445-1451, 2005; Hwang W. et al., Methods. 36 (1):

35-42, 2005; Dall'Acqua W F, et al., *Methods* 36 (1): 43-60, 2005; and Clark, M., *Immunology Today*. 21 (8): 397-402, 2000.

[0167] A molecule of the present invention may also comprise regions of a fully human monoclonal antibody. Fully human monoclonal antibodies may be generated by any number of techniques with which those having ordinary skill in the art will be familiar. Such methods include, but are not limited to, Epstein Barr Virus (EBV) transformation of human peripheral blood cells (e.g., containing B lymphocytes), in vitro immunization of human B-cells, fusion of spleen cells from immunized transgenic mice carrying inserted human immunoglobulin genes, isolation from human immunoglobulin V region phage libraries, or other procedures as known in the art and based on the disclosure herein.

[0168] Procedures have been developed for generating human monoclonal antibodies in non-human animals. For example, mice in which one or more endogenous immunoglobulin genes have been inactivated by various means have been prepared. Human immunoglobulin genes have been introduced into the mice to replace the inactivated mouse genes. In this technique, elements of the human heavy and light chain locus are introduced into strains of mice derived from embryonic stem cell lines that contain targeted disruptions of the endogenous heavy chain and light chain loci (see also Bruggemann et al., *Curr. Opin. Biotechnol.* 8:455-58 (1997)). For example, human immunoglobulin transgenes may be mini-gene constructs, or transloci on yeast artificial chromosomes, which undergo B-cell-specific DNA rearrangement and hypermutation in the mouse lymphoid tissue.

[0169] Antibodies produced in the animal incorporate human immunoglobulin polypeptide chains encoded by the human genetic material introduced into the animal. In one embodiment, a non-human animal, such as a transgenic mouse, is immunized with a suitable immunogen.

[0170] Examples of techniques for production and use of transgenic animals for the production of human or partially human antibodies are described in U.S. Pat. Nos. 5,814,318, 5,569,825, and 5,545,806, Davis et al., *Production of human antibodies from transgenic mice* in Lo, ed. *Antibody Engineering: Methods and Protocols*, Humana Press, NJ: 191-200 (2003), Kellermann et al., 2002, *Curr Opin Biotechnol.* 13:593-97, Russel et al., 2000, *Infect Immun.* 68:1820-26, Gallo et al., 2000, *Eur J Immun.* 30:534-40, Davis et al., 1999, *Cancer Metastasis Rev.* 18:421-25, Green, 1999, *J Immunol Methods.* 231:11-23, Jakobovits, 1998, *Advanced Drug Delivery Reviews* 31:33-42, Green et al., 1998, *J Exp Med.* 188:483-95, Jakobovits A, 1998, *Exp. Opin. Invest. Drugs.* 7:607-14, Tsuda et al., 1997, *Genomics.* 42:413-21, Mendez et al., 1997, *Nat Genet.* 15:146-56, Jakobovits, 1994, *Curr Biol.* 4:761-63, Arbones et al., 1994, *Immunity.* 1:247-60, Green et al., 1994, *Nat Genet.* 7:13-21, Jakobovits et al., 1993, *Nature.* 362:255-58, Jakobovits et al., 1993, *Proc Natl Acad Sci USA.* 90:2551-55. Chen, J., M. Trounstein, F. W. Alt, F. Young, C. Kurahara, J. Loring, D. Huszar. "Immunoglobulin gene rearrangement in B-cell deficient mice generated by targeted deletion of the JH locus." *International Immunology* 5 (1993): 647-656, Choi et al., 1993, *Nature Genetics* 4:117-23, Fishwild et al., 1996, *Nature Biotechnology* 14:845-51, Harding et al., 1995, *Annals of the New York Academy of Sciences*, Lonberg et al., 1994, *Nature* 368:856-59, Lonberg, 1994, *Transgenic Approaches to Human Monoclonal Antibodies in Handbook*

of Experimental Pharmacology 113:49-101, Lonberg et al., 1995, *Internal Review of Immunology* 13:65-93, Neuberger, 1996, *Nature Biotechnology* 14:826, Taylor et al., 1992, *Nucleic Acids Research* 20:6287-95, Taylor et al., 1994, *International Immunology* 6:579-91, Tomizuka et al., 1997, *Nature Genetics* 16:133-43, Tomizuka et al., 2000, *Proceedings of the National Academy of Sciences USA* 97:722-27, Tuaillon et al., 1993, *Proceedings of the National Academy of Sciences USA* 90:3720-24, and Tuaillon et al., 1994, *Journal of Immunology* 152:2912-20.; Lonberg et al., *Nature* 368:856, 1994; Taylor et al., *Int. Immun.* 6:579, 1994; U.S. Pat. No. 5,877,397; Bruggemann et al., 1997 *Curr. Opin. Biotechnol.* 8:455-58; Jakobovits et al., 1995 *Ann. N. Y. Acad. Sci.* 764:525-35. In addition, protocols involving the XenoMouse® (Abgenix, now Amgen, Inc.) are described, for example in U.S. Ser. No. 05/011,8643 and WO 05/694879, WO 98/24838, WO 00/76310, and U.S. Pat. No. 7,064,244.

[0171] Lymphoid cells from the immunized transgenic mice are fused with myeloma cells for example to produce hybridomas. Myeloma cells for use in hybridoma-producing fusion procedures preferably are non-antibody-producing, have high fusion efficiency, and enzyme deficiencies that render them incapable of growing in certain selective media which support the growth of only the desired fused cells (hybridomas). Examples of suitable cell lines for use in such fusions include Sp-20, P3-X63/Ag8, P3-X63-Ag8.653, NS1/1.Ag 4 1, Sp210-Ag14, FO, NSO/U, MPC-11, MPC11-X45-GTG 1.7 and S194/5XX0 Bul; examples of cell lines used in rat fusions include R210.RCY3, Y3-Ag 1.2.3, IR983F and 4B210. Other cell lines useful for cell fusions are U-266, GM1500-GRG2, LICR-LON-HMy2 and UC729-6.

[0172] The lymphoid (e.g., spleen) cells and the myeloma cells may be combined for a few minutes with a membrane fusion-promoting agent, such as polyethylene glycol or a nonionic detergent, and then plated at low density on a selective medium that supports the growth of hybridoma cells but not unfused myeloma cells. One selection media is HAT (hypoxanthine, aminopterin, thymidine). After a sufficient time, usually about one to two weeks, colonies of cells are observed. Single colonies are isolated, and antibodies produced by the cells may be tested for binding activity to desired targets using any one of a variety of immunoassays known in the art and described herein. The hybridomas are cloned (e.g., by limited dilution cloning or by soft agar plaque isolation) and positive clones that produce a molecule specific to a desired target is selected and cultured. The binding molecules from the hybridoma cultures may be isolated from the supernatants of hybridoma cultures. Thus, the present invention provides hybridomas that comprise polynucleotides encoding the binding molecules of the invention in the chromosomes of the cell. These hybridomas can be cultured according to methods described herein and known in the art.

[0173] Another method for generating human antibodies useful for generating the binding molecules of the invention includes immortalizing human peripheral blood cells by EBV transformation. See, e.g., U.S. Pat. No. 4,464,456. Such an immortalized B-cell line (or lymphoblastoid cell line) producing a monoclonal antibody that specifically binds to a desired target can be identified by immunodetection methods as provided herein, for example, an ELISA, and then isolated by standard cloning techniques. The sta-

bility of the lymphoblastoid cell line producing an antibody may be improved by fusing the transformed cell line with a murine myeloma to produce a mouse-human hybrid cell line according to methods known in the art (see, e.g., Glasky et al., *Hybridoma* 8:377-89 (1989)). Still another method to generate human monoclonal antibodies is in vitro immunization, which includes priming human splenic B-cells with antigen, followed by fusion of primed B-cells with a heterohybrid fusion partner. See, e.g., Boerner et al., 1991 *J. Immunol.* 147:86-95.

[0174] In certain embodiments, a B-cell that is producing a desired antibody is selected and the light chain and heavy chain variable regions are cloned from the B-cell according to molecular biology techniques known in the art (WO 92/02551; U.S. Pat. No. 5,627,052; Babcook et al., *Proc. Natl. Acad. Sci. USA* 93:7843-48 (1996)) and described herein. B-cells from an immunized animal may be isolated from the spleen, lymph node, or peripheral blood sample by selecting a cell that is producing a desired antibody. B-cells may also be isolated from humans, for example, from a peripheral blood sample. Methods for detecting single B-cells that are producing an antibody with the desired specificity are well known in the art, for example, by plaque formation, fluorescence-activated cell sorting, in vitro stimulation followed by detection of specific antibody, and the like. Methods for selection of specific antibody-producing B-cells include, for example, preparing a single cell suspension of B-cells in soft agar that contains antigen. Binding of the specific antibody produced by the B-cell to the antigen results in the formation of a complex, which may be visible as an immunoprecipitate. After the B-cells producing the desired antibody are selected, the specific antibody genes may be cloned by isolating and amplifying DNA or mRNA and used to generate the molecules of the present invention according to methods known in the art and described herein.

[0175] An additional method for obtaining antibodies useful for generating the molecules of the invention is by phage display. See, e.g., Winter et al., 1994 *Annu. Rev. Immunol.* 12:433-55; Burton et al., 1994 *Adv. Immunol.* 57:191-280. Human or murine immunoglobulin variable region gene combinatorial libraries may be created in phage vectors that can be screened to select Ig fragments (Fab, Fv, sFv, or multimers thereof) that bind specifically to TGF-beta binding protein or variant or fragment thereof. See, e.g., U.S. Pat. No. 5,223,409; Huse et al., 1989 *Science* 246:1275-81; Sastry et al., *Proc. Natl. Acad. Sci. USA* 86:5728-32 (1989); Altling-Mees et al., *Strategies in Molecular Biology* 3:1-9 (1990); Kang et al., 1991 *Proc. Natl. Acad. Sci. USA* 88:4363-66; Hoogenboom et al., 1992 *J. Molec. Biol.* 227:381-388; Schlebusch et al., 1997 *Hybridoma* 16:47-52 and references cited therein. For example, a library containing a plurality of polynucleotide sequences encoding Ig variable region fragments may be inserted into the genome of a filamentous bacteriophage, such as M13 or a variant thereof, in frame with the sequence encoding a phage coat protein. A fusion protein may be a fusion of the coat protein with the light chain variable region domain and/or with the heavy chain variable region domain. According to certain embodiments, immunoglobulin Fab fragments may also be displayed on a phage particle (see, e.g., U.S. Pat. No. 5,698,426).

[0176] Heavy and light chain immunoglobulin cDNA expression libraries may also be prepared in lambda phage,

for example, using λ ImmunoZapTM(H) and λ ImmunoZapTM(L) vectors (Stratagene, La Jolla, California). Briefly, mRNA is isolated from a B-cell population, and used to create heavy and light chain immunoglobulin cDNA expression libraries in the λ ImmunoZap(H) and λ ImmunoZap(L) vectors. These vectors may be screened individually or co-expressed to form Fab fragments or antibodies (see Huse et al., *supra*; see also Sastry et al., *supra*). Positive plaques may subsequently be converted to a non-lytic plasmid that allows high level expression of monoclonal antibody fragments from *E. coli*.

[0177] In one embodiment, in a hybridoma the variable regions of a gene expressing a monoclonal antibody of interest are amplified using nucleotide primers. These primers may be synthesized by one of ordinary skill in the art, or may be purchased from commercially available sources. (See, e.g., Stratagene (La Jolla, California), which sells primers for mouse and human variable regions including, among others, primers for VH_a, VH_b, VH_c, VH_d, CH1, VL and CL regions.) These primers may be used to amplify heavy or light chain variable regions, which may then be inserted into vectors such as ImmunoZAPTMH or ImmunoZAPTML (Stratagene), respectively. These vectors may then be introduced into *E. coli*, yeast, or mammalian-based systems for expression. Large amounts of a single-chain protein containing a fusion of the VH and VL domains may be produced using these methods (see Bird et al., *Science* 242:423-426, 1988).

[0178] In certain embodiments, the binding molecules of the invention are obtained from transgenic animals (e.g., mice) that produce "heavy chain only" antibodies or "HCABs." HCABs are analogous to naturally occurring camel and llama single-chain VHH antibodies. See, for example, U.S. Pat. Nos. 8,507,748 and 8,502,014, and U.S. Patent Application Publication Nos. US2009/0285805A1, US2009/0169548A1, US2009/0307787A1, US2011/0314563A1, US2012/0151610A1, WO2008/122886A2, and WO2009/013620A2.

[0179] Once cells producing antibodies according to the invention have been obtained using any of the above-described immunization and other techniques, the specific antibody genes may be cloned by isolating and amplifying DNA or mRNA therefrom according to standard procedures as described herein and then used to generate the molecules of the present invention. The antibodies produced therefrom may be sequenced and the CDRs identified and the DNA coding for the CDRs may be manipulated as described previously to generate other molecules according to the invention.

[0180] Molecular evolution of the complementarity determining regions (CDRs) in the center of the antibody binding site also has been used to isolate antibodies with increased affinity, for example, those as described by Schier et al., 1996, *J. Mol. Biol.* 263:551. Accordingly, such techniques are useful in preparing binding molecules of the invention.

[0181] Although human, partially human, or humanized antibodies will be suitable for many applications, particularly those of the present invention, other types of binding molecules will be suitable for certain applications. These non-human antibodies can be, for example, derived from any antibody-producing animal, such as mouse, rat, rabbit, goat, donkey, or non-human primate (for example, monkey such as cynomolgous or rhesus monkey) or ape (e.g., chimpanzee)). An antibody from a particular species can be

made by, for example, immunizing an animal of that species with the desired immunogen or using an artificial system for generating antibodies of that species (e.g., a bacterial or phage display-based system for generating antibodies of a particular species), or by converting an antibody from one species into an antibody from another species by replacing, e.g., the constant region of the antibody with a constant region from the other species, or by replacing one or more amino acid residues of the antibody so that it more closely resembles the sequence of an antibody from the other species. In one embodiment, the antibody is a chimeric antibody comprising amino acid sequences derived from antibodies from two or more different species. Then, the desired binding region sequences can be used to generate the molecules of the present invention.

[0182] Where it is desired to improve the affinity of binding molecules according to the invention containing one or more of the above-mentioned CDRs can be obtained by a number of affinity maturation protocols including maintaining the CDRs (Yang et al., *J. Mol. Biol.*, 254, 392-403, 1995), chain shuffling (Marks et al., *Bio/Technology*, 10, 779-783, 1992), use of mutation strains of *E. coli*. (Low et al., *J. Mol. Biol.*, 250, 350-368, 1996), DNA shuffling (Patten et al., *Curr. Opin. Biotechnol.*, 8, 724-733, 1997), phage display (Thompson et al., *J. Mol. Biol.*, 256, 7-88, 1996) and additional PCR techniques (Cramer et al., *Nature*, 391, 288-291, 1998). All of these methods of affinity maturation are discussed by Vaughan et al. (*Nature Biotechnology*, 16, 535-539, 1998).

[0183] In certain embodiments, to generate the (HHLL)² molecules of the present invention it may first be desirable to generate a more typical single chain antibody which may be formed by linking heavy and light chain variable domain (Fv region) fragments via an amino acid bridge (short peptide linker), resulting in a single polypeptide chain. Such single-chain Fvs (scFvs) have been prepared by fusing DNA encoding a peptide linker between DNAs encoding the two variable domain polypeptides (VL and VH). The resulting polypeptides can fold back on themselves to form antigen-binding monomers, or they can form multimers (e.g., dimers, trimers, or tetramers), depending on the length of a flexible linker between the two variable domains (Kortt et al., 1997, *Prot. Eng.* 10:423; Kortt et al., 2001, *Biomol. Eng.* 18:95-108). Techniques developed for the production of single chain antibodies include those described in U.S. Pat. No. 4,946,778; Bird, 1988, *Science* 242:423; Huston et al., 1988, *Proc. Natl. Acad. Sci. USA* 85:5879; Ward et al., 1989, *Nature* 334:544; de Graaf et al., 2002, *Methods Mol Biol.* 178:379-87. These single chain antibodies are distinct from and differ from the molecules of the invention.

[0184] Antigen binding fragments derived from an antibody can also be obtained, for example, by proteolytic hydrolysis of the antibody, for example, pepsin or papain digestion of whole antibodies according to conventional methods. By way of example, antibody fragments can be produced by enzymatic cleavage of antibodies with pepsin to provide a 5S fragment termed F(ab')₂. This fragment can be further cleaved using a thiol reducing agent to produce 3.5S Fab' monovalent fragments. Optionally, the cleavage reaction can be performed using a blocking group for the sulfhydryl groups that result from cleavage of disulfide linkages. As an alternative, an enzymatic cleavage using papain produces two monovalent Fab fragments and an Fc fragment directly. These methods are described, for

example, by Goldenberg, U.S. Pat. No. 4,331,647; Nisonoff et al., *Arch. Biochem. Biophys.* 89:230, 1960; Porter, *Biochem. J.* 73:119, 1959; Edelman et al., in *Methods in Enzymology* 1:422 (Academic Press 1967); and by Andrews, S. M. and Titus, J. A. in *Current Protocols in Immunology* (Coligan J. E., et al., eds), John Wiley & Sons, New York (2003), pages 2.8.1-2.8.10 and 2.10A.1-2.10A.5. Other methods for cleaving antibodies, such as separating heavy chains to form monovalent light-heavy chain fragments (Fd), further cleaving of fragments, or other enzymatic, chemical, or genetic techniques may also be used, so long as the fragments bind to the antigen that is recognized by the intact antibody.

[0185] In certain embodiments, the molecules comprise one or more complementarity determining regions (CDRs) of an antibody. CDRs can be obtained by constructing polynucleotides that encode the CDR of interest. Such polynucleotides are prepared, for example, by using the polymerase chain reaction to synthesize the variable region using mRNA of antibody-producing cells as a template (see, for example, Larrick et al., *Methods: A Companion to Methods in Enzymology* 2:106, 1991; Courtenay-Luck, "Genetic Manipulation of Monoclonal Antibodies," in *Monoclonal Antibodies: Production, Engineering and Clinical Application*, Ritter et al. (eds.), page 166 (Cambridge University Press 1995); and Ward et al., "Genetic Manipulation and Expression of Antibodies," in *Monoclonal Antibodies: Principles and Applications*, Birch et al. (eds.), page 137 (Wiley-Liss, Inc. 1995)). The antibody fragment further may comprise at least one variable region domain of an antibody described herein. Thus, for example, the V region domain may be monomeric and be a VH or VL domain, which is capable of independently binding a desired target (e.g., human CD3) with an affinity at least equal to 10⁻⁷M or less as described herein.

[0186] The variable region may be any naturally occurring variable domain or an engineered version thereof. By engineered version is meant a variable region that has been created using recombinant DNA engineering techniques. Such engineered versions include those created, for example, from a specific antibody variable region by insertions, deletions, or changes in or to the amino acid sequences of the specific antibody. One of ordinary skill in the art can use any known methods for identifying amino acid residues appropriate for engineering. Additional examples include engineered variable regions containing at least one CDR and optionally one or more framework amino acids from a first antibody and the remainder of the variable region domain from a second antibody. Engineered versions of antibody variable domains may be generated by any number of techniques with which those having ordinary skill in the art will be familiar.

[0187] The variable region may be covalently attached at a C-terminal amino acid to at least one other antibody domain or a fragment thereof. Thus, for example, a VH that is present in the variable region may be linked to an immunoglobulin CH1 domain. Similarly, a VL domain may be linked to a CK domain. In this way, for example, the antibody may be a Fab fragment wherein the antigen binding domain contains associated VH and VL domains covalently linked at their C-termini to a CH1 and CK domain, respectively. The CH1 domain may be extended with further amino acids, for example to provide a hinge region or a portion of

a hinge region domain as found in a Fab' fragment, or to provide further domains, such as antibody CH2 and CH3 domains.

Binding Specificity

[0188] An antibody or (HLL)² molecule “specifically binds” to an antigen if it binds to the antigen with a tight binding affinity as determined by an equilibrium dissociation constant (KD, or corresponding KD, as defined below) value of 10⁻⁷ M or less.

[0189] Affinity can be determined using a variety of techniques known in the art, for example but not limited to, equilibrium methods (e.g., enzyme-linked immunosorbent assay (ELISA); KinExA, Rathanaswami et al. Analytical Biochemistry, Vol. 373:52-60, 2008; or radioimmunoassay (RIA)), or by a surface plasmon resonance assay or other mechanism of kinetics-based assay (e.g., BIACORE® analysis or Octet® analysis (forteBIO)), and other methods such as indirect binding assays, competitive binding assays fluorescence resonance energy transfer (FRET), gel electrophoresis and chromatography (e.g., gel filtration). These and other methods may utilize a label on one or more of the components being examined and/or employ a variety of detection methods including but not limited to chromogenic, fluorescent, luminescent, or isotopic labels. A detailed description of binding affinities and kinetics can be found in Paul, W. E., ed., Fundamental Immunology, 4th Ed., Lippincott-Raven, Philadelphia (1999), which focuses on antibody-immunogen interactions. One example of a competitive binding assay is a radioimmunoassay comprising the incubation of labeled antigen with the antibody of interest in the presence of increasing amounts of unlabeled antigen, and the detection of the antibody bound to the labeled antigen. The affinity of the antibody of interest for a particular antigen and the binding off-rates can be determined from the data by scatchard plot analysis. Competition with a second antibody can also be determined using radioimmunoassays. In this case, the antigen is incubated with antibody of interest conjugated to a labeled compound in the presence of increasing amounts of an unlabeled second antibody. This type of assay can be readily adapted for use with the molecules of the present invention.

[0190] Further embodiments of the invention provide molecules that bind to desired targets with an equilibrium dissociation constant or KD (koff/kon) of less than 10⁻⁷ M, or of less than 10⁻⁸ M, or of less than 10⁻⁹ M, or of less than 10⁻¹⁰ M, or of less than 10⁻¹¹ M, or of less than 10⁻¹² M, or of less than 10⁻¹³ M, or of less than 5×10⁻¹³ M (lower values indicating tighter binding affinity). Yet further embodiments of the invention are molecules that bind to desired targets with an with an equilibrium dissociation constant or KD (koff/kon) of less than about 10⁻⁷ M, or of less than about 10⁻⁸ M, or of less than about 10⁻⁹ M, or of less than about 10⁻¹⁰ M, or of less than about 10⁻¹¹ M, or of less than about 10⁻¹² M, or of less than about 10⁻¹³ M, or of less than about 5×10⁻¹³ M.

[0191] In still another embodiment, molecules that bind to desired targets have an equilibrium dissociation constant or KD (koff/kon) of between about 10⁻⁷ M and about 10⁻⁸ M, between about 10⁻⁸ M and about 10⁻⁹ M, between about 10⁻⁹ M and about 10⁻¹⁰ M, between about 10⁻¹⁰ M and about 10⁻¹¹ M, between about 10⁻¹¹ M and about 10⁻¹² M, between about 10⁻¹² M and about 10⁻¹³ M. In still another embodiment, a molecule of the invention have an equilib-

rium dissociation constant or KD (koff/kon) of between 10⁻⁷ M and 10⁻⁸ M, between 10⁻⁸ M and 10⁻⁹ M, between 10⁻⁹ M and 10⁻¹⁰ M, between 10⁻¹⁰ M and 10⁻¹¹ M, between 10⁻¹¹ M and 10⁻¹² M, between 10⁻¹² M and 10⁻¹³ M.

Molecule Stability

[0192] Various aspects of molecule stability may be desired, particularly in the context of a biopharmaceutical therapeutic molecule. For example, stability at various temperatures (“thermostability”) may be desired. In some embodiments, this can encompass stability at physiologic temperature ranges, e.g., at or about 37° C., or from 32° C. to 42° C. In other embodiments, this can encompass stability at higher temperature ranges, e.g., 42° C. to 60° C. In other embodiments, this can encompass stability at cooler temperature ranges, e.g. 20° C. to 32° C. In yet other embodiments, this can encompass stability while in the frozen state, e.g. 0° C. or lower.

[0193] Assays to determine thermostability of protein molecules are known in the art. For example, the fully automated UNcle platform (Unchained Labs) which allowed for simultaneous acquisition of intrinsic protein fluorescence and static light scattering (SLS) data during thermal ramp was used and is further described in the Examples. Additionally, thermal stability and aggregation assays described herein in the Examples, such as differential scanning fluorimetry (DSF) and static light scattering (SLS), can also be used to measure both thermal melting (Tm) and thermal aggregation (Tagg) respectively.

[0194] Alternatively, accelerated stress studies can be performed on the molecules. Briefly, this involves incubating the protein molecules at a particular temperature (e.g., 40° C.) and then measuring aggregation by size exclusion chromatography (SEC) at various timepoints, where lower levels of aggregation indicate better protein stability.

[0195] Alternatively, the thermostability parameter can be determined in terms of molecule aggregation temperature as follows: molecule solution at a concentration 250 µg/ml is transferred into a single use cuvette and placed in a Dynamic Light Scattering (DLS) device. The sample is heated from 40° C. to 70° C. at a heating rate of 0.5° C./min with constant acquisition of the measured radius. Increase of radius indicating melting of the protein and aggregation is used to calculate the aggregation temperature of the molecule.

[0196] Alternatively, temperature melting curves can be determined by Differential Scanning calorimetry (DSC) to determine intrinsic biophysical protein stabilities of the binding molecules. These experiments are performed using a MicroCal LLC (Northampton, MA, U.S.A) VP-DSC device. The energy uptake of a sample containing a binding molecule is recorded from 20° C. to 90° C. compared to a sample containing only the formulation buffer. The binding molecules are adjusted to a final concentration of 250 µg/ml e.g. in SEC running buffer. For recording of the respective melting curve, the overall sample temperature is increased stepwise. At each temperature T energy uptake of the sample and the formulation buffer reference is recorded. The difference in energy uptake Cp (kcal/mole/° C.) of the sample minus the reference is plotted against the respective temperature. The melting temperature is defined as the temperature at the first maximum of energy uptake.

[0197] In a further embodiment the molecules according to the invention is stable at or about physiologic pH, i.e., about pH 7.4. In other embodiments, the molecules are

stable at a lower pH, e.g., down to pH 6.0. In other embodiments, the molecules are stable at a higher pH, e.g., up to pH 9.0. In one embodiment, the molecules are stable at a pH of 6.0 to 9.0. In another embodiment, the molecules are stable at a pH of 6.0 to 8.0. In another embodiment, the molecules are stable at a pH of 7.0 to 9.0.

[0198] In certain embodiments, the more tolerant the molecule is to unphysiologic pH (e.g., pH 6.0), the higher the recovery of the molecule eluted from an ion exchange column is relative to the total amount of loaded protein. In one embodiment, recovery of the molecule from an ion (e.g., cation) exchange column is $\geq 30\%$. In another embodiment, recovery of the molecule from an ion (e.g., cation) exchange column is $\geq 40\%$. In another embodiment, recovery of the molecule from an ion (e.g., cation) exchange column is $\geq 50\%$. In another embodiment, recovery of the molecule from an ion (e.g., cation) exchange column is $\geq 60\%$. In another embodiment, recovery of the molecule from an ion (e.g., cation) exchange column is $\geq 70\%$. In another embodiment, recovery of the molecule from an ion (e.g., cation) exchange column is $\geq 80\%$. In another embodiment, recovery of the molecule from an ion (e.g., cation) exchange column is $\geq 90\%$. In another embodiment, recovery of the molecule from an ion (e.g., cation) exchange column is $\geq 95\%$. In another embodiment, recovery of the molecule from an ion (e.g., cation) exchange column is $\geq 99\%$.

[0199] In certain embodiments, it may be desired to determine the chemical stability of the molecules. Determination of molecule chemical stability can be carried out via isothermal chemical denaturation ("ICD") by monitoring intrinsic protein fluorescence, as further described herein in the Examples. ICD yields $C1/2$ and ΔG which can be good metrics for protein stability. $C1/2$ is the amount of chemical denaturant required to denature 50% of the protein and is used to derive ΔG (or unfolding energy).

[0200] Clipping of protein chains is another critical product quality attribute that is carefully monitored and reported for biologic drugs. Typically, a longer and/or a less structured linker is expected to result in increased clipping as a function of incubation time and temperature. Clipping is a critical issue for molecules as clips to linkers connecting either the target or T-cell engaging domains have terminal detrimental impact on drug potency and efficacy. Clips to additional sites including the scFc may impact pharmacodynamic/kinetic properties. Increased clipping is an attribute to be avoided in a pharmaceutical product. Accordingly, in certain embodiments, protein clipping can be assayed as described herein in the Examples.

Immune Effector Cells and Effector Cell Proteins

[0201] A molecule can bind to a molecule expressed on the surface of an immune effector cell (called "effector cell protein" herein) and to another molecule expressed on the surface of a target cell (called a "target cell protein" herein). The immune effector cell can be a T cell, an NK cell, a macrophage, or a neutrophil. In some embodiments the effector cell protein is a protein included in the T cell receptor (TCR)-CD3 complex. The TCR-CD3 complex is a heteromultimer comprising TCR α and TCR β or TCR γ and TCR δ plus various CD3 chains from among the CD3 zeta (CD3 ζ) chain, CD3 epsilon (CD3 ϵ) chain, CD3 gamma (CD3 γ) chain, and CD3 delta (CD3 δ) chain.

[0202] The CD3 receptor complex is a protein complex and is composed of four chains. In mammals, the complex contains a CD3 γ (gamma) chain, a CD3 δ (delta) chain, and two CD3 ϵ (epsilon) chains. These chains associate with the T cell receptor (TCR) and the so-called ((zeta) chain to form the T cell receptor CD3 complex and to generate an activation signal in T lymphocytes. The CD3 γ (gamma), CD3 δ (delta), and CD3 ϵ (epsilon) chains are highly related cell-surface proteins of the immunoglobulin superfamily containing a single extracellular immunoglobulin domain. The intracellular tails of the CD3 molecules contain a single conserved motif known as an immunoreceptor tyrosine-based activation motif or ITAM for short, which is essential for the signaling capacity of the TCR. The CD3 epsilon molecule is a polypeptide which in humans is encoded by the CD3E gene which resides on chromosome 11. The most preferred epitope of CD3 epsilon is comprised within amino acid residues 1-27 of the human CD3 epsilon extracellular domain. It is envisaged that the molecules according to the present invention typically and advantageously show less unspecific T cell activation, which is not desired in specific immunotherapy. This translates to a reduced risk of side effects.

[0203] In some embodiments the effector cell protein can be the human CD3 epsilon (CD3 ϵ) chain (the mature amino acid sequence of which is disclosed in SEQ ID NO: 40), which can be part of a multimeric protein. Alternatively, the effector cell protein can be human and/or cynomolgus monkey TCR α , TCR β , TCR δ , TCR γ , CD3 beta (CD3 β) chain, CD3 gamma (CD3 γ) chain, CD3 delta (CD3 δ) chain, or CD3 zeta (CD3 ζ) chain.

[0204] Moreover, in some embodiments, a molecule can also bind to a CD3 δ chain from a non-human species, such as mouse, rat, rabbit, new world monkey, and/or old world monkey species. Such species include, without limitation, the following mammalian species: *Mus musculus*; *Rattus rattus*; *Rattus norvegicus*; the cynomolgus monkey, *Macaca fascicularis*; the hamadryas baboon, *Papio hamadryas*; the Guinea baboon, *Papio papio*; the olive baboon, *Papio anubis*; the yellow baboon, *Papio cynocephalus*; the Chacma baboon, *Papio ursinus*; *Callithrix jacchus*; *Saguinus oedipus*; and *Saimiri sciureus*. The mature amino acid sequence of the CD3 δ chain of cynomolgus monkey is provided in SEQ ID NO: 34. Having a therapeutic molecule that has comparable activity in humans and species commonly used for preclinical testing, such as mice and monkeys, can simplify, accelerate, and ultimately provide improved outcomes in drug development. In the long and expensive process of bringing a drug to market, such advantages can be critical.

[0205] In certain embodiments, the (HHLL)² molecule can bind to an epitope within the first 27 amino acids of the CD3 ϵ chain (SEQ ID NO: 36), which may be a human CD3 ϵ chain or a CD3 ϵ chain from different species, particularly one of the mammalian species listed above. The epitope can contain the amino acid sequence Gln-Asp-Gly-Asn-Glu. The advantages of a molecule that binds such an epitope are explained in detail in U.S. Patent Application Publication 2010/0183615A1, the relevant portions of which are incorporated herein by reference. The epitope to which an antibody or molecule binds can be determined by alanine scanning, which is described in, e.g., U.S. Patent Application Publication 2010/0183615A1, the relevant portions of which are incorporated herein by reference. In other embodi-

ments, the molecule can bind to an epitope within the extracellular domain of CD3 ϵ (SEQ ID NO: 35).

[0206] In embodiments where a T cell is the immune effector cell, effector cell proteins to which a molecule can bind include, without limitation, the CD3 ϵ chain, the CD3 γ , the CD3 δ chain, the CD3 ζ chain, TCR α , TCR β , TCR γ , and TCR δ . In embodiments where an NK cell or a cytotoxic T cell is an immune effector cell, NKG2D, CD352, NKp46, or CD16a can, for example, be an effector cell protein. In embodiments where a CD8 $^{+}$ T cell is an immune effector cell, 4-1BB or NKG2D, for example, can be an effector cell protein. Alternatively, in other embodiments a molecule could bind to other effector cell proteins expressed on T cells, NK cells, macrophages, or neutrophils.

Target Cells and Target Cell Proteins Expressed on Target Cells

[0207] As explained above, a molecule can bind to an effector cell protein and a target cell protein. The target cell protein can, for example, be expressed on the surface of a cancer cell, a cell infected with a pathogen, or a cell that mediates a disease, for example an inflammatory, autoimmune, and/or fibrotic condition. In some embodiments, the target cell protein can be highly expressed on the target cell, although high levels of expression are not necessarily required.

[0208] Where the target cell is a cancer cell, a molecule as described herein can bind to a cancer cell antigen as described above. A cancer cell antigen, or tumor associated antigen ("TAA") can be a human protein or a protein from another species. For example, a molecule may bind to a target cell protein from a mouse, rat, rabbit, new world monkey, and/or old world monkey species, among many others. Such species include, without limitation, the following species: *Mus musculus*; *Rattus rattus*; *Rattus norvegicus*; cynomolgus monkey, *Macaca fascicularis*; the hamadryas baboon, *Papio hamadryas*; the Guinea baboon, *Papio papio*; the olive baboon, *Papio anubis*; the yellow baboon, *Papio cynocephalus*; the Chacma baboon, *Papio ursinus*, *Callithrix jacchus*, *Saguinus oedipus*, and *Saimiri sciureus*. Preferred target cell surface antigens in the context of the present invention are, MSLN, CDH3, FLT3, CLL1, EpCAM, CD20, and CD22. Typically, target cell surface antigens in the context of the present invention are tumor associated antigens (TAA). B-lymphocyte antigen CD20 or CD20 is expressed on the surface of all B-cells beginning at the pro-B phase (CD45R $^{+}$, CD117 $^{+}$) and progressively increasing in concentration until maturity. CD22, or cluster of differentiation-22, is a molecule belonging to the SIGLEC family of lectins. It is found on the surface of mature B cells and to a lesser extent on some immature B cells. Fms like tyrosine kinase 3 (FLT3) is also known as Cluster of differentiation antigen 135 (CD135), receptor-type tyrosine-protein kinase FLT3, or fetal liver kinase-2 (Flk2). FLT3 is a cytokine receptor which belongs to the receptor tyrosine kinase class III. CD135 is the receptor for the cytokine Flt3 ligand (FLT3L). The FLT3 gene is frequently mutated in acute myeloid leukemia (AML). C-type lectin-like receptor (CLL1), also known as CLEC12A, or as MICL. It contains an ITIM motif in cytoplasmic tail that can associate with signaling phosphatases SHP-1 and SHP-2. Human MICL is expressed as a monomer primarily on myeloid cells, including granulocytes, monocytes, macrophages and dendritic cells and is associated with AML. Mesothelin (MSLN) is a

40 kDa protein that is expressed in mesothelial cells and overexpressed in several human tumors. Cadherin-3 (CDH3), also known as P-Cadherin, is a calcium-dependent cell-cell adhesion glycoprotein composed of five extracellular cadherin repeats, a transmembrane region and a highly conserved cytoplasmic tail. It is associated with some types of tumors. Epithelial cell adhesion molecule (EpCAM) is a transmembrane glycoprotein mediating Ca $^{2+}$ -independent homotypic cell-cell adhesion in epithelia. EpCAM has oncogenic potential and appears to play a role in tumorigenesis and metastasis of carcinomas.

[0209] In some examples, the target cell protein can be a protein selectively expressed on an infected cell. For example, in the case of an HBV or HCV infection, the target cell protein can be an envelope protein of HBV or HCV that is expressed on the surface of an infected cell. In other embodiments, the target cell protein can be gp 120 encoded by human immunodeficiency virus (HIV) on HIV-infected cells.

[0210] In other aspects, a target cell can be a cell that mediates an autoimmune or inflammatory disease. For example, human eosinophils in asthma can be target cells, in which case, EGF-like module containing mucin-like hormone receptor (EMR1), for example, can be a target cell protein. Alternatively, excess human B cells in a systemic lupus erythematosus patient can be target cells, in which case CD19 or CD20, for example, can be a target cell protein. In other autoimmune conditions, excess human Th2 T cells can be target cells, in which case CCR4 can, for example, be a target cell protein. Similarly, a target cell can be a fibrotic cell that mediates a disease such as atherosclerosis, chronic obstructive pulmonary disease (COPD), cirrhosis, scleroderma, kidney transplant fibrosis, kidney allograft nephropathy, or a pulmonary fibrosis, including idiopathic pulmonary fibrosis and/or idiopathic pulmonary hypertension. For such fibrotic conditions, fibroblast activation protein alpha (FAP alpha) can, for example, be a target cell protein.

Therapeutic Methods and Compositions

[0211] Molecules can be used to treat a wide variety of conditions including, for example, various forms of cancer, infections, autoimmune or inflammatory conditions, and/or fibrotic conditions.

[0212] Accordingly, in an embodiment provided herein are molecules for use in the prevention, treatment, or amelioration of a disease.

[0213] Another embodiment provides the use of the binding molecule of the invention (or of the binding molecule produced according to the process of the invention) in the manufacture of a medicament for the prevention, treatment or amelioration of a disease.

[0214] Provided herein are pharmaceutical compositions comprising molecules. These pharmaceutical compositions comprise a therapeutically effective amount of a molecule and one or more additional components such as a physiologically acceptable carrier, excipient, or diluent. In some embodiments, these additional components can include buffers, carbohydrates, polyols, amino acids, chelating agents, stabilizers, and/or preservatives, among many possibilities.

[0215] In some embodiments, a molecule can be used to treat cell proliferative diseases, including cancer, which involve the unregulated and/or inappropriate proliferation of cells, sometimes accompanied by destruction of adjacent

tissue and growth of new blood vessels, which can allow invasion of cancer cells into new areas, i.e. metastasis. Included within conditions treatable with a molecule are non-malignant conditions that involve inappropriate cell growth, including colorectal polyps, cerebral ischemia, gross cystic disease, polycystic kidney disease, benign prostatic hyperplasia, and endometriosis. A preferred method of targeting cancer is to target a molecule to a cancer cell surface antigen, i.e., a tumor associated antigen (TAA). It may be a protein, preferably the extracellular portion of a protein, or a carbohydrate structure, preferably a carbohydrate structure of a protein, such as a glycoprotein.

[0216] A molecule of the invention can be used to treat a hematologic or solid tumor malignancy. More specifically, cell proliferative diseases that can be treated using a molecule are, for example, cancers including mesotheliomas, squamous cell carcinomas, myelomas, osteosarcomas, glioblastomas, gliomas, carcinomas, adenocarcinomas, melanomas, sarcomas, acute and chronic leukemias, lymphomas, and meningiomas, Hodgkin's disease, Sézary syndrome, multiple myeloma, and lung, non-small cell lung, small cell lung, laryngeal, breast, head and neck, bladder, ovarian, skin, prostate, cervical, vaginal, gastric, renal cell, kidney, pancreatic, colorectal, endometrial, and esophageal, hepatobiliary, bone, skin, and hematologic cancers, as well as cancers of the nasal cavity and paranasal sinuses, the nasopharynx, the oral cavity, the oropharynx, the larynx, the hypopharynx, the salivary glands, the mediastinum, the stomach, the small intestine, the colon, the rectum and anal region, the ureter, the urethra, the penis, the testis, the vulva, the endocrine system, the central nervous system, and plasma cells.

[0217] Among the texts providing guidance for cancer therapy is *Cancer, Principles and Practice of Oncology*, 4th Edition, DeVita et al., Eds. J. B. Lippincott Co., Philadelphia, PA (1993). An appropriate therapeutic approach is chosen according to the particular type of cancer, and other factors such as the general condition of the patient, as is recognized in the pertinent field. A molecule can be added to a therapy regimen using other anti-neoplastic agents in treating a cancer patient.

[0218] In some embodiments, a molecule can be administered concurrently with, before, or after a variety of drugs and treatments widely employed in cancer treatment such as, for example, chemotherapeutic agents, non-chemotherapeutic, anti-neoplastic agents, and/or radiation. For example, chemotherapy and/or radiation can occur before, during, and/or after any of the treatments described herein. Examples of chemotherapeutic agents are discussed above and include, but are not limited to, cisplatin, taxol, etoposide, mitoxantrone (Novantrone®), actinomycin D, cycloheximide, camptothecin (or water soluble derivatives thereof), methotrexate, mitomycin (e.g., mitomycin C), dacarbazine (DTIC), anti-neoplastic antibiotics such as adriamycin (doxorubicin) and daunomycin, and all the chemotherapeutic agents mentioned above.

[0219] A molecule can also be used to treat infectious disease, for example a chronic hepatitis B virus (HBV) infection, a hepatitis C virus (HCV) infection, a human immunodeficiency virus (HIV) infection, an Epstein-Barr virus (EBV) infection, or a cytomegalovirus (CMV) infection, among many others.

[0220] A molecule can find further use in other kinds of conditions where it is beneficial to deplete certain cell types.

For example, depletion of human eosinophils in asthma, excess human B cells in systemic lupus erythematosus, excess human Th2 T cells in autoimmune conditions, or pathogen-infected cells in infectious diseases can be beneficial. In a fibrotic condition, it can be useful to deplete cells forming fibrotic tissue.

[0221] Therapeutically effective doses of a molecule can be administered. The amount of molecule that constitutes a therapeutically dose may vary with the indication treated, the weight of the patient, the calculated skin surface area of the patient. Dosing of a molecule can be adjusted to achieve the desired effects. In many cases, repeated dosing may be required.

[0222] A molecule, or a pharmaceutical composition containing such a molecule, can be administered by any feasible method. Protein therapeutics will ordinarily be administered by a parenteral route, for example by injection, since oral administration, in the absence of some special formulation or circumstance, would lead to hydrolysis of the protein in the acid environment of the stomach. Subcutaneous, intramuscular, intravenous, intraarterial, intralesional, or peritoneal bolus injection are possible routes of administration. A molecule can also be administered via infusion, for example intravenous or subcutaneous infusion. Topical administration is also possible, especially for diseases involving the skin. Alternatively, a molecule can be administered through contact with a mucus membrane, for example by intra-nasal, sublingual, vaginal, or rectal administration or administration as an inhalant. Alternatively, certain appropriate pharmaceutical compositions comprising a molecule can be administered orally.

[0223] The term "treatment" encompasses alleviation of at least one symptom or other embodiment of a disorder, or reduction of disease severity, and the like. A molecule according to the present invention need not effect a complete cure, or eradicate every symptom or manifestation of a disease, to constitute a viable therapeutic agent. As is recognized in the pertinent field, drugs employed as therapeutic agents may reduce the severity of a given disease state, but need not abolish every manifestation of the disease to be regarded as useful therapeutic agents. Simply reducing the impact of a disease (for example, by reducing the number or severity of its symptoms, or by increasing the effectiveness of another treatment, or by producing another beneficial effect), or reducing the likelihood that the disease will occur or worsen in a subject, is sufficient. One embodiment of the invention is directed to a method comprising administering to a patient a molecule of the invention in an amount and for a time sufficient to induce a sustained improvement over baseline of an indicator that reflects the severity of the particular disorder.

[0224] The term "prevention" encompasses prevention of at least one symptom or other embodiment of a disorder, and the like. A prophylactically administered treatment incorporating a molecule according to the present invention need not be completely effective in preventing the onset of a condition in order to constitute a viable prophylactic agent. Simply reducing the likelihood that the disease will occur or worsen in a subject, is sufficient.

[0225] As is understood in the pertinent field, pharmaceutical compositions comprising the molecule are administered to a subject in a manner appropriate to the indication and the composition. Pharmaceutical compositions may be administered by any suitable technique, including but not

limited to parenterally, topically, or by inhalation. If injected, the pharmaceutical composition can be administered, for example, via intra-articular, intravenous, intramuscular, intralesional, intraperitoneal or subcutaneous routes, by bolus injection, or continuous infusion. Delivery by inhalation includes, for example, nasal or oral inhalation, use of a nebulizer, inhalation of the binding molecule in aerosol form, and the like. Other alternatives include oral preparations including pills, syrups, or lozenges.

[0226] The molecules can be administered in the form of a composition comprising one or more additional components such as a physiologically acceptable carrier, excipient or diluent. Optionally, the composition additionally comprises one or more physiologically active agents. In various particular embodiments, the composition comprises one, two, three, four, five, or six physiologically active agents in addition to one or more molecules.

[0227] Kits for use by medical practitioners are provided including one or more molecule and a label or other instructions for use in treating any of the conditions discussed herein. In one embodiment, the kit includes a sterile preparation of one or more molecules which may be in the form of a composition as disclosed herein, and may be in one or more vials.

[0228] Dosages and the frequency of administration may vary according to such factors as the route of administration, the particular molecule employed, the nature and severity of the disease to be treated, whether the condition is acute or chronic, and the size and general condition of the subject.

[0229] Having described the invention in general terms above, the following examples are offered by way of illustration and not limitation.

EXAMPLES

Example 1

Generation and Expression of (HHLL)² Binding Molecules

[0230] FIG. 1 has representative structures for both the (HLHL)² and (HHLL)² formats. Versions of both of these molecules were generated.

[0231] The (HHLL)² version (T6M) that was generated comprises the following domains from N- to C-terminus: Anti MSLN VH-(GGGS)₄ Linker-Anti CD3 VH-(GGGS)₄ Linker-Anti MSLN VL-(GGGS)₄ Linker-Anti CD3 VL-(GGGS)₃ Linker-scFc-(GGGS)₃ Linker-Anti CDH3 VH-(GGGS)₄ Linker-Anti CD3 VH-(GGGS)₄ Linker-Anti CDH3 VL-(GGGS)₄ Linker-Anti CD3 VL.

[0232] The (HLHL)² version (G7Q) that was generated comprises the following domains from N- to C-terminus: Anti MSLN VH-(GGGS)₃ Linker-Anti MSLN VL-(SGGGS)₁ Linker-Anti CD3 VH-(GGGS)₃ Linker-Anti CD3 VL-(GGGS)₃ Linker-scFc-(GGGS)₃ Linker-Anti CDH3 VH-(GGGS)₃ Linker-Anti CDH3 VL-(SGGGS)₁ Linker-Anti CD3 VH-(GGGS)₃ Linker-Anti CD3 VL.

[0233] The open reading frames of two different formats as shown in FIG. 1 were ordered as gene syntheses and subcloned into a mammalian expression vector containing an IgG derived signal peptide for secreted expression into the cell culture supernatant. Sequence verified plasmid clones were transfected stably transfected into CHO cells, cell culture supernatant was harvested after 6 days and stored at -80° C. until protein purification. Summaries of the production runs for both T6M and G7Q are provided in

Tables 3 and 4, respectively, and demonstrate comparable protein yields for both molecules.

TABLE 3

T6M CHO CS9 800 ml Production Runs	
Material Type	Purified
Target	CDH3-MSLN
Molecule Name	(T6M) MS 15-B12 × 6H10.09 HHLL × scFc × CH3 15-E11 × 6H10.09
Concentration (mg/ml)	0.24
Mass Produced (mg)	13.8
General Comment	4.47588455735631E-06M

TABLE 4

G7Q CHO CS9 800 ml Production Runs	
Material Type	Purified
Target	CDH3-MSLN
Molecule Name	(G7Q) MS 15-B12 CC × 6H10.09 × G4S3 × scFc × G4S3 × CH3 15-E11 CC × 6H10.09
Concentration (mg/ml)	0.252
Mass Produced (mg)	13.442
General Comment	1.58417287195211E-06M

Example 2

Chromatography Analysis

[0234] Protein purification was done by Protein A (GE Healthcare, mAb SelectSuRE) affinity chromatography of the filtrated cell culture supernatant, followed by size exclusion chromatography in a buffer solution at pH 7 (Error! Reference source not found.e 3). According to the OD280 nm signal peaks were pooled and MW was analyzed by SDS-PAGE. Protein monomer peaks (peaks at 159.9 ml for G7Q or 166.08 ml for T6M respectively) were formulated in a buffer solution and aliquoted for storage at -80° C.

SDS-PAGE Analysis

[0235] The purified monomer sample was applied to SDS PAGE analysis to check purity and correct molecular weight (MW) (FIG. 4). 60 µl sample was mixed with 20 µl (4×) LDS Sample buffer and 10 µl IM DTT and incubated at 70° C. for 10 min. 15 µl sample was loaded on each lane using a Bolt 4-12% Bis-Tris Plus 12Well gel (NW04122BOX, Invitrogen). For the marker 7.5 µl (Sharp Pre-Stained Protein Standard (LC5800, Invitrogen)) was loaded onto a separate lane. The running buffer was 1×MES (20×MES SDS Running Buffer, Invitrogen, NP0002-02) and the gel was run at 200V-120 mA max-60 min. Final results demonstrate T6M running at the expected molecular weight.

Example 3

Cytotoxicity Assay (TDCC) with Unstimulated Human PBMC

Isolation of Effector Cells

[0236] Human peripheral blood mononuclear cells (PBMC) were prepared by Ficoll density gradient centrifugation from enriched lymphocyte preparations (buffy coats), a side product of blood banks collecting blood for transfusions. Buffy coats were supplied by a local blood bank and PBMC were prepared on the day after blood collection. After Ficoll density centrifugation and extensive washes with Dulbecco's PBS (Gibco), remaining erythrocytes were removed from PBMC via incubation with erythrocyte lysis buffer (155 mM NH₄Cl, 10 mM KHCO₃, 100 μM EDTA). Remaining lymphocytes mainly encompass B and T lymphocytes, NK cells and monocytes. PBMC were kept in culture at 37° C./5% CO₂ in RPMI medium (Gibco) with 10% FCS (Gibco).

Depletion of CD14+ and CD56+ Cells

[0237] For depletion of CD14+ cells, human CD14 MicroBeads (Milteny Biotec, MACS, #130-050-201) were used, for depletion of NK cells human CD56 MicroBeads (MACS, #130-050-401). PBMC were counted and centrifuged for 10 minutes at room temperature with 300×g. The supernatant was discarded and the cell pellet resuspended in MACS isolation buffer (60 μL/10⁷ cells). CD14 MicroBeads and CD56 MicroBeads (20 μL/10⁷ cells) were added and incubated for 15 min at 4–8° C. The cells were washed with AutoMACS rinsing buffer (Milteny #130-091-222) (1–2 mL/10⁷ cells). After centrifugation (see above), supernatant was discarded and cells were resuspended in MACS isolation buffer (500 μL/10⁸ cells). CD14/CD56 negative cells were then isolated using LS Columns (Milteny Biotec, #130-042-401). PBMC w/o CD14+/CD56+ cells were adjusted to 1.2×10⁶ cells/mL and cultured in RPMI complete medium i.e. RPMI1640 (Biochrom AG, #FG1215) supplemented with 10% FBS (Bio West, #S1810), 1× non-essential amino acids (Biochrom AG, #K0293), 10 mM Hepes buffer (Biochrom AG, #L1613), 1 mM sodium pyruvate (Biochrom AG, #L0473) and 100 U/mL penicillin/streptomycin (Biochrom AG, #A2213) at 37° C. in an incubator until needed.

Target Cell Preparation

[0238] Cells were harvested, centrifugally spun down and adjusted to 1.2×10⁵ cells/mL in complete RPMI medium. The vitality of cells was determined using Nucleocounter NC-250 (Chemometec) and Solution 18 Dye containing Acridine Orange and DAPI (Chemometec).

Luciferase Based Analysis

[0239] This assay was designed to quantify the lysis of target cells in the presence of serial dilutions of multi-specific antibody constructs. Equal volumes of Luciferase-

positive target cells and effector cells (i.e., PBMC w/o CD14+; CD56+ cells) were mixed, resulting in an E:T cell ratio of 10:1. 42 μL of this suspension were transferred to each well of a 384-well plate. 8 μL of serial dilutions of the corresponding molecules and a negative control molecules (a CD3-based molecule that also recognizes an irrelevant target antigen) or RPMI complete medium as an additional negative control were added. The molecule-mediated cytotoxic reaction proceeded for 48 hours in a 5% CO₂ humidified incubator. Then 25 L substrate (Steady-Glo® Reagent, Promega) were transferred to the 384-well plate. Only living, Luciferase-positive cells react to the substrate and thus create a luminescence signal. Samples were measured with a SPARK microplate reader (TECAN) and analyzed by Spark Control Magellan software (TECAN).

[0240] Percentage of cytotoxicity was calculated as follows:

$$\text{Cytotoxicity [\%]} = \left(1 - \frac{RLU_{\text{Sample}}}{RLU_{\text{Negative-Control}}} \right) \times 100$$

[0241] RLU=relative light units

[0242] Negative-Control=cells without multi-specific antibody construct

[0243] Using GraphPad Prism 7.04 software (Graph Pad Software, San Diego), the percentage of cytotoxicity was plotted against the corresponding multi-specific antibody construct concentrations. Dose response curves were analyzed with the four parametric logistic regression models for evaluation of sigmoid dose response curves with fixed hill slope and EC₅₀ values were calculated. Results of this experiment are depicted in FIGS. 5 and 6, and demonstrate in vitro functionality of the molecules tested, with the (HHLL)² molecules showing superior activity at both 48 hours (FIG. 5) and 72 hours (FIG. 6).

[0244] The following target cell lines were used for the Luciferase-based cytotoxicity assay:

[0245] GSU-LUC wt (CDH3+ and MSLN+)

[0246] GSU-LUC KO CDH3 (CDH3– and MSLN+)

[0247] GSU-LUC KO MSLN (CDH3+ and MSLN–)

[0248] Each and every reference cited herein is incorporated herein by reference in its entirety for all purposes.

[0249] The present invention is not to be limited in scope by the specific embodiments described herein, which are intended as single illustrations of individual embodiments of the invention, and functionally equivalent methods and components of the invention. Indeed, various modifications of the invention, in addition to those shown and described herein will become apparent to those skilled in the art from the foregoing description and accompanying drawings. Such modifications are intended to fall within the scope of the claims.

SEQUENCES

Exemplary Linker Sequences

GGGS (SEQ ID NO: 1)

GGGSGGGGS (SEQ ID NO: 2)

GGGSGGGSGGGGS (SEQ ID NO: 3)

GGGSGGGSGGGSGGGGS (SEQ ID NO: 4)

-continued

GGGGSGGGSGGGSGGGSGGGGS (SEQ ID NO: 5)

GGGGQ (SEQ ID NO: 6)

GGGGQGGGGQ (SEQ ID NO: 7)

GGGGQGGGGQGGGGQ (SEQ ID NO: 8)

GGGGQGGGGQGGGGQGGGGQ (SEQ ID NO: 9)

GGGGQGGGGQGGGGQGGGGQGGGGQ (SEQ ID NO: 10)

GGGGSAAA (SEQ ID NO: 11)

TVAAP (SEQ ID NO: 12)

ASTKGP (SEQ ID NO: 13)

AAA (SEQ ID NO: 14)

GGNGT (SEQ ID NO: 15)

YNGT (SEQ ID NO: 16)

SGGGGS (SEQ ID NO: 17)

SGGGGQ (SEQ ID NO: 18)

GGGG (SEQ ID NO: 19)

(GGGG)2 (SEQ ID NO: 20)

(GGGG)3 (SEQ ID NO: 21)

(GGGG)4 (SEQ ID NO: 22)

(GGGG)5 (SEQ ID NO: 23)

(GGGG)1-10 (SEQ ID NO: 24)

(GGGG)2-10 (SEQ ID NO: 25)

(GGGG)3-10 (SEQ ID NO: 26)

(GGGGS)1-10 (SEQ ID NO: 27)

(GGGGS)2-10 (SEQ ID NO: 28)

(GGGGS)3-10 (SEQ ID NO: 29)

(GGGGQ)1-10 (SEQ ID NO: 30)

(GGGGQ)2-10 (SEQ ID NO: 31)

(GGGGQ)3-10 (SEQ ID NO: 32)

Amino acid sequence of the mature human CD38 (SEQ ID NO: 33)

QDGN EEMGGITQTPYKVSISGTTVILTCPQYPGSEILWQHNDKNIGDEDDKNIGSDE
DHL SLKEFSELEQSGYYVCYPRGSKPEDANFYLYLRARVCENCMEMDMVMSVATIVI
VDICITGGLLLVYYWSKNRKAKAKPVTRGAGAGGRQGRQNKERPPPPVNPDPYEPI
RKQQRDLYSGLNQRI

Amino acid sequence of the mature CD38 of cynomolgus monkey (SEQ ID NO: 34)

QDGN EEMGSITQTPYQVSISGTTVILTCSQHLGSEAQWQHNGKNKGD SGDLFLPEF
SEMEQSGYYVCYPRGSPEDASHHLYLKARVCENCMEMDMVAVATIVIVDICTLG
LLLLVYYWSKNRKAKAKPVTRGAGAGGRQGRQNKERPPPPVNPDPYEPIRKQQDL
YSGLNQRI

Amino acid sequence of the extracellular domain of human CD38 (SEQ ID NO: 35)

QDGN EEMGGITQTPYKVSISGTTVILTCPQYPGSEILWQHNDKNIGDEDDKNIGSDE
DHL SLKEFSELEQSGYYVCYPRGSKPEDANFYLYLRARVCENCMEMDMVMS

-continued

Amino acids 1-27 of human CD38 (SEQ ID NO: 36)
QDGN EEMGGITQTPYKVSISGTTVILT

T6M mature (SEQ ID NO: 37)
QVQLQESGPGLVKPSSETLSLTCTVSGGSISSSSSYFWGWI RQPPGKCLEWIGNIYYSGSS
NYPNLSKSRVTISVDTSKNQFSLKLSSVTAADTAVYYCARLPRGDRDAFDI WGGQT
MVTVSSGGGSGGGSGGGSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFN
KYAMNWVRQAPGKGM EWVARIRSKYNNYATYYADAVKDRFTISRDDS KNTLYLQ
MNNLKTEDTAVYYCV RAGNFGSSYISYFAYWGQGT LVTVSSGGGSGGGSGGGG
SGGGGSDIVMTQSPSSLSASV GDRVITITCRASQGISNYLAWYQKPGKVPKLLIYAA
STLQSGVPSRFGSGSGTDFTLTIS SLQPEDFATYYCQQSYSTPFTFGCGTKVEIKGGG
SGGGGSGGGSGGGSGGT VVTQEPSLTVSPGGT VTI TCGSSTGAVTSGNYPNWIQK
KPGQAPRGLIGGTFKFLAPGTPARFSGSLEGGKAALTLSGVQPEDEAEYYCVLYYSNR
WVFGSGTKLTVLSSGGGSGGGSGGGGSDKTHTCPPCPAPELLGGPSVFLFPPKPKD
TLMI SRTPEVT CVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPCEEQYGS TYRCVSV
LTVLHQDWLNGKEYKCKVSNKALPAPI EKTISKAKGQPREPQVYTLPPSREMTKNQ
VSLTCLVKGFPSPDI AVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQ
GNVFS C SVMHEALHNHYTQKSLSLSPGKGGGSGGGSGGGSGGGSGGGSGGGSGG
GGSDKTHTCPPCPAPELLGGPSVFLFPPKPKD TLMI SRTPEVT CVVVDVSHEDPEVKF
NWYVDGVEVHNAKTKPCEEQYGS TYRCVSVLTVLHQDWLNGKEYKCKVSNKALP
API EKTISKAKGQPREPQVYTLPPSREMTKNQVSLTCLVKGFPSPDI AVEWESNGQP
ENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFS C SVMHEALHNHYTQKSLSL
SPGKGGGSGGGSGGGSGGVQLVQSGAEVKKPGASVKVSCASGYTFTNYWMN
WVRQAPGQCLEWMGNIA YGVKGTN YNQKFPQGRVTMTVDTSSSTAYMELSR LRS
DTAVYYCATRYFYVMDYWGQGT LVTVSSGGGSGGGSGGGSGGGSGGGSEVQLVE
SGGGLVQPGGSLKLSCAASGFTFNKYAMNWVRQAPGKGM EWVARIRSKYNNYAT
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GQGT LVTVSSGGGSGGGSGGGSGGGSDIQMTQSPSSLSASV GDRVITITCRASQ
DISNYLWYQKPGKVPKLLIYYTSRLHSGVPSRFGSGSGTDFTLTIS SLQPEDVAT
YYCVQYAFPLT FPGCGTKVEIKGGGSGGGSGGGSGGGSGGGSGGT VVTQEPSLTVSP
GGTVTITCGSSTGAVTSGNYPNWIQKPGQAPRGLIGGTFKFLAPGTPARFSGSLEGGK
AALTLSGVQPEDEAEYYCVLYYSNRWVFGSGTKLTVL

G7Q mature (SEQ ID NO: 38)
QVQLQESGPGLVKPSSETLSLTCTVSGGSISSSSSYFWGWI RQPPGKCLEWIGNIYYSGSS
NYPNLSKSRVTISVDTSKNQFSLKLSSVTAADTAVYYCARLPRGDRDAFDI WGGQT
MVTVSSGGGSGGGSGGGSGGGSDIVMTQSPSSLSASV GDRVITITCRASQGISNYLAWY
QKPGKVPKLLIYAASTLQSGVPSRFGSGSGTDFTLTIS SLQPEDFATYYCQQSYSTP
FTFGCGTKVEIKSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAMNWVR
QAPGKGM EWVARIRSKYNNYATYYADAVKDRFTISRDDS KNTLYLQMN LKTEDT
AVYYCV RAGNFGSSYISYFAYWGQGT LVTVSSGGGSGGGSGGGSGGT VVTQEPS
LTVSPGGT VTI TCGSSTGAVTSGNYPNWIQKPGQAPRGLIGGTFKFLAPGTPARFSGS
LEGGKAALTLSGVQPEDEAEYYCVLYYSNRWVFGSGTKLTVLGGGSGGGSGGG
GSDKTHTCPPCPAPELLGGPSVFLFPPKPKD TLMI SRTPEVT CVVVDVSHEDPEVKFN
WYVDGVEVHNAKTKPCEEQYGS TYRCVSVLTVLHQDWLNGKEYKCKVSNKALPA
PI EKTISKAKGQPREPQVYTLPPSREMTKNQVSLTCLVKGFPSPDI AVEWESNGQPE
NNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFS C SVMHEALHNHYTQKSLSLS
PGKGGGSGGGSGGGSGGGSGGGSGGGSGGSDKTHTCPPCPAPELLGGPSVFLF
PPKPKD TLMI SRTPEVT CVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPCEEQYGS
TYRCVSVLTVLHQDWLNGKEYKCKVSNKALPAPI EKTISKAKGQPREPQVYTLPPSRE
EMTKNQVSLTCLVKGFPSPDI AVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVD
KSRWQQGNVFS C SVMHEALHNHYTQKSLSLSPGKGGGSGGGSGGGSGGVQLVQ
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YNQKFPQGRVTMTVDTSSSTAYMELSR LRSDDTAVYYCATRYFYVMDYWGQGT LVT
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KPGKVPKLLIYYTSRLHSGVPSRFGSGSGTDFTLTIS SLQPEDVATYYCVQYAFPLT
FGCGTKVEIKSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAMNWVRQ
APGKGM EWVARIRSKYNNYATYYADAVKDRFTISRDDS KNTLYLQMN LKTEDTA
VYYCV RAGNFGSSYISYFAYWGQGT LVTVSSGGGSGGGSGGGSGGT VVTQEPSL
TVSPGGT VTI TCGSSTGAVTSGNYPNWIQKPGQAPRGLIGGTFKFLAPGTPARFSGSL
EGGKAALTLSGVQPEDEAEYYCVLYYSNRWVFGSGTKLTVL

Anti-Mesothelin 15-B12 CC VH (SEQ ID NO: 39)
QVQLQESGPGLVKPSSETLSLTCTVSGGSISSSSSYFWGWI RQPPGKCLEWIGNIYYSGSS
NYPNLSKSRVTISVDTSKNQFSLKLSSVTAADTAVYYCARLPRGDRDAFDI WGGQT
MVTVSS

Anti-Mesothelin 15-B12 CC VL (SEQ ID NO: 40)
DIVMTQSPSSLSASV GDRVITITCRASQGISNYLAWYQKPGKVPKLLIYAASTLQSGV
PSRFGSGSGTDFTLTIS SLQPEDFATYYCQQSYSTPFTFGCGTKVEIK

Anti-CD3 6H10.09 VH (SEQ ID NO: 41)
EVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAMNWVRQAPGKGM EWVARIRSKY
NNYATYYADAVKDRFTISRDDS KNTLYLQMN LKTEDTAVYYCV RAGNFGSSYISY
FAYWGQGT LVTVSS

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Anti-CD3 6H10.09 VL (SEQ ID NO: 42)
QTVVTQEPSTLTVSPGGTVITTCGSSGTAVTSGNYPNWIQKKPGQAPRGLIGGTFKLAP
GTPARFSGSLGEGKAALTLSGVQPEDEAEYCVLYYSNRWVFGSGTKLTVL

Anti-CDH3 15-E11 CC VH (SEQ ID NO: 43)
QVQLVQSGAEVKKPGASVKVSCASGYTFTNYWMNWRQAPGQCLEWMGNIAI
GVKGTNYNQKQGRVTMTVDTSSSTAYMELSRRLSDDTAVYCATRYFYVMDYW
GQGTLLVTVSS

Anti-CDH3 15-E11 CC VL (SEQ ID NO: 44)
DIQMTQSPSSLSASVGDRVTITCRASQDISNYLNWYQQKPKGVKLLIYYTSRLHSGV
PSRFGSGSGGTDFTLTISLQPEDVATYYCVQYAFPLTFGCGTKVEIK

scFv (SEQ ID NO: 45)
DKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPCEEQYGSTYRCVSVLTVLHQDWLNGKEYKCKVSNKALPAPIE
KTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENN
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KGGGSGGGGSGGGGSGGGGSGGGGSGGGGSDKTHTCPPCPAPELLGGPSVFLFPP
KPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPCEEQYGSTYR
CVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEM
TKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKS
RWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK

scFv-2 (SEQ ID NO: 46)
DKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPCEEQYGSTYRCVSVLTVLHQDWLNGKEYKCKVSNKALPAPIE
KTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENN
YKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPG
GGGSGGGGSGGGGSGGGGSGGGGSGGGGSDKTHTCPPCPAPELLGGPSVFLFPPKPK
DTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPCEEQYGSTYRCVS
VLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKN
QVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQ
QGNVFCSCVMHEALHNHYTQKSLSLSP

scFv-3 (SEQ ID NO: 47)
DKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
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KPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYR
VVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEM
TKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKS
RWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK

scFv-4 (SEQ ID NO: 48)
DKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDG
VEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQ
PREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFL
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FNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIE
KTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPV
LDSDGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSP

scFv-5 (SEQ ID NO: 49)
DKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
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KTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENN
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KPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYGSTYR
VVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEM
TKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKS
RWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK

scFv-6 (SEQ ID NO: 50)
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VDGVEVHNAKTKPREEQYGSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIE
KTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENN
YKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPG
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Human Serum Albumin (HSA) (SEQ ID NO: 57)
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PRLVYRPEVDVMCTAFHNDNETFLKKYLLEYARRHPYFYAPELLFFAKRYKAATECC
QAADKAACVLPLKDLRDEGKASSAKQRLKASLQFKGAFKAWAVARLSQRFP
KAEPAEVSKLVDTLTKVHTCECHGDLLECADDRADLAKYI CENQDSISSKLKECECK
PLLEKSHCIAEVENDEMPADLPALADFVSKDVCNKYAEAKDFVGLMFLPYEYARR
HPDYSVULLLRLAKTYETTLKCSAAADPHCEYAKVDFEKPFLVEBPONLIKONCEL

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FEQLGEYKFQNALLVRYTKKVPQVSTPTLVEVSRNLGKVGSKCKHPEAKRMPCAE
 DYLSVVLNQLCVLHEKTPVSDRVTKCCTESLVNRRPCFSALEVDETYVPKEFNAETF
 TFHADICTLSEKERQIKKQTALVELVKHKPKATKEQLKAVMDDFAAFVEKCKKADD
 KETCFAEEGKKLVAASQAALGL

Additional CD3 Binder Sequences

SEQ ID NO:	Designation	A.A. Sequence
58	I2C-HCDR1	KYAMN
59	I2C-HCDR2	RIRSKYNNYATYYADSVKD
60	I2C-HCDR3	HGNFGNSYISYWAY
61	I2C-LCDR1	GSSTGAVTSGNYPN
62	I2C-LCDR2	GTKFLAP
63	I2C-LCDR3	VLWYSNRWV
64	I2C-VH	EVQLVESGGGLVQPGGSLKLSCAASGFT FNKYAMNWVRQAPGKGLEWVARIRSK YNNYATYYADSVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHGNFGNS YISYWAYWGQGLTVTVSS
65	I2C-VL	QTVVTQEPSLTVSPGGTVTLTCGSSTGA VTSNGYNWVQQKPGQAPRGLIGGTFK LAPGTPARFSGSLLGGKAALTLSGVQPE DEAEYYCVLWYSNRWVFGGKTLTVL
66	I2C_44/100cc-HCDR1	KYAMN
67	I2C_44/100cc-HCDR2	RIRSKYNNYATYYADSVKD
68	I2C_44/100cc-HCDR3	HGNFGNSYISYWAY
69	I2C_44/100cc-LCDR1	GSSTGAVTSGNYPN
70	I2C_44/100cc-LCDR2	GTKFLAP
71	I2C_44/100cc-LCDR3	VLWYSNRWV
72	I2C_44/100cc-VH	EVQLVESGGGLVQPGGSLKLSCAASGFT FNKYAMNWVRQAPGKCLEWVARIRSK YNNYATYYADSVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHGNFGNS YISYWAYWGQGLTVTVSS
73	I2C_44/100cc-VL	QTVVTQEPSLTVSPGGTVTLTCGSSTGA VTSNGYNWVQQKPGQAPRGLIGGTFK LAPGTPARFSGSLLGGKAALTLSGVQPE DEAEYYCVLWYSNRWVFGCGTTLTVL
74	I2E-HCDR1	KYAIN
75	I2E-HCDR2	RIRSKYNNYATYYADAVKD
76	I2E-HCDR3	AGNFGSSYISYWAY
77	I2E-LCDR1	GSSTGAVTSGNYPN
78	I2E-LCDR2	GTKFLAP
79	I2E-LCDR3	VLWYSNRWV

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SEQ ID NO:	Designation	A.A. Sequence
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81	I2E-VL	QTVVTQEPSLTVSPGGTVTITCGSSTGAV TSGNYPNWVQKKPGQAPRGLIGGTKFL APGTPARFSGSLSGGKAALTLSGVQPED EAEYYCVLWYSNRWVFGSGTKLTVL
82	I2L-HCDR1	KYAMN
83	I2L-HCDR2	RIRSKYNNYATYYADAVKD
84	I2L-HCDR3	AGNFGSSYISYFAY
85	I2L-LCDR1	GSSTGAVTSGNYPN
86	I2L-LCDR2	GTKFLAP
87	I2L-LCDR3	VLYYSNRWV
88	I2L-VH	EVQLVESGGGLVQPGGSLKLSCAASGFT FNKYAMNWVRQAPGKGMWVARIRSK YNNYATYYADAVKDRFTISRDDSKNTL YLQMNNLKTEDTAVYYCVRAGNFGSSY ISYFAYWGQGLTVTVSS
89	I2L-VL	QTVVTQEPSLTVSPGGTVTITCGSSTGAV TSGNYPNWVQKKPGQAPRGLIGGTKFLA PGTPARFSGSLEGGKAALTLSGVQPEDE AEYYCVLYYSNRWVFGSGTKLTVL
90	I2M2-HCDR1	KYAIN
91	I2M2-HCDR2	RIRSKYNNYATYYADAVKD
92	I2M2-HCDR3	NANFGTSYISYFAY
93	I2M2-LCDR1	GSSTGAVTSGNYPN
94	I2M2-LCDR2	GTKFLAP
95	I2M2-LCDR3	VLWYSNRWV
96	I2M2-VH	EVQLVESGGGLVQPGGSLKLSCAASGFT FNKYAINWVREAPGKGLEWVARIRSKY NNYATYYADAVKDRFTISRDDSKNTAY LQMNNLKTEDTAVYYCVRNANFGTSYI SYFAYWGQGLTVTVSS
97	I2M2-VL	QTVVTQEPSLTVSPGGTVTLTSGSSTGA VTSNYPNWVQKKPGQAPRGLIGGTKF LAPGTPARFSGSLGGKAALTLSGVQPE DEAEYYCVLWYSNRWVFGSGTKLTVL

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 97

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<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

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<210> SEQ ID NO 2
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<210> SEQ ID NO 3
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<210> SEQ ID NO 4
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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

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Gly Gly Gly Ser
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<210> SEQ ID NO 5
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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

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Gly Gly Gly Ser Gly Gly Gly Gly Ser
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<210> SEQ ID NO 6
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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
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<210> SEQ ID NO 8
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 8

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<210> SEQ ID NO 9
<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 9

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1 5 10 15

Gly Gly Gly Gln
20

<210> SEQ ID NO 10
<211> LENGTH: 25
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 10

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1 5 10 15

Gly Gly Gly Gln Gly Gly Gly Gly Gln
20 25

<210> SEQ ID NO 11
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 11

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<210> SEQ ID NO 12
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
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<223> OTHER INFORMATION: Synthetic

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<400> SEQUENCE: 12

Thr Val Ala Ala Pro
1 5

<210> SEQ ID NO 13
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<400> SEQUENCE: 13

Ala Ser Thr Lys Gly Pro
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<210> SEQ ID NO 14

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<213> ORGANISM: Artificial Sequence
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Gly Gly Asn Gly Thr
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<210> SEQ ID NO 16
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 16

Tyr Gly Asn Gly Thr
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<210> SEQ ID NO 17
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 17

Ser Gly Gly Gly Gly Ser
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<210> SEQ ID NO 18
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 18

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<210> SEQ ID NO 19
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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 19

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<211> LENGTH: 12
<212> TYPE: PRT
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<400> SEQUENCE: 21

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<210> SEQ ID NO 22
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 22

Gly Gly Gly Gly Gly Gly Gly Gly Gly Gly Gly Gly Gly Gly Gly Gly
1 5 10 15

<210> SEQ ID NO 23
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<213> ORGANISM: Artificial Sequence
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1 5 10 15

Gly Gly Gly Gly
20

<210> SEQ ID NO 24
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<223> OTHER INFORMATION: Repeat 2-10 times

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<210> SEQ ID NO 26
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<212> TYPE: PRT
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<223> OTHER INFORMATION: Synthetic
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<223> OTHER INFORMATION: Repeat 3-10 times

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<210> SEQ ID NO 27
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<223> OTHER INFORMATION: Repeat 1-10 times

<400> SEQUENCE: 27

Gly Gly Gly Gly Ser
1 5

<210> SEQ ID NO 28
<211> LENGTH: 5
<212> TYPE: PRT
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic
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<221> NAME/KEY: REPEAT
<222> LOCATION: (1)..(5)
<223> OTHER INFORMATION: Repeat 2-10 times

<400> SEQUENCE: 28

-continued

Gly Gly Gly Gly Ser
1 5

<210> SEQ ID NO 29
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(5)
<223> OTHER INFORMATION: Repeat 3-10 times

<400> SEQUENCE: 29

Gly Gly Gly Gly Ser
1 5

<210> SEQ ID NO 30
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(5)
<223> OTHER INFORMATION: Repeat 1-10 times

<400> SEQUENCE: 30

Gly Gly Gly Gly Gln
1 5

<210> SEQ ID NO 31
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic
<220> FEATURE:
<221> NAME/KEY: REPEAT
<222> LOCATION: (1)..(5)
<223> OTHER INFORMATION: Repeat 2-10 times

<400> SEQUENCE: 31

Gly Gly Gly Gly Gln
1 5

<210> SEQ ID NO 32
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic
<220> FEATURE:
<221> NAME/KEY: REPEAT
<222> LOCATION: (1)..(5)
<223> OTHER INFORMATION: Repeat 3-10 times

<400> SEQUENCE: 32

Gly Gly Gly Gly Gln
1 5

<210> SEQ ID NO 33
<211> LENGTH: 186
<212> TYPE: PRT
<213> ORGANISM: Homo Sapiens

-continued

<400> SEQUENCE: 33

Gln Asp Gly Asn Glu Glu Met Gly Gly Ile Thr Gln Thr Pro Tyr Lys
 1 5 10 15
 Val Ser Ile Ser Gly Thr Thr Val Ile Leu Thr Cys Pro Gln Tyr Pro
 20 25 30
 Gly Ser Glu Ile Leu Trp Gln His Asn Asp Lys Asn Ile Gly Gly Asp
 35 40 45
 Glu Asp Asp Lys Asn Ile Gly Ser Asp Glu Asp His Leu Ser Leu Lys
 50 55 60
 Glu Phe Ser Glu Leu Glu Gln Ser Gly Tyr Tyr Val Cys Tyr Pro Arg
 65 70 75 80
 Gly Ser Lys Pro Glu Asp Ala Asn Phe Tyr Leu Tyr Leu Arg Ala Arg
 85 90 95
 Val Cys Glu Asn Cys Met Glu Met Asp Val Met Ser Val Ala Thr Ile
 100 105 110
 Val Ile Val Asp Ile Cys Ile Thr Gly Gly Leu Leu Leu Leu Val Tyr
 115 120 125
 Tyr Trp Ser Lys Asn Arg Lys Ala Lys Ala Lys Pro Val Thr Arg Gly
 130 135 140
 Ala Gly Ala Gly Gly Arg Gln Arg Gly Gln Asn Lys Glu Arg Pro Pro
 145 150 155 160
 Pro Val Pro Asn Pro Asp Tyr Glu Pro Ile Arg Lys Gly Gln Arg Asp
 165 170 175
 Leu Tyr Ser Gly Leu Asn Gln Arg Arg Ile
 180 185

<210> SEQ ID NO 34

<211> LENGTH: 177

<212> TYPE: PRT

<213> ORGANISM: cynomolgus

<400> SEQUENCE: 34

Gln Asp Gly Asn Glu Glu Met Gly Ser Ile Thr Gln Thr Pro Tyr Gln
 1 5 10 15
 Val Ser Ile Ser Gly Thr Thr Val Ile Leu Thr Cys Ser Gln His Leu
 20 25 30
 Gly Ser Glu Ala Gln Trp Gln His Asn Gly Lys Asn Lys Gly Asp Ser
 35 40 45
 Gly Asp Gln Leu Phe Leu Pro Glu Phe Ser Glu Met Glu Gln Ser Gly
 50 55 60
 Tyr Tyr Val Cys Tyr Pro Arg Gly Ser Asn Pro Glu Asp Ala Ser His
 65 70 75 80
 His Leu Tyr Leu Lys Ala Arg Val Cys Glu Asn Cys Met Glu Met Asp
 85 90 95
 Val Met Ala Val Ala Thr Ile Val Ile Val Asp Ile Cys Ile Thr Leu
 100 105 110
 Gly Leu Leu Leu Leu Val Tyr Tyr Trp Ser Lys Asn Arg Lys Ala Lys
 115 120 125
 Ala Lys Pro Val Thr Arg Gly Ala Gly Ala Gly Gly Arg Gln Arg Gly
 130 135 140
 Gln Asn Lys Glu Arg Pro Pro Pro Val Pro Asn Pro Asp Tyr Glu Pro
 145 150 155 160

-continued

Ile

```
<210> SEQ ID NO 35
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Homo Sapiens
```

<400> SEQUENCE: 35

Val Cys Glu Asn Cys Met Glu Met Asp Val Met Ser
100 105

```
<210> SEQ ID NO 36
<211> LENGTH: 27
<212> TYPE: PRT
<213> ORGANISM: Homo Sapiens
```

<400> SEQUENCE: 36

Val Ser Ile Ser Gly Thr Thr Val Ile Leu Thr
20 25

```
<210> SEQ ID NO 37
<211> LENGTH: 1555
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic
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<400> SEQUENCE: 37

Ser Leu Lys Leu Ser Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Tyr

-continued

Cys	Ala	Arg	Leu	Pro	Arg	Gly	Asp	Arg	Asp	Ala	Phe	Asp	Ile	Trp	Gly	
			100					105					110			
Gln	Gly	Thr	Met	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	
			115				120					125				
Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Val	Gln	
			130				135					140				
Leu	Val	Glu	Ser	Gly	Gly	Gly	Leu	Val	Gln	Pro	Gly	Gly	Ser	Leu	Lys	
						150				155					160	
Leu	Ser	Cys	Ala	Ala	Ser	Gly	Phe	Thr	Phe	Asn	Lys	Tyr	Ala	Met	Asn	
				165					170					175		
Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Met	Glu	Trp	Val	Ala	Arg	Ile	
				180				185					190			
Arg	Ser	Lys	Tyr	Asn	Asn	Tyr	Ala	Thr	Tyr	Tyr	Ala	Asp	Ala	Val	Lys	
			195				200					205				
Asp	Arg	Phe	Thr	Ile	Ser	Arg	Asp	Asp	Ser	Lys	Asn	Thr	Leu	Tyr	Leu	
			210			215					220					
Gln	Met	Asn	Asn	Leu	Lys	Thr	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	Val	
					230					235					240	
Arg	Ala	Gly	Asn	Phe	Gly	Ser	Ser	Tyr	Ile	Ser	Tyr	Phe	Ala	Tyr	Trp	
				245					250					255		
Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	
			260					265					270			
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Asp	Ile	
			275				280					285				
Val	Met	Thr	Gln	Ser	Pro	Ser	Ser	Leu	Ser	Ala	Ser	Val	Gly	Asp	Arg	
			290			295					300					
Val	Thr	Ile	Thr	Cys	Arg	Ala	Ser	Gln	Gly	Ile	Ser	Asn	Tyr	Leu	Ala	
					310					315					320	
Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Lys	Val	Pro	Lys	Leu	Leu	Ile	Tyr	Ala	
				325					330					335		
Ala	Ser	Thr	Leu	Gln	Ser	Gly	Val	Pro	Ser	Arg	Phe	Ser	Gly	Ser	Gly	
				340				345					350			
Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Ser	Leu	Gln	Pro	Glu	Asp	
			355				360					365				
Phe	Ala	Thr	Tyr	Tyr	Cys	Gln	Gln	Ser	Tyr	Ser	Thr	Pro	Phe	Thr	Phe	
			370			375					380					
Gly	Cys	Gly	Thr	Lys	Val	Glu	Ile	Lys	Gly	Gly	Gly	Gly	Ser	Gly	Gly	
					390				395					400		
Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gln	Thr	Val	
				405					410					415		
Val	Thr	Gln	Glu	Pro	Ser	Leu	Thr	Val	Ser	Pro	Gly	Gly	Thr	Val	Thr	
				420				425					430			
Ile	Thr	Cys	Gly	Ser	Ser	Thr	Gly	Ala	Val	Thr	Ser	Gly	Asn	Tyr	Pro	
				435			440					445				
Asn	Trp	Ile	Gln	Lys	Lys	Pro	Gly	Gln	Ala	Pro	Arg	Gly	Leu	Ile	Gly	
						455					460					
Gly	Thr	Lys	Phe	Leu	Ala	Pro	Gly	Thr	Pro	Ala	Arg	Phe	Ser	Gly	Ser	
					470					475					480	
Leu	Glu	Gly	Gly	Lys	Ala	Ala	Leu	Thr	Leu	Ser	Gly	Val	Gln	Pro	Glu	
				485					490					495		

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Asp	Glu	Ala	Glu	Tyr	Tyr	Cys	Val	Leu	Tyr	Tyr	Ser	Asn	Arg	Trp	Val
			500					505					510		
Phe	Gly	Ser	Gly	Thr	Lys	Leu	Thr	Val	Leu	Ser	Gly	Gly	Gly	Gly	Ser
		515					520					525			
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Ser	Asp	Lys	Thr	His	Thr	Cys	
	530					535				540					
Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val	Phe	Leu
545					550					555					560
Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu
			565						570					575	
Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Lys
			580					585					590		
Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys
		595					600					605			
Pro	Cys	Glu	Glu	Gln	Tyr	Gly	Ser	Thr	Tyr	Arg	Cys	Val	Ser	Val	Leu
	610					615					620				
Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys
625				630						635					640
Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys
			645						650					655	
Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser
			660					665					670		
Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys
	675						680					685			
Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln
	690					695					700				
Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser	Asp	Gly
705				710						715					720
Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln
			725						730					735	
Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn
		740						745					750		
His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys	Gly	Gly	Gly
	755						760					765			
Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly
	770					775					780				
Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Asp	Lys	Thr	His	Thr
785					790					795					800
Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val	Phe
			805						810					815	
Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro
			820					825					830		
Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val
		835					840					845			
Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr
	850					855					860				
Lys	Pro	Cys	Glu	Glu	Gln	Tyr	Gly	Ser	Thr	Tyr	Arg	Cys	Val	Ser	Val
865					870					875					880
Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys
			885						890					895	
Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser

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900					905					910					
Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro
		915						920				925			
Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val
		930				935						940			
Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly
945						950				955				960	
Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser	Asp
				965						970				975	
Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp
		980						985				990			
Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His
		995						1000				1005			
Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys	Gly	
1010						1015						1020			
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gln	
1025						1030						1035			
Val	Gln	Leu	Val	Gln	Ser	Gly	Ala	Glu	Val	Lys	Lys	Pro	Gly	Ala	
1040						1045						1050			
Ser	Val	Lys	Val	Ser	Cys	Lys	Ala	Ser	Gly	Tyr	Thr	Phe	Thr	Asn	
1055						1060						1065			
Tyr	Trp	Met	Asn	Trp	Val	Arg	Gln	Ala	Pro	Gly	Gln	Cys	Leu	Glu	
1070						1075						1080			
Trp	Met	Gly	Asn	Ile	Ala	Tyr	Gly	Val	Lys	Gly	Thr	Asn	Tyr	Asn	
1085						1090						1095			
Gln	Lys	Phe	Gln	Gly	Arg	Val	Thr	Met	Thr	Val	Asp	Thr	Ser	Ser	
1100						1105						1110			
Ser	Thr	Ala	Tyr	Met	Glu	Leu	Ser	Arg	Leu	Arg	Ser	Asp	Asp	Thr	
1115						1120						1125			
Ala	Val	Tyr	Tyr	Cys	Ala	Thr	Arg	Tyr	Phe	Tyr	Val	Met	Asp	Tyr	
1130						1135						1140			
Trp	Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	
1145						1150						1155			
Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	
1160						1165						1170			
Ser	Glu	Val	Gln	Leu	Val	Glu	Ser	Gly	Gly	Gly	Leu	Val	Gln	Pro	
1175						1180						1185			
Gly	Gly	Ser	Leu	Lys	Leu	Ser	Cys	Ala	Ala	Ser	Gly	Phe	Thr	Phe	
1190						1195						1200			
Asn	Lys	Tyr	Ala	Met	Asn	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	
1205						1210						1215			
Met	Glu	Trp	Val	Ala	Arg	Ile	Arg	Ser	Lys	Tyr	Asn	Asn	Tyr	Ala	
1220						1225						1230			
Thr	Tyr	Tyr	Ala	Asp	Ala	Val	Lys	Asp	Arg	Phe	Thr	Ile	Ser	Arg	
1235						1240						1245			
Asp	Asp	Ser	Lys	Asn	Thr	Leu	Tyr	Leu	Gln	Met	Asn	Asn	Leu	Lys	
1250						1255						1260			
Thr	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	Val	Arg	Ala	Gly	Asn	Phe	
1265						1270						1275			
Gly	Ser	Ser	Tyr	Ile	Ser	Tyr	Phe	Ala	Tyr	Trp	Gly	Gln	Gly	Thr	
1280						1285						1290			

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Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly
 1295 1300 1305
 Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Asp Ile Gln Met
 1310 1315 1320
 Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly Asp Arg Val
 1325 1330 1335
 Thr Ile Thr Cys Arg Ala Ser Gln Asp Ile Ser Asn Tyr Leu Asn
 1340 1345 1350
 Trp Tyr Gln Gln Lys Pro Gly Lys Val Pro Lys Leu Leu Ile Tyr
 1355 1360 1365
 Tyr Thr Ser Arg Leu His Ser Gly Val Pro Ser Arg Phe Ser Gly
 1370 1375 1380
 Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln
 1385 1390 1395
 Pro Glu Asp Val Ala Thr Tyr Tyr Cys Val Gln Tyr Ala Gln Phe
 1400 1405 1410
 Pro Leu Thr Phe Gly Cys Gly Thr Lys Val Glu Ile Lys Gly Gly
 1415 1420 1425
 Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly
 1430 1435 1440
 Gly Gly Ser Gln Thr Val Val Thr Gln Glu Pro Ser Leu Thr Val
 1445 1450 1455
 Ser Pro Gly Gly Thr Val Thr Ile Thr Cys Gly Ser Ser Thr Gly
 1460 1465 1470
 Ala Val Thr Ser Gly Asn Tyr Pro Asn Trp Ile Gln Lys Lys Pro
 1475 1480 1485
 Gly Gln Ala Pro Arg Gly Leu Ile Gly Gly Thr Lys Phe Leu Ala
 1490 1495 1500
 Pro Gly Thr Pro Ala Arg Phe Ser Gly Ser Leu Glu Gly Gly Lys
 1505 1510 1515
 Ala Ala Leu Thr Leu Ser Gly Val Gln Pro Glu Asp Glu Ala Glu
 1520 1525 1530
 Tyr Tyr Cys Val Leu Tyr Tyr Ser Asn Arg Trp Val Phe Gly Ser
 1535 1540 1545
 Gly Thr Lys Leu Thr Val Leu
 1550 1555

<210> SEQ ID NO 38
 <211> LENGTH: 1506
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 38

Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Glu
 1 5 10 15
 Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Gly Ser Ile Ser Ser Ser
 20 25 30
 Ser Tyr Phe Trp Gly Trp Ile Arg Gln Pro Pro Gly Lys Cys Leu Glu
 35 40 45
 Trp Ile Gly Asn Ile Tyr Tyr Ser Gly Ser Ser Asn Tyr Asn Pro Ser
 50 55 60

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Leu	Lys	Ser	Arg	Val	Thr	Ile	Ser	Val	Asp	Thr	Ser	Lys	Asn	Gln	Phe	65	70	75	80
Ser	Leu	Lys	Leu	Ser	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Tyr	85	90	95	
Cys	Ala	Arg	Leu	Pro	Arg	Gly	Asp	Arg	Asp	Ala	Phe	Asp	Ile	Trp	Gly	100	105	110	
Gln	Gly	Thr	Met	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	115	120	125	
Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Asp	Ile	Val	Met	Thr	Gln	Ser	Pro	130	135	140	
Ser	Ser	Leu	Ser	Ala	Ser	Val	Gly	Asp	Arg	Val	Thr	Ile	Thr	Cys	Arg	145	150	155	160
Ala	Ser	Gln	Gly	Ile	Ser	Asn	Tyr	Leu	Ala	Trp	Tyr	Gln	Gln	Lys	Pro	165	170	175	
Gly	Lys	Val	Pro	Lys	Leu	Leu	Ile	Tyr	Ala	Ala	Ser	Thr	Leu	Gln	Ser	180	185	190	
Gly	Val	Pro	Ser	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	195	200	205	
Leu	Thr	Ile	Ser	Ser	Leu	Gln	Pro	Glu	Asp	Phe	Ala	Thr	Tyr	Tyr	Cys	210	215	220	
Gln	Gln	Ser	Tyr	Ser	Thr	Pro	Phe	Thr	Phe	Gly	Cys	Gly	Thr	Lys	Val	225	230	235	240
Glu	Ile	Lys	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Val	Gln	Leu	Val	Glu	Ser	245	250	255	
Gly	Gly	Gly	Leu	Val	Gln	Pro	Gly	Gly	Ser	Leu	Lys	Leu	Ser	Cys	Ala	260	265	270	
Ala	Ser	Gly	Phe	Thr	Phe	Asn	Lys	Tyr	Ala	Met	Asn	Trp	Val	Arg	Gln	275	280	285	
Ala	Pro	Gly	Lys	Gly	Met	Glu	Trp	Val	Ala	Arg	Ile	Arg	Ser	Lys	Tyr	290	295	300	
Asn	Asn	Tyr	Ala	Thr	Tyr	Tyr	Ala	Asp	Ala	Val	Lys	Asp	Arg	Phe	Thr	305	310	315	320
Ile	Ser	Arg	Asp	Asp	Ser	Lys	Asn	Thr	Leu	Tyr	Leu	Gln	Met	Asn	Asn	325	330	335	
Leu	Lys	Thr	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	Val	Arg	Ala	Gly	Asn	340	345	350	
Phe	Gly	Ser	Ser	Tyr	Ile	Ser	Tyr	Phe	Ala	Tyr	Trp	Gly	Gln	Gly	Thr	355	360	365	
Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	370	375	380	
Gly	Gly	Gly	Gly	Ser	Gln	Thr	Val	Val	Thr	Gln	Glu	Pro	Ser	Leu	Thr	385	390	395	400
Val	Ser	Pro	Gly	Gly	Thr	Val	Thr	Ile	Thr	Cys	Gly	Ser	Ser	Thr	Gly	405	410	415	
Ala	Val	Thr	Ser	Gly	Asn	Tyr	Pro	Asn	Trp	Ile	Gln	Lys	Lys	Pro	Gly	420	425	430	
Gln	Ala	Pro	Arg	Gly	Leu	Ile	Gly	Gly	Thr	Lys	Phe	Leu	Ala	Pro	Gly	435	440	445	
Thr	Pro	Ala	Arg	Phe	Ser	Gly	Ser	Leu	Glu	Gly	Gly	Lys	Ala	Ala	Leu	450	455	460	

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Thr	Leu	Ser	Gly	Val	Gln	Pro	Glu	Asp	Glu	Ala	Glu	Tyr	Tyr	Cys	Val	465	470	475	480
Leu	Tyr	Tyr	Ser	Asn	Arg	Trp	Val	Phe	Gly	Ser	Gly	Thr	Lys	Leu	Thr	485	490	495	
Val	Leu	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	500	505	510	
Ser	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	Leu	515	520	525	
Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	530	535	540	
Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	545	550	555	560
His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	565	570	575	
Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Cys	Glu	Glu	Gln	Tyr	Gly	Ser	Thr	580	585	590	
Tyr	Arg	Cys	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	595	600	605	
Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	610	615	620	
Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	625	630	635	640
Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	645	650	655	
Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	660	665	670	
Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	675	680	685	
Pro	Val	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	690	695	700	
Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	705	710	715	720
Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	725	730	735	
Ser	Pro	Gly	Lys	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	740	745	750	
Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	755	760	765	
Gly	Ser	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	770	775	780	
Leu	Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	785	790	795	800
Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	805	810	815	
Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	820	825	830	
Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Cys	Glu	Glu	Gln	Tyr	Gly	Ser	835	840	845	
Thr	Tyr	Arg	Cys	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	850	855	860	
Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala				

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865	870	875	880
Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro	885	890	895
Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln	900	905	910
Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala	915	920	925
Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr	930	935	940
Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu	945	950	955
Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser	965	970	975
Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser	980	985	990
Leu Ser Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly	995	1000	1005
Gly Gly Gly Ser Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val	1010	1015	1020
Lys Lys Pro Gly Ala Ser Val Lys Val Ser Cys Lys Ala Ser Gly	1025	1030	1035
Tyr Thr Phe Thr Asn Tyr Trp Met Asn Trp Val Arg Gln Ala Pro	1040	1045	1050
Gly Gln Cys Leu Glu Trp Met Gly Asn Ile Ala Tyr Gly Val Lys	1055	1060	1065
Gly Thr Asn Tyr Asn Gln Lys Phe Gln Gly Arg Val Thr Met Thr	1070	1075	1080
Val Asp Thr Ser Ser Ser Thr Ala Tyr Met Glu Leu Ser Arg Leu	1085	1090	1095
Arg Ser Asp Asp Thr Ala Val Tyr Tyr Cys Ala Thr Arg Tyr Phe	1100	1105	1110
Tyr Val Met Asp Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser	1115	1120	1125
Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly	1130	1135	1140
Ser Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser	1145	1150	1155
Val Gly Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Asp Ile	1160	1165	1170
Ser Asn Tyr Leu Asn Trp Tyr Gln Gln Lys Pro Gly Lys Val Pro	1175	1180	1185
Lys Leu Leu Ile Tyr Tyr Thr Ser Arg Leu His Ser Gly Val Pro	1190	1195	1200
Ser Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr	1205	1210	1215
Ile Ser Ser Leu Gln Pro Glu Asp Val Ala Thr Tyr Tyr Cys Val	1220	1225	1230
Gln Tyr Ala Gln Phe Pro Leu Thr Phe Gly Cys Gly Thr Lys Val	1235	1240	1245
Glu Ile Lys Ser Gly Gly Gly Gly Ser Glu Val Gln Leu Val Glu	1250	1255	1260

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Ser Gly Gly Gly Leu Val Gln Pro Gly Gly Ser Leu Lys Leu Ser
 1265 1270 1275
 Cys Ala Ala Ser Gly Phe Thr Phe Asn Lys Tyr Ala Met Asn Trp
 1280 1285 1290
 Val Arg Gln Ala Pro Gly Lys Gly Met Glu Trp Val Ala Arg Ile
 1295 1300 1305
 Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp Ala Val
 1310 1315 1320
 Lys Asp Arg Phe Thr Ile Ser Arg Asp Asp Ser Lys Asn Thr Leu
 1325 1330 1335
 Tyr Leu Gln Met Asn Asn Leu Lys Thr Glu Asp Thr Ala Val Tyr
 1340 1345 1350
 Tyr Cys Val Arg Ala Gly Asn Phe Gly Ser Ser Tyr Ile Ser Tyr
 1355 1360 1365
 Phe Ala Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Gly
 1370 1375 1380
 Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln
 1385 1390 1395
 Thr Val Val Thr Gln Glu Pro Ser Leu Thr Val Ser Pro Gly Gly
 1400 1405 1410
 Thr Val Thr Ile Thr Cys Gly Ser Ser Thr Gly Ala Val Thr Ser
 1415 1420 1425
 Gly Asn Tyr Pro Asn Trp Ile Gln Lys Lys Pro Gly Gln Ala Pro
 1430 1435 1440
 Arg Gly Leu Ile Gly Gly Thr Lys Phe Leu Ala Pro Gly Thr Pro
 1445 1450 1455
 Ala Arg Phe Ser Gly Ser Leu Glu Gly Gly Lys Ala Ala Leu Thr
 1460 1465 1470
 Leu Ser Gly Val Gln Pro Glu Asp Glu Ala Glu Tyr Tyr Cys Val
 1475 1480 1485
 Leu Tyr Tyr Ser Asn Arg Trp Val Phe Gly Ser Gly Thr Lys Leu
 1490 1495 1500
 Thr Val Leu
 1505

<210> SEQ ID NO 39
 <211> LENGTH: 121
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 39

Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Glu
 1 5 10 15
 Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Gly Ser Ile Ser Ser
 20 25 30
 Ser Tyr Phe Trp Gly Trp Ile Arg Gln Pro Pro Gly Lys Cys Leu Glu
 35 40 45
 Trp Ile Gly Asn Ile Tyr Tyr Ser Gly Ser Ser Asn Tyr Asn Pro Ser
 50 55 60
 Leu Lys Ser Arg Val Thr Ile Ser Val Asp Thr Ser Lys Asn Gln Phe
 65 70 75 80

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Ser Leu Lys Leu Ser Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Tyr
 85 90 95

Cys Ala Arg Leu Pro Arg Gly Asp Arg Asp Ala Phe Asp Ile Trp Gly
 100 105 110

Gln Gly Thr Met Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 40
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 40

Asp Ile Val Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
 1 5 10 15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Gly Ile Ser Asn Tyr
 20 25 30

Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Val Pro Lys Leu Leu Ile
 35 40 45

Tyr Ala Ala Ser Thr Leu Gln Ser Gly Val Pro Ser Arg Phe Ser Gly
 50 55 60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
 65 70 75 80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Ser Tyr Ser Thr Pro Phe
 85 90 95

Thr Phe Gly Cys Gly Thr Lys Val Glu Ile Lys
 100 105

<210> SEQ ID NO 41
 <211> LENGTH: 125
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 41

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Lys Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asn Lys Tyr
 20 25 30

Ala Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Met Glu Trp Val
 35 40 45

Ala Arg Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp
 50 55 60

Ala Val Lys Asp Arg Phe Thr Ile Ser Arg Asp Asp Ser Lys Asn Thr
 65 70 75 80

Leu Tyr Leu Gln Met Asn Asn Leu Lys Thr Glu Asp Thr Ala Val Tyr
 85 90 95

Tyr Cys Val Arg Ala Gly Asn Phe Gly Ser Ser Tyr Ile Ser Tyr Phe
 100 105 110

Ala Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
 115 120 125

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<210> SEQ ID NO 42
 <211> LENGTH: 109
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 42

Gln Thr Val Val Thr Gln Glu Pro Ser Leu Thr Val Ser Pro Gly Gly
 1 5 10 15
 Thr Val Thr Ile Thr Cys Gly Ser Ser Thr Gly Ala Val Thr Ser Gly
 20 25 30
 Asn Tyr Pro Asn Trp Ile Gln Lys Lys Pro Gly Gln Ala Pro Arg Gly
 35 40 45
 Leu Ile Gly Gly Thr Lys Phe Leu Ala Pro Gly Thr Pro Ala Arg Phe
 50 55 60
 Ser Gly Ser Leu Glu Gly Gly Lys Ala Ala Leu Thr Leu Ser Gly Val
 65 70 75 80
 Gln Pro Glu Asp Glu Ala Glu Tyr Tyr Cys Val Leu Tyr Tyr Ser Asn
 85 90 95
 Arg Trp Val Phe Gly Ser Gly Thr Lys Leu Thr Val Leu
 100 105

<210> SEQ ID NO 43
 <211> LENGTH: 117
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 43

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
 1 5 10 15
 Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Asn Tyr
 20 25 30
 Trp Met Asn Trp Val Arg Gln Ala Pro Gly Gln Cys Leu Glu Trp Met
 35 40 45
 Gly Asn Ile Ala Tyr Gly Val Lys Gly Thr Asn Tyr Asn Gln Lys Phe
 50 55 60
 Gln Gly Arg Val Thr Met Thr Val Asp Thr Ser Ser Ser Thr Ala Tyr
 65 70 75 80
 Met Glu Leu Ser Arg Leu Arg Ser Asp Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Ala Thr Arg Tyr Phe Tyr Val Met Asp Tyr Trp Gly Gln Gly Thr Leu
 100 105 110
 Val Thr Val Ser Ser
 115

<210> SEQ ID NO 44
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 44

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
 1 5 10 15

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Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Asp Ile Ser Asn Tyr
 20 25 30
 Leu Asn Trp Tyr Gln Gln Lys Pro Gly Lys Val Pro Lys Leu Leu Ile
 35 40 45
 Tyr Tyr Thr Ser Arg Leu His Ser Gly Val Pro Ser Arg Phe Ser Gly
 50 55 60
 Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
 65 70 75 80
 Glu Asp Val Ala Thr Tyr Tyr Cys Val Gln Tyr Ala Gln Phe Pro Leu
 85 90 95
 Thr Phe Gly Cys Gly Thr Lys Val Glu Ile Lys
 100 105

<210> SEQ ID NO 45
 <211> LENGTH: 484
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 45

Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly
 1 5 10 15
 Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met
 20 25 30
 Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His
 35 40 45
 Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val
 50 55 60
 His Asn Ala Lys Thr Lys Pro Cys Glu Glu Gln Tyr Gly Ser Thr Tyr
 65 70 75 80
 Arg Cys Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly
 85 90 95
 Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile
 100 105 110
 Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val
 115 120 125
 Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser
 130 135 140
 Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu
 145 150 155 160
 Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro
 165 170 175
 Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val
 180 185 190
 Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met
 195 200 205
 His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser
 210 215 220
 Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
 225 230 235 240
 Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly
 245 250 255

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Ser Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu
 260 265 270
 Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu
 275 280 285
 Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser
 290 295 300
 His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu
 305 310 315 320
 Val His Asn Ala Lys Thr Lys Pro Cys Glu Glu Gln Tyr Gly Ser Thr
 325 330 335
 Tyr Arg Cys Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn
 340 345 350
 Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro
 355 360 365
 Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln
 370 375 380
 Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val
 385 390 395 400
 Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val
 405 410 415
 Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro
 420 425 430
 Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr
 435 440 445
 Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val
 450 455 460
 Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu
 465 470 475 480
 Ser Pro Gly Lys

 <210> SEQ ID NO 46
 <211> LENGTH: 480
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

 <400> SEQUENCE: 46
 Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly
 1 5 10 15
 Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met
 20 25 30
 Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His
 35 40 45
 Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val
 50 55 60
 His Asn Ala Lys Thr Lys Pro Cys Glu Glu Gln Tyr Gly Ser Thr Tyr
 65 70 75 80
 Arg Cys Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly
 85 90 95
 Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile
 100 105 110

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Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val
	115						120					125			
Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser
	130					135					140				
Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu
145					150					155					160
Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro
			165						170					175	
Val	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val
		180						185					190		
Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met
	195						200					205			
His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser
	210				215						220				
Pro	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser
225					230					235					240
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Asp
				245					250					255	
Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	Leu	Gly	Gly
			260					265					270		
Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile
	275						280					285			
Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu
	290					295					300				
Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His
305					310					315					320
Asn	Ala	Lys	Thr	Lys	Pro	Cys	Glu	Glu	Gln	Tyr	Gly	Ser	Thr	Tyr	Arg
				325					330					335	
Cys	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys
			340					345					350		
Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu
	355						360					365			
Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr
	370					375					380				
Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu
385					390					395					400
Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp
				405					410					415	
Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val
			420					425					430		
Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp
	435						440					445			
Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His
	450					455					460				
Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro
465					470					475					480

<210> SEQ ID NO 47

<211> LENGTH: 484

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic

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<400> SEQUENCE: 47

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Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly
1      5      10      15
Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met
20      25      30
Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His
35      40      45
Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val
50      55      60
His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr
65      70      75      80
Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly
85      90      95
Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile
100     105     110
Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val
115     120     125
Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser
130     135     140
Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu
145     150     155     160
Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro
165     170     175
Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val
180     185     190
Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met
195     200     205
His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser
210     215     220
Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
225     230     235     240
Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly
245     250     255
Ser Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu
260     265     270
Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu
275     280     285
Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser
290     295     300
His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu
305     310     315     320
Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr
325     330     335
Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn
340     345     350
Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro
355     360     365
Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln
370     375     380
Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val

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385	390	395	400
Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val	405	410	415
Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro	420	425	430
Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr	435	440	445
Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val	450	455	460
Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu	465	470	475
480			
Ser Pro Gly Lys			
<210> SEQ ID NO 48			
<211> LENGTH: 480			
<212> TYPE: PRT			
<213> ORGANISM: Artificial Sequence			
<220> FEATURE:			
<223> OTHER INFORMATION: Synthetic			
<400> SEQUENCE: 48			
Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly	1	5	10
Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met	20	25	30
Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His	35	40	45
Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val	50	55	60
His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr	65	70	75
Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly	85	90	95
Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile	100	105	110
Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val	115	120	125
Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser	130	135	140
Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu	145	150	155
Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro	165	170	175
Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val	180	185	190
Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met	195	200	205
His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser	210	215	220
Pro Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser	225	230	235
Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Asp	245	250	255

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Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly
 260 265 270
 Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile
 275 280 285
 Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu
 290 295 300
 Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His
 305 310 315 320
 Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg
 325 330 335
 Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys
 340 345 350
 Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu
 355 360 365
 Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr
 370 375 380
 Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser Leu
 385 390 395 400
 Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp
 405 410 415
 Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val
 420 425 430
 Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp
 435 440 445
 Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His
 450 455 460
 Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro
 465 470 475 480

<210> SEQ ID NO 49

<211> LENGTH: 484

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 49

Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly
 1 5 10 15
 Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met
 20 25 30
 Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His
 35 40 45
 Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val
 50 55 60
 His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Gly Ser Thr Tyr
 65 70 75 80
 Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly
 85 90 95
 Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile
 100 105 110
 Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val
 115 120 125

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Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser
 130                               135                     140

Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu
145                               150                     155                     160

Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro
      165                               170                     175

Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val
      180                               185                     190

Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met
      195                               200                     205

His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser
      210                               215                     220

Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
225                               230                     235                     240

Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly
      245                               250                     255

Ser Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu
      260                               265                     270

Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu
      275                               280                     285

Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser
      290                               295                     300

His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu
305                               310                     315                     320

Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Gly Ser Thr
      325                               330                     335

Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn
      340                               345                     350

Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro
      355                               360                     365

Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln
      370                               375                     380

Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val
385                               390                     395                     400

Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val
      405                               410                     415

Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro
      420                               425                     430

Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr
      435                               440                     445

Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val
      450                               455                     460

Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu
465                               470                     475                     480

Ser Pro Gly Lys

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<210> SEQ ID NO 50

<211> LENGTH: 480

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic

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<400> SEQUENCE: 50

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Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly
1      5      10      15
Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met
20      25      30
Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His
35      40      45
Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val
50      55      60
His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Gly Ser Thr Tyr
65      70      75      80
Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly
85      90      95
Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile
100     105     110
Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val
115     120     125
Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser
130     135     140
Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu
145     150     155     160
Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro
165     170     175
Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val
180     185     190
Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met
195     200     205
His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser
210     215     220
Pro Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
225     230     235     240
Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Asp
245     250     255
Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly
260     265     270
Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile
275     280     285
Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu
290     295     300
Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His
305     310     315     320
Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Gly Ser Thr Tyr Arg
325     330     335
Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys
340     345     350
Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu
355     360     365
Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr
370     375     380
Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser Leu

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385		390		395		400
Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp						
	405			410		415
Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val						
	420			425		430
Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp						
	435			440		445
Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His						
	450			455		460
Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro						
465		470		475		480

<210> SEQ ID NO 51
 <211> LENGTH: 484
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 51

Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly						
1		5		10		15
Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met						
	20			25		30
Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His						
	35			40		45
Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val						
	50			55		60
His Asn Ala Lys Thr Lys Pro Cys Glu Glu Gln Tyr Asn Ser Thr Tyr						
65		70		75		80
Arg Cys Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly						
	85			90		95
Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile						
	100			105		110
Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val						
	115			120		125
Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser						
	130			135		140
Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu						
145		150		155		160
Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro						
	165			170		175
Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val						
	180			185		190
Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met						
	195			200		205
His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser						
	210			215		220
Pro Gly Lys Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly						
225		230		235		240
Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly						
	245			250		255
Ser Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu						

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260					265					270					
Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu
	275						280					285			
Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser
	290					295					300				
His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu
305						310					315				320
Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Cys	Glu	Glu	Gln	Tyr	Asn	Ser	Thr
				325					330					335	
Tyr	Arg	Cys	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn
			340					345					350		
Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro
	355						360					365			
Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln
370						375					380				
Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val
385						390					395				400
Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val
			405						410					415	
Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro
			420					425					430		
Pro	Val	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr
	435						440					445			
Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val
	450					455					460				
Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu
465						470					475				480
Ser	Pro	Gly	Lys												

<210> SEQ ID NO 52

<211> LENGTH: 480

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 52

Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	Leu	Gly
1				5					10					15	
Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met
	20						25						30		
Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His
	35					40						45			
Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val
	50				55					60					
His	Asn	Ala	Lys	Thr	Lys	Pro	Cys	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr
65					70					75				80	
Arg	Cys	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly
		85					90						95		
Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile
	100						105					110			
Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val
	115					120						125			

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Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser
130                      135                      140

Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu
145                      150                      155                      160

Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro
165                      170                      175

Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val
180                      185                      190

Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met
195                      200                      205

His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser
210                      215                      220

Pro Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
225                      230                      235                      240

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Asp
245                      250                      255

Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly
260                      265                      270

Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile
275                      280                      285

Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu
290                      295                      300

Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His
305                      310                      315                      320

Asn Ala Lys Thr Lys Pro Cys Glu Glu Gln Tyr Asn Ser Thr Tyr Arg
325                      330                      335

Cys Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys
340                      345                      350

Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu
355                      360                      365

Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr
370                      375                      380

Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser Leu
385                      390                      395                      400

Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp
405                      410                      415

Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val
420                      425                      430

Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp
435                      440                      445

Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His
450                      455                      460

Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro
465                      470                      475                      480

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<210> SEQ ID NO 53

<211> LENGTH: 474

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 53

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Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val	Phe	1	5	10	15
Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	20	25	30	
Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Glu	Pro	Glu	Val	35	40	45	
Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	50	55	60	
Lys	Pro	Cys	Glu	Glu	Gln	Tyr	Gly	Ser	Thr	Tyr	Arg	Cys	Val	Ser	Val	65	70	75	80
Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	85	90	95	
Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	100	105	110	
Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	115	120	125	
Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	130	135	140	
Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	145	150	155	160
Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser	Asp	165	170	175	
Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	180	185	190	
Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	195	200	205	
Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys	Gly	Gly	210	215	220	
Gly	Gly	Gln	Gly	Gly	Gly	Gly	Gln	Gly	Gly	Gly	Gly	Gln	Gly	Gly	Gly	225	230	235	240
Gly	Gln	Gly	Gly	Gly	Gly	Gln	Gly	Gly	Gly	Gly	Gln	Cys	Pro	Pro	Cys	245	250	255	
Pro	Ala	Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	260	265	270	
Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	275	280	285	
Val	Val	Val	Asp	Val	Ser	His	Glu	Glu	Pro	Glu	Val	Lys	Phe	Asn	Trp	290	295	300	
Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Cys	Glu	305	310	315	320
Glu	Gln	Tyr	Gly	Ser	Thr	Tyr	Arg	Cys	Val	Ser	Val	Leu	Thr	Val	Leu	325	330	335	
His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	340	345	350	
Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	355	360	365	
Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	370	375	380	
Met	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	385	390	395	400

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Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn
      405                      410                      415

Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe
      420                      425                      430

Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn
      435                      440                      445

Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr
      450                      455                      460

Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
465                      470

<210> SEQ ID NO 54
<211> LENGTH: 968
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 54

Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly
1      5      10      15

Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met
      20      25      30

Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His
      35      40      45

Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val
      50      55      60

His Asn Ala Lys Thr Lys Pro Cys Glu Glu Gln Tyr Gly Ser Thr Tyr
      65      70      75      80

Arg Cys Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly
      85      90      95

Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile
      100     105     110

Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val
      115     120     125

Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser
      130     135     140

Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu
      145     150     155     160

Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro
      165     170     175

Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val
      180     185     190

Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met
      195     200     205

His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser
      210     215     220

Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
      225     230     235     240

Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly
      245     250     255

Ser Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu
      260     265     270

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Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	275	280	285
Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	290	295	300
His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	305	310	315
Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Cys	Glu	Glu	Gln	Tyr	Gly	Ser	Thr	325	330	335
Tyr	Arg	Cys	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	340	345	350
Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	355	360	365
Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	370	375	380
Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	385	390	395
Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	405	410	415
Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	420	425	430
Pro	Val	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	435	440	445
Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	450	455	460
Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	465	470	475
Ser	Pro	Gly	Lys	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	485	490	495
Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	500	505	510
Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	515	520	525
Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	530	535	540
Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Cys	Glu	Glu	Gln	Tyr	545	550	555
Gly	Ser	Thr	Tyr	Arg	Cys	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	565	570	575
Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	580	585	590
Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	595	600	605
Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	610	615	620
Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	625	630	635
Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	645	650	655
Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	660	665	670
Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser			

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675	680	685
Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser		
690	695	700
Leu Ser Leu Ser Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly		
705	710	715
Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser		
	725	730
		735
Gly Gly Gly Gly Ser Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala		
	740	745
		750
Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro		
	755	760
		765
Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val		
	770	775
		780
Val Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val		
	785	790
		795
Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Cys Glu Glu Gln		
	805	810
		815
Tyr Gly Ser Thr Tyr Arg Cys Val Ser Val Leu Thr Val Leu His Gln		
	820	825
		830
Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala		
	835	840
		845
Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro		
	850	855
		860
Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr		
	865	870
		875
Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser		
	885	890
		895
Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr		
	900	905
		910
Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr		
	915	920
		925
Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe		
	930	935
		940
Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys		
	945	950
		955
		960
Ser Leu Ser Leu Ser Pro Gly Lys		
	965	

<210> SEQ ID NO 55

<211> LENGTH: 227

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 55

Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly		
1	5	10
		15
Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met		
	20	25
		30
Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His		
	35	40
		45
Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val		

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50	55	60
His Asn Ala Lys Thr Lys Pro Cys Glu Glu Gln Tyr Gly Ser Thr Tyr		
65	70	75 80
Arg Cys Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly		
	85	90 95
Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile		
	100	105 110
Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val		
	115	120 125
Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser		
	130	135 140
Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu		
145	150	155 160
Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Asp Thr Thr Pro Pro		
	165	170 175
Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Asp Leu Thr Val		
	180	185 190
Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met		
	195	200 205
His Glu Ala Leu His Asn His Tyr Thr Gln Asp Ser Leu Ser Leu Ser		
	210	215 220
Pro Gly Lys		
225		

<210> SEQ ID NO 56

<211> LENGTH: 227

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 56

Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly		
1	5	10 15
Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met		
	20	25 30
Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His		
	35	40 45
Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val		
50	55	60
His Asn Ala Lys Thr Lys Pro Cys Glu Glu Gln Tyr Gly Ser Thr Tyr		
65	70	75 80
Arg Cys Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly		
	85	90 95
Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile		
	100	105 110
Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val		
	115	120 125
Tyr Thr Leu Pro Pro Ser Arg Lys Glu Met Thr Lys Asn Gln Val Ser		
	130	135 140
Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu		
145	150	155 160
Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro		

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165					170					175					
Val	Leu	Lys	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val
			180					185					190		
Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met
		195					200					205			
His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser
	210					215					220				
Pro Gly Lys															
225															
<210> SEQ ID NO 57															
<211> LENGTH: 585															
<212> TYPE: PRT															
<213> ORGANISM: Homo Sapiens															
<400> SEQUENCE: 57															
Asp	Ala	His	Lys	Ser	Glu	Val	Ala	His	Arg	Phe	Lys	Asp	Leu	Gly	Glu
1				5					10					15	
Glu	Asn	Phe	Lys	Ala	Leu	Val	Leu	Ile	Ala	Phe	Ala	Gln	Tyr	Leu	Gln
		20						25					30		
Gln	Cys	Pro	Phe	Glu	Asp	His	Val	Lys	Leu	Val	Asn	Glu	Val	Thr	Glu
		35					40				45				
Phe	Ala	Lys	Thr	Cys	Val	Ala	Asp	Glu	Ser	Ala	Glu	Asn	Cys	Asp	Lys
	50					55				60					
Ser	Leu	His	Thr	Leu	Phe	Gly	Asp	Lys	Leu	Cys	Thr	Val	Ala	Thr	Leu
65				70					75					80	
Arg	Glu	Thr	Tyr	Gly	Glu	Met	Ala	Asp	Cys	Cys	Ala	Lys	Gln	Glu	Pro
			85					90					95		
Glu	Arg	Asn	Glu	Cys	Phe	Leu	Gln	His	Lys	Asp	Asp	Asn	Pro	Asn	Leu
		100					105					110			
Pro	Arg	Leu	Val	Arg	Pro	Glu	Val	Asp	Val	Met	Cys	Thr	Ala	Phe	His
		115				120					125				
Asp	Asn	Glu	Glu	Thr	Phe	Leu	Lys	Lys	Tyr	Leu	Tyr	Glu	Ile	Ala	Arg
	130					135				140					
Arg	His	Pro	Tyr	Phe	Tyr	Ala	Pro	Glu	Leu	Leu	Phe	Phe	Ala	Lys	Arg
145				150					155					160	
Tyr	Lys	Ala	Ala	Phe	Thr	Glu	Cys	Cys	Gln	Ala	Ala	Asp	Lys	Ala	Ala
		165					170					175			
Cys	Leu	Leu	Pro	Lys	Leu	Asp	Glu	Leu	Arg	Asp	Glu	Gly	Lys	Ala	Ser
		180					185					190			
Ser	Ala	Lys	Gln	Arg	Leu	Lys	Cys	Ala	Ser	Leu	Gln	Lys	Phe	Gly	Glu
		195				200					205				
Arg	Ala	Phe	Lys	Ala	Trp	Ala	Val	Ala	Arg	Leu	Ser	Gln	Arg	Phe	Pro
	210				215					220					
Lys	Ala	Glu	Phe	Ala	Glu	Val	Ser	Lys	Leu	Val	Thr	Asp	Leu	Thr	Lys
225				230					235				240		
Val	His	Thr	Glu	Cys	Cys	His	Gly	Asp	Leu	Leu	Glu	Cys	Ala	Asp	Asp
		245					250					255			
Arg	Ala	Asp	Leu	Ala	Lys	Tyr	Ile	Cys	Glu	Asn	Gln	Asp	Ser	Ile	Ser
		260					265				270				
Ser	Lys	Leu	Lys	Glu	Cys	Cys	Glu	Lys	Pro	Leu	Leu	Glu	Lys	Ser	His
	275					280					285				

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Cys	Ile	Ala	Glu	Val	Glu	Asn	Asp	Glu	Met	Pro	Ala	Asp	Leu	Pro	Ser	290	295	300
Leu	Ala	Ala	Asp	Phe	Val	Glu	Ser	Lys	Asp	Val	Cys	Lys	Asn	Tyr	Ala	305	310	315
Glu	Ala	Lys	Asp	Val	Phe	Leu	Gly	Met	Phe	Leu	Tyr	Glu	Tyr	Ala	Arg	325	330	335
Arg	His	Pro	Asp	Tyr	Ser	Val	Val	Leu	Leu	Leu	Arg	Leu	Ala	Lys	Thr	340	345	350
Tyr	Glu	Thr	Thr	Leu	Glu	Lys	Cys	Cys	Ala	Ala	Ala	Asp	Pro	His	Glu	355	360	365
Cys	Tyr	Ala	Lys	Val	Phe	Asp	Glu	Phe	Lys	Pro	Leu	Val	Glu	Glu	Pro	370	375	380
Gln	Asn	Leu	Ile	Lys	Gln	Asn	Cys	Glu	Leu	Phe	Glu	Gln	Leu	Gly	Glu	385	390	395
Tyr	Lys	Phe	Gln	Asn	Ala	Leu	Leu	Val	Arg	Tyr	Thr	Lys	Lys	Val	Pro	405	410	415
Gln	Val	Ser	Thr	Pro	Thr	Leu	Val	Glu	Val	Ser	Arg	Asn	Leu	Gly	Lys	420	425	430
Val	Gly	Ser	Lys	Cys	Cys	Lys	His	Pro	Glu	Ala	Lys	Arg	Met	Pro	Cys	435	440	445
Ala	Glu	Asp	Tyr	Leu	Ser	Val	Val	Leu	Asn	Gln	Leu	Cys	Val	Leu	His	450	455	460
Glu	Lys	Thr	Pro	Val	Ser	Asp	Arg	Val	Thr	Lys	Cys	Cys	Thr	Glu	Ser	465	470	475
Leu	Val	Asn	Arg	Arg	Pro	Cys	Phe	Ser	Ala	Leu	Glu	Val	Asp	Glu	Thr	485	490	495
Tyr	Val	Pro	Lys	Glu	Phe	Asn	Ala	Glu	Thr	Phe	Thr	Phe	His	Ala	Asp	500	505	510
Ile	Cys	Thr	Leu	Ser	Glu	Lys	Glu	Arg	Gln	Ile	Lys	Lys	Gln	Thr	Ala	515	520	525
Leu	Val	Glu	Leu	Val	Lys	His	Lys	Pro	Lys	Ala	Thr	Lys	Glu	Gln	Leu	530	535	540
Lys	Ala	Val	Met	Asp	Asp	Phe	Ala	Ala	Phe	Val	Glu	Lys	Cys	Cys	Lys	545	550	555
Ala	Asp	Asp	Lys	Glu	Thr	Cys	Phe	Ala	Glu	Glu	Gly	Lys	Lys	Leu	Val	565	570	575
Ala	Ala	Ser	Gln	Ala	Ala	Leu	Gly	Leu								580	585	

<210> SEQ ID NO 58
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 58

Lys Tyr Ala Met Asn
 1 5

<210> SEQ ID NO 59
 <211> LENGTH: 19
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:

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<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 59

Arg Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp Ser
1 5 10 15

Val Lys Asp

<210> SEQ ID NO 60

<211> LENGTH: 14

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 60

His Gly Asn Phe Gly Asn Ser Tyr Ile Ser Tyr Trp Ala Tyr
1 5 10

<210> SEQ ID NO 61

<211> LENGTH: 14

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 61

Gly Ser Ser Thr Gly Ala Val Thr Ser Gly Asn Tyr Pro Asn
1 5 10

<210> SEQ ID NO 62

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 62

Gly Thr Lys Phe Leu Ala Pro
1 5

<210> SEQ ID NO 63

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 63

Val Leu Trp Tyr Ser Asn Arg Trp Val
1 5

<210> SEQ ID NO 64

<211> LENGTH: 125

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 64

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15

Ser Leu Lys Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asn Lys Tyr
20 25 30

-continued

Gln	Thr	Val	Val	Thr	Gln	Glu	Pro	Ser	Leu	Thr	Val	Ser	Pro	Gly	Gly
1				5					10					15	
Thr	Val	Thr	Leu	Thr	Cys	Gly	Ser	Ser	Thr	Gly	Ala	Val	Thr	Ser	Gly
			20					25					30		
Asn	Tyr	Pro	Asn	Trp	Val	Gln	Gln	Lys	Pro	Gly	Gln	Ala	Pro	Arg	Gly
		35					40					45			
Leu	Ile	Gly	Gly	Thr	Lys	Phe	Leu	Ala	Pro	Gly	Thr	Pro	Ala	Arg	Phe
	50					55					60				
Ser	Gly	Ser	Leu	Leu	Gly	Gly	Lys	Ala	Ala	Leu	Thr	Leu	Ser	Gly	Val
65					70					75				80	
Gln	Pro	Glu	Asp	Glu	Ala	Glu	Tyr	Tyr	Cys	Val	Leu	Trp	Tyr	Ser	Asn
				85					90					95	
Arg	Trp	Val	Phe	Gly	Gly	Gly	Thr	Lys	Leu	Thr	Val	Leu			
			100					105							

Lys Tyr Ala Met Asn
1 5

Arg Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp Ser
1 5 10 15

-continued

Val Lys Asp

<210> SEQ ID NO 68
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 68

His Gly Asn Phe Gly Asn Ser Tyr Ile Ser Tyr Trp Ala Tyr
1 5 10

<210> SEQ ID NO 69
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 69

Gly Ser Ser Thr Gly Ala Val Thr Ser Gly Asn Tyr Pro Asn
1 5 10

<210> SEQ ID NO 70
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 70

Gly Thr Lys Phe Leu Ala Pro
1 5

<210> SEQ ID NO 71
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 71

Val Leu Trp Tyr Ser Asn Arg Trp Val
1 5

<210> SEQ ID NO 72
<211> LENGTH: 125
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 72

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15

Ser Leu Lys Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asn Lys Tyr
20 25 30

Ala Met Asn Trp Val Arg Gln Ala Pro Gly Lys Cys Leu Glu Trp Val
35 40 45

Ala Arg Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp
50 55 60

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Ser Val Lys Asp Arg Phe Thr Ile Ser Arg Asp Asp Ser Lys Asn Thr
65 70 75 80

Ala Tyr Leu Gln Met Asn Asn Leu Lys Thr Glu Asp Thr Ala Val Tyr
85 90 95

Tyr Cys Val Arg His Gly Asn Phe Gly Asn Ser Tyr Ile Ser Tyr Trp
100 105 110

Ala Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115 120 125

<210> SEQ ID NO 73
 <211> LENGTH: 109
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 73

Gln Thr Val Val Thr Gln Glu Pro Ser Leu Thr Val Ser Pro Gly Gly
1 5 10 15

Thr Val Thr Leu Thr Cys Gly Ser Ser Thr Gly Ala Val Thr Ser Gly
20 25 30

Asn Tyr Pro Asn Trp Val Gln Gln Lys Pro Gly Gln Ala Pro Arg Gly
35 40 45

Leu Ile Gly Gly Thr Lys Phe Leu Ala Pro Gly Thr Pro Ala Arg Phe
50 55 60

Ser Gly Ser Leu Leu Gly Gly Lys Ala Ala Leu Thr Leu Ser Gly Val
65 70 75 80

Gln Pro Glu Asp Glu Ala Glu Tyr Tyr Cys Val Leu Trp Tyr Ser Asn
85 90 95

Arg Trp Val Phe Gly Cys Gly Thr Lys Leu Thr Val Leu
100 105

<210> SEQ ID NO 74
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 74

Lys Tyr Ala Ile Asn
1 5

<210> SEQ ID NO 75
 <211> LENGTH: 19
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 75

Arg Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp Ala
1 5 10 15

Val Lys Asp

<210> SEQ ID NO 76
 <211> LENGTH: 14
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence

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<220> FEATURE:

<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 76

Ala Gly Asn Phe Gly Ser Ser Tyr Ile Ser Tyr Trp Ala Tyr
1 5 10

<210> SEQ ID NO 77

<211> LENGTH: 14

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 77

Gly Ser Ser Thr Gly Ala Val Thr Ser Gly Asn Tyr Pro Asn
1 5 10

<210> SEQ ID NO 78

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 78

Gly Thr Lys Phe Leu Ala Pro
1 5

<210> SEQ ID NO 79

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 79

Val Leu Trp Tyr Ser Asn Arg Trp Val
1 5

<210> SEQ ID NO 80

<211> LENGTH: 125

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 80

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15

Ser Leu Lys Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asn Lys Tyr
20 25 30

Ala Ile Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ala Arg Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp
50 55 60

Ala Val Lys Asp Arg Phe Thr Ile Ser Arg Asp Ser Lys Asn Thr
65 70 75 80

Val Tyr Leu Gln Met Asn Asn Leu Lys Thr Glu Asp Thr Ala Val Tyr
85 90 95

Tyr Cys Ala Arg Ala Gly Asn Phe Gly Ser Ser Tyr Ile Ser Tyr Trp

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100	105	110
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Ala Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115 120 125

<210> SEQ ID NO 81
<211> LENGTH: 109
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 81

Gln Thr Val Val Thr Gln Glu Pro Ser Leu Thr Val Ser Pro Gly Gly
1 5 10 15

Thr Val Thr Ile Thr Cys Gly Ser Ser Thr Gly Ala Val Thr Ser Gly
20 25 30

Asn Tyr Pro Asn Trp Val Gln Lys Lys Pro Gly Gln Ala Pro Arg Gly
35 40 45

Leu Ile Gly Gly Thr Lys Phe Leu Ala Pro Gly Thr Pro Ala Arg Phe
50 55 60

Ser Gly Ser Leu Ser Gly Gly Lys Ala Ala Leu Thr Leu Ser Gly Val
65 70 75 80

Gln Pro Glu Asp Glu Ala Glu Tyr Tyr Cys Val Leu Trp Tyr Ser Asn
85 90 95

Arg Trp Val Phe Gly Ser Gly Thr Lys Leu Thr Val Leu
100 105

<210> SEQ ID NO 82
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 82

Lys Tyr Ala Met Asn
1 5

<210> SEQ ID NO 83
<211> LENGTH: 19
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 83

Arg Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp Ala
1 5 10 15

Val Lys Asp

<210> SEQ ID NO 84
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 84

Ala Gly Asn Phe Gly Ser Ser Tyr Ile Ser Tyr Phe Ala Tyr
1 5 10

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<210> SEQ ID NO 85
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 85

Gly Ser Ser Thr Gly Ala Val Thr Ser Gly Asn Tyr Pro Asn
1 5 10

<210> SEQ ID NO 86
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 86

Gly Thr Lys Phe Leu Ala Pro
1 5

<210> SEQ ID NO 87
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 87

Val Leu Tyr Tyr Ser Asn Arg Trp Val
1 5

<210> SEQ ID NO 88
<211> LENGTH: 125
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 88

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15
Ser Leu Lys Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asn Lys Tyr
20 25 30
Ala Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Met Glu Trp Val
35 40 45
Ala Arg Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp
50 55 60
Ala Val Lys Asp Arg Phe Thr Ile Ser Arg Asp Asp Ser Lys Asn Thr
65 70 75 80
Leu Tyr Leu Gln Met Asn Asn Leu Lys Thr Glu Asp Thr Ala Val Tyr
85 90 95
Tyr Cys Val Arg Ala Gly Asn Phe Gly Ser Ser Tyr Ile Ser Tyr Phe
100 105 110
Ala Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115 120 125

<210> SEQ ID NO 89

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<211> LENGTH: 109
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 89

Gln Thr Val Val Thr Gln Glu Pro Ser Leu Thr Val Ser Pro Gly Gly
1 5 10 15

Thr Val Thr Ile Thr Cys Gly Ser Ser Thr Gly Ala Val Thr Ser Gly
 20 25 30

Asn Tyr Pro Asn Trp Ile Gln Lys Lys Pro Gly Gln Ala Pro Arg Gly
 35 40 45

Leu Ile Gly Gly Thr Lys Phe Leu Ala Pro Gly Thr Pro Ala Arg Phe
50 55 60

Ser Gly Ser Leu Glu Gly Gly Lys Ala Ala Leu Thr Leu Ser Gly Val
65 70 75 80

Gln Pro Glu Asp Glu Ala Glu Tyr Tyr Cys Val Leu Tyr Tyr Ser Asn
 85 90 95

Arg Trp Val Phe Gly Ser Gly Thr Lys Leu Thr Val Leu
100 105

<210> SEQ ID NO 90
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 90

Lys Tyr Ala Ile Asn
1 5

<210> SEQ ID NO 91
<211> LENGTH: 19
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 91

Arg Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp Ala
1 5 10 15

Val Lys Asp

<210> SEQ ID NO 92
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 92

Asn Ala Asn Phe Gly Thr Ser Tyr Ile Ser Tyr Phe Ala Tyr
1 5 10

<210> SEQ ID NO 93
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 93

Gly Ser Ser Thr Gly Ala Val Thr Ser Gly Asn Tyr Pro Asn
1 5 10

<210> SEQ ID NO 94

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 94

Gly Thr Lys Phe Leu Ala Pro
1 5

<210> SEQ ID NO 95

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 95

Val Leu Trp Tyr Ser Asn Arg Trp Val
1 5

<210> SEQ ID NO 96

<211> LENGTH: 125

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 96

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15

Ser Leu Lys Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asn Lys Tyr
20 25 30

Ala Ile Asn Trp Val Arg Glu Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ala Arg Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp
50 55 60

Ala Val Lys Asp Arg Phe Thr Ile Ser Arg Asp Asp Ser Lys Asn Thr
65 70 75 80

Ala Tyr Leu Gln Met Asn Asn Leu Lys Thr Glu Asp Thr Ala Val Tyr
85 90 95

Tyr Cys Val Arg Asn Ala Asn Phe Gly Thr Ser Tyr Ile Ser Tyr Phe
100 105 110

Ala Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115 120 125

<210> SEQ ID NO 97

<211> LENGTH: 109

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 97

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Gln	Thr	Val	Val	Thr	Gln	Glu	Pro	Ser	Leu	Thr	Val	Ser	Pro	Gly	Gly
1				5					10					15	
Thr	Val	Thr	Leu	Thr	Cys	Gly	Ser	Ser	Thr	Gly	Ala	Val	Thr	Ser	Gly
			20					25					30		
Asn	Tyr	Pro	Asn	Trp	Val	Gln	Lys	Lys	Pro	Gly	Gln	Ala	Pro	Arg	Gly
		35					40					45			
Leu	Ile	Gly	Gly	Thr	Lys	Phe	Leu	Ala	Pro	Gly	Thr	Pro	Ala	Arg	Phe
	50					55					60				
Ser	Gly	Ser	Leu	Leu	Gly	Gly	Lys	Ala	Ala	Leu	Thr	Leu	Ser	Gly	Val
65					70					75					80
Gln	Pro	Glu	Asp	Glu	Ala	Glu	Tyr	Tyr	Cys	Val	Leu	Trp	Tyr	Ser	Asn
				85					90					95	
Arg	Trp	Val	Phe	Gly	Ser	Gly	Thr	Lys	Leu	Thr	Val	Leu			
			100					105							

1. (canceled)
 2. A molecule comprising a polypeptide chain having the structure:
 - a) VH1-L1-VH2-L2-VL1-L3-VL2-L4-VH3-L1-VH4-L2-VL3-L3-VL4, or
 - b) VH1-L1-VH2-L2-VL1-L3-VL2-L4-Half-Life Extending moiety-L5-VH3-L1-VH4-L2-VL3-L3-VL4,wherein VH1, VH2, VH3, and VH4 are immunoglobulin heavy chain variable regions, VL1, VL2, VL3, and VL4 are immunoglobulin light chain variable regions, and L1, L2 and L3 are linkers, wherein L1 is at least 10 amino acids, L2 is at least 10 amino acids and L3 is at least 10 amino acids, and wherein the total amino acids of L1, L2 and L3 is at least 35 amino acids, and wherein the molecule can bind to an immune effector cell and a target cell, and wherein the half-life extending moiety is a single chain immunoglobulin Fc region ("scFc").
 3. (canceled)
 4. The molecule of claim 2, wherein the half-life extending moiety is an scFc from a human IgG1, IgG2, or IgG4 antibody.
 5. The molecule of claim 4, wherein the scFc polypeptide chain comprises one or more alterations that inhibit Fc gamma receptor (FcγR) binding and/or one or more alterations that extends half life
 6. The molecule of claim 2, wherein the VH1, VH2, VH3, VH4, VL1, VL2, VL3, and VL4 all have different sequences.
 7. The molecule of claim 2, wherein the VH2 and VH4 sequence comprise SEQ ID NO: 41 and the VL2 and VH4 sequence comprise SEQ ID NO: 42.
 8. The molecule of claim 2, wherein L1, L2 and L3 are different or the same lengths.
 9. (canceled)
 10. The molecule of claim 2, wherein L1 and L2 are the same length.
 11. The molecule of claim 2, wherein L1 and L3 are the same length.
 12. The molecule of claim 2, wherein L2 and L3 are the same length.
 13. The molecule of claim 2, wherein the amino acid sequence of L1 is at least 10 amino acids long, the amino acid sequence of L2 is at least 15 amino acids long, and the amino acid sequence of L3 is at least 15 amino acids long.
 14. The molecule of claim 2, wherein the molecule exhibits enhanced stability as compared to a molecule having a structure of VH1-Linker-VL1-Linker-VH2-Linker-VL2-Linker-VH3-Linker-VL3-Linker-VH4-Linker-VL4, or VH1-Linker-VL1-Linker-VH2-Linker-VL2-Linker-Half-Life Extending Moiety-Linker-VH3-Linker-VL3-Linker-VH4-Linker-VL4.
 15. The molecule of claim 1, wherein the molecule exhibits enhanced in vitro expression as compared to a molecule having a structure of VH1-Linker-VL1-Linker-VH2-Linker-VL2-Linker-VH3-Linker-VL3-Linker-VH4-Linker-VL4 or VH1-Linker-VL1-Linker-VH2-Linker-VL2-Linker-Half-Life Extending Moiety-Linker-VH3-Linker-VL3-Linker-VH4-Linker-VL4.
 16. The molecule of claim 2, wherein the effector cell expresses an effector cell protein that is part of a human T cell receptor (TCR)-CD3 complex.
 17. The molecule of claim 16, wherein the effector cell protein is the CD3& chain.
 - 18.-20. (canceled)
 21. A method of manufacturing the molecule of claim 2 comprising (1) culturing a host cell comprising one or more nucleic acid(s) encoding the molecule under conditions so as to express the molecule and (2) recovering the molecule from the cell mass or cell culture supernatant.
 22. A method of treating a cancer patient comprising administering to the patient a therapeutically effective amount of the molecule of claim 2.
 23. The method of claim 22, wherein a chemotherapeutic agent, a non-chemotherapeutic anti-neoplastic agent, and/or radiation is administered to the patient concurrently with, before, or after administration of the molecule.
 - 24.-25. (canceled)
 26. A pharmaceutical composition comprising the molecule of claim 2.
 27. (canceled)
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